CSEE 3827: Fundamentals of Computer Systems, Spring 2026

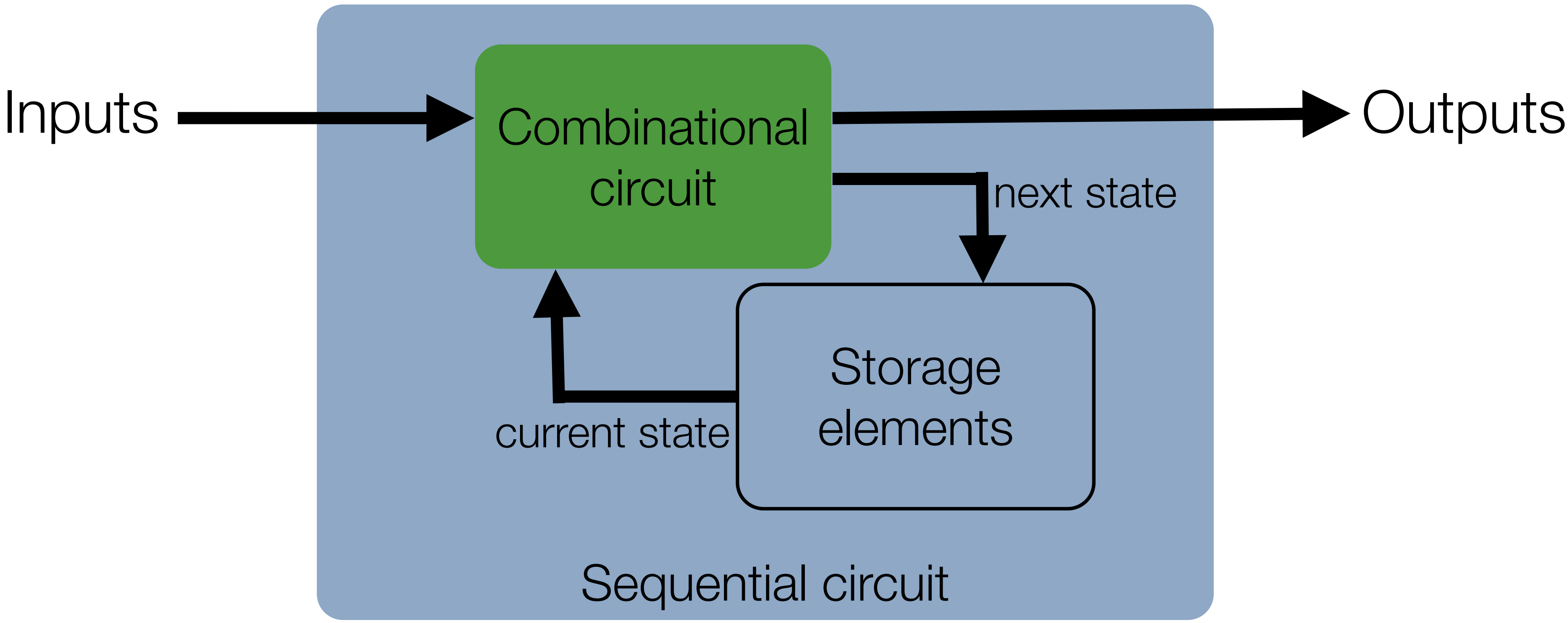
Lecture 6

Prof. Dan Rubenstein (danr@cs.columbia.edu)

Agenda (M&K 4.4-4.7, 6.1-6.3)

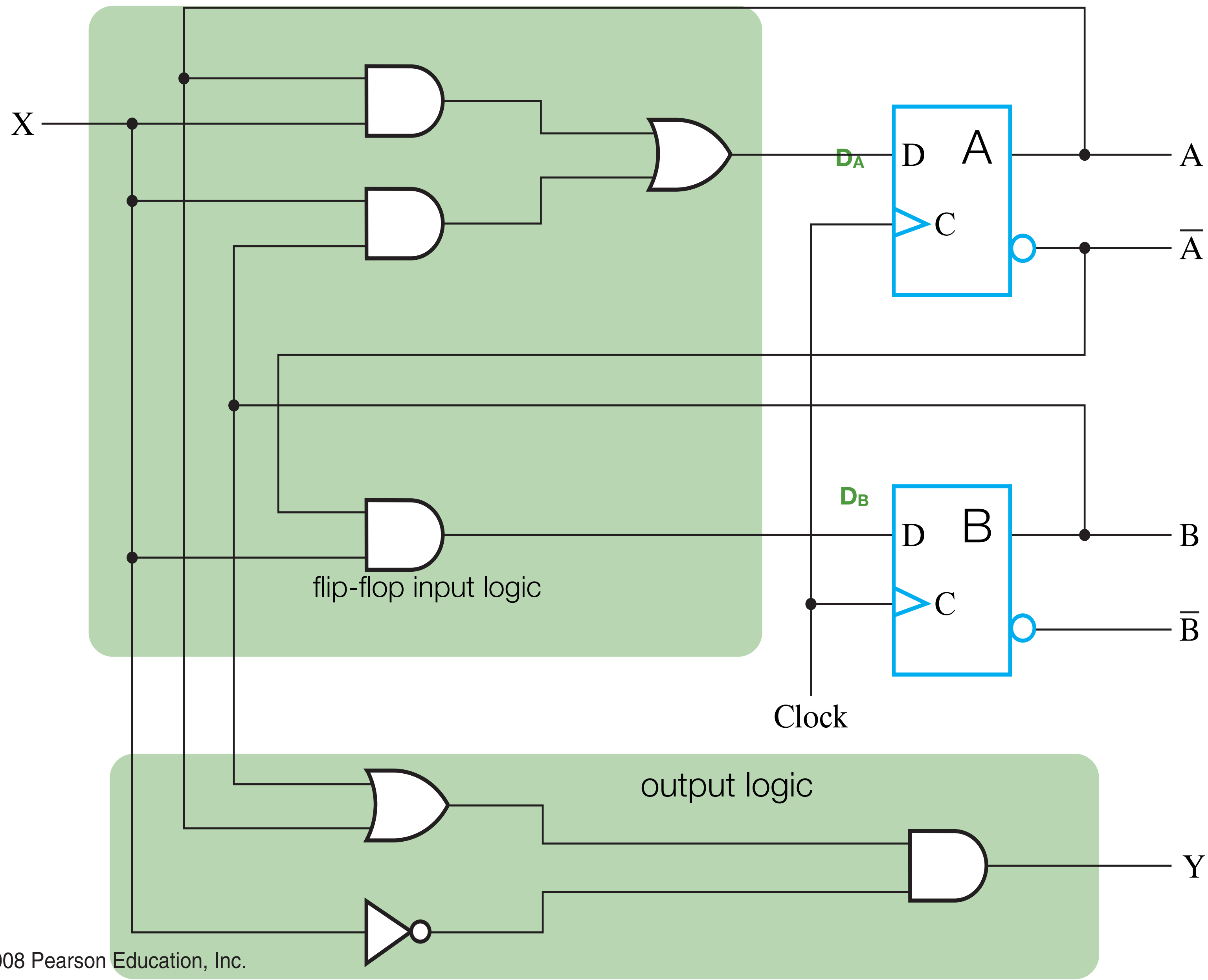
- Sequent Circuit Analysis & Design: State Machines
  - Circuit Analysis (w.r.t. time)
  - Mealy State Machines
  - Sequential Circuit Design using Flip-Flops
    - D, T, JK
- PLAs; ROM
- Register Design: Load and Transfer
  - Shifter
  - Counter
  - Adder

Recall: Sequential circuit

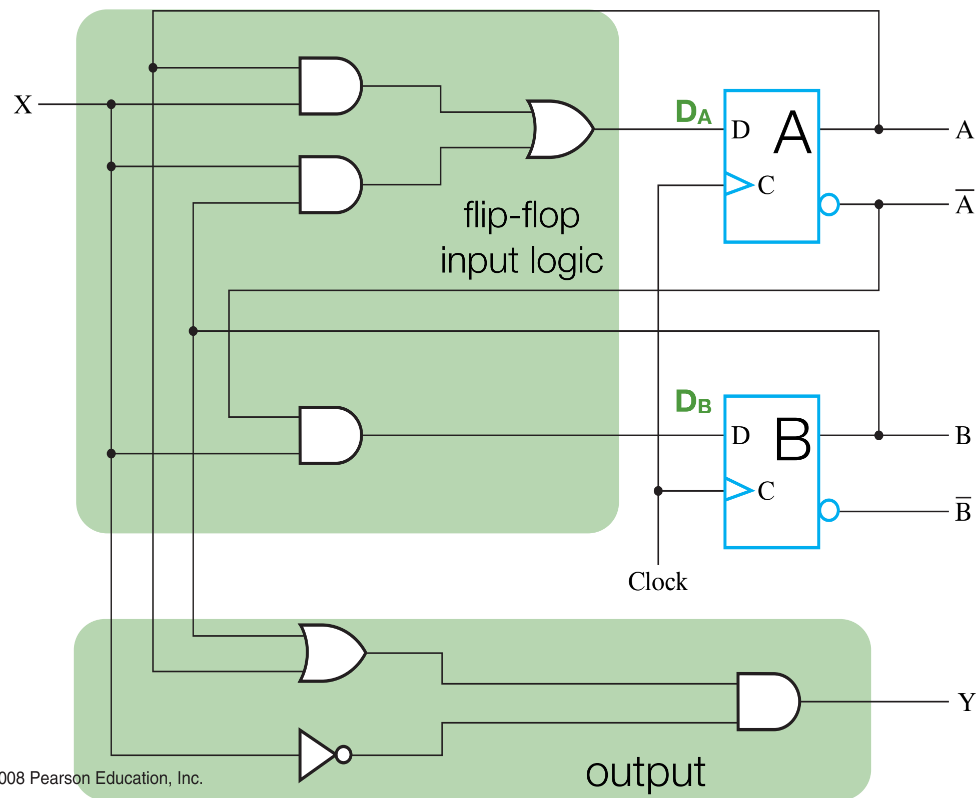


Using Flip-Flops

Sequential circuit (schematic)



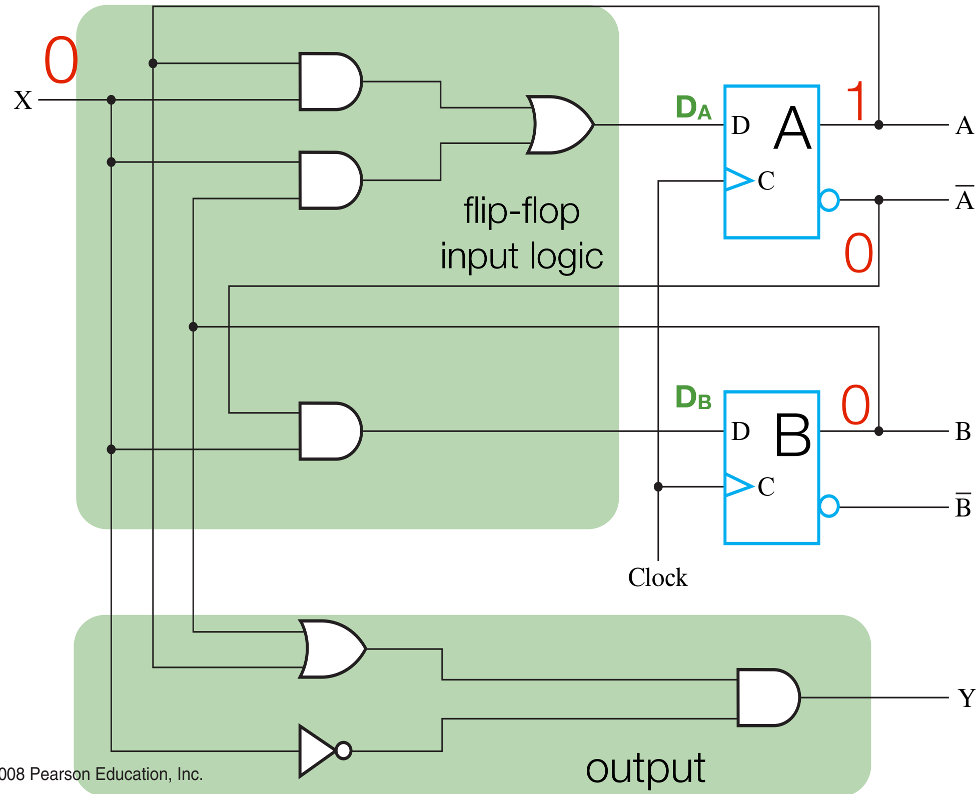
Sequential circuit (schematic)



What happens in Clock cycle t?

- X(t): an arbitrary binary input
  - can assume stays fixed for entire clock cycle
- A(t): output of top D flip-flop (fixed)
- B(t): output of bottom D flip-flop (fixed)
- Y(t): output of circuit is function of X(t), A(t), B(t)
- A(t+1): gets set (as a function of X(t), A(t), B(t))
- B(t+1): gets computed (as a function of X(t), A(t))

Sequential circuit (schematic)



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e.g., suppose A(t)=1, B(t)=0, X(t)=0

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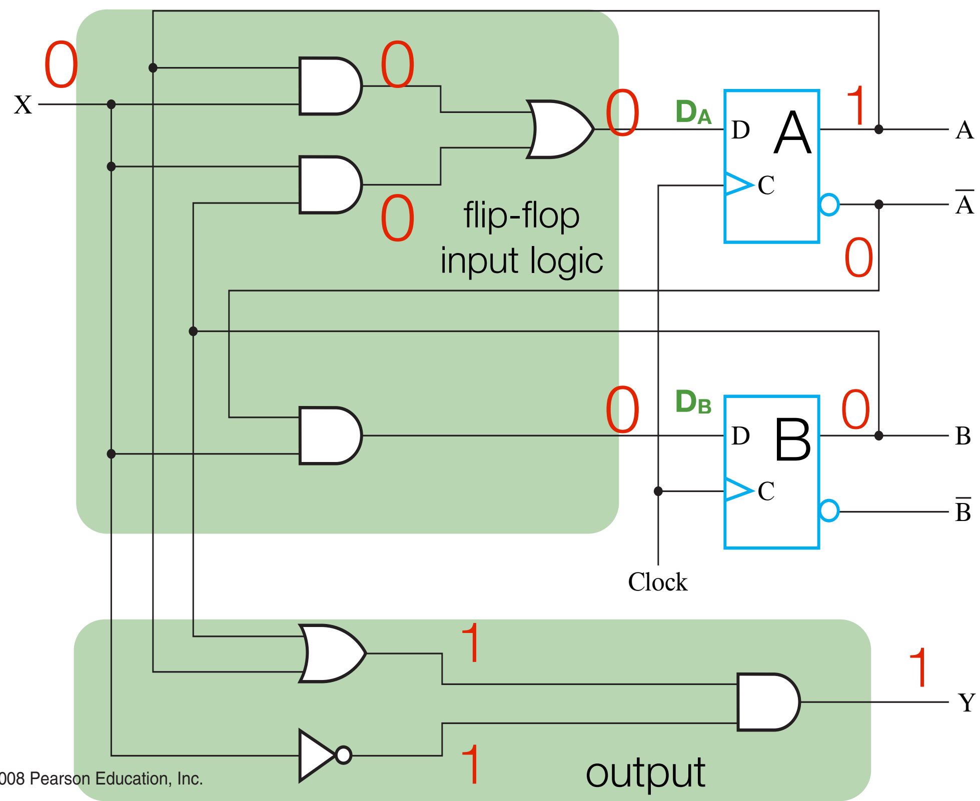

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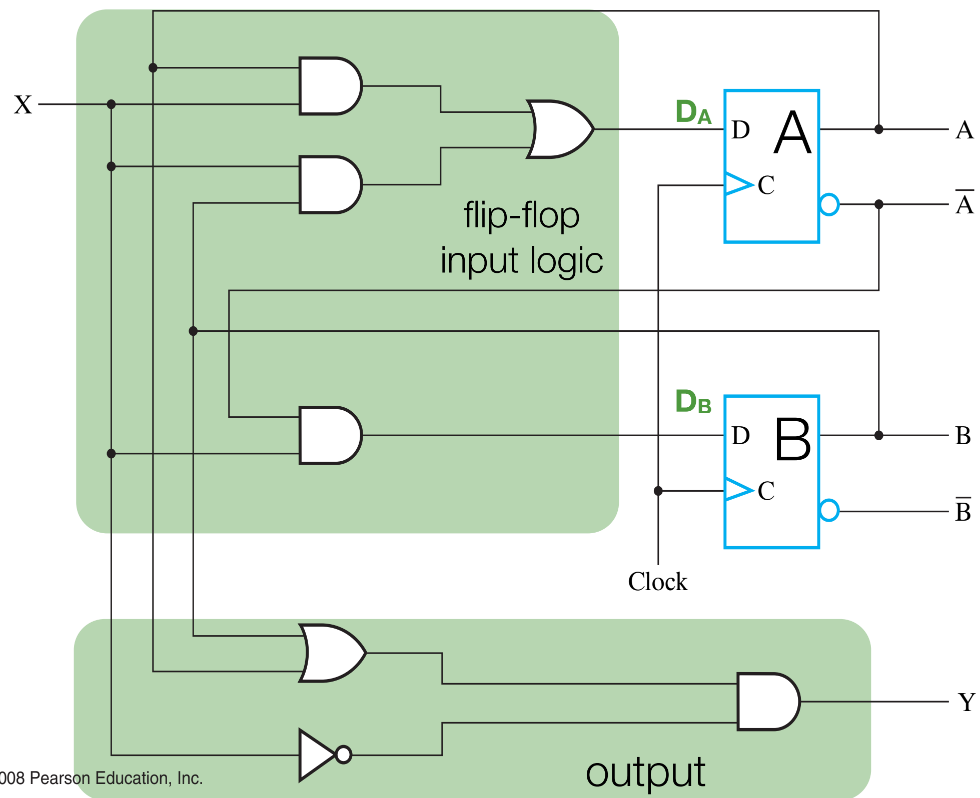
then Y(t)=1, A(t+1)=0, B(t+1)=0

Sequential circuit (schematic)



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Algebraically:

$$A(t+1) = A(t)X(t) + X(t)B(t) \quad [A = AX+XB]$$

$$B(t+1) = \overline{A(t)}X(t) \quad [B = \overline{A}X]$$

$$Y(t) = (A(t) + B(t))\overline{X}(t) \quad [Y = (A+B)\overline{X}]$$

Sequential circuit (schematic)

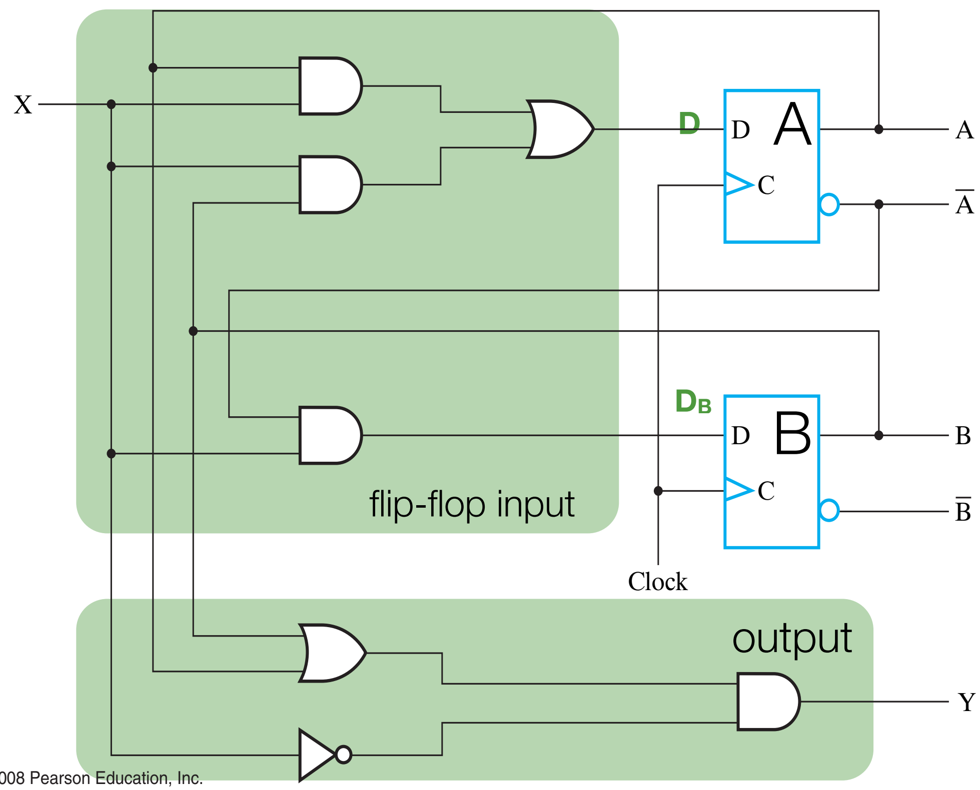
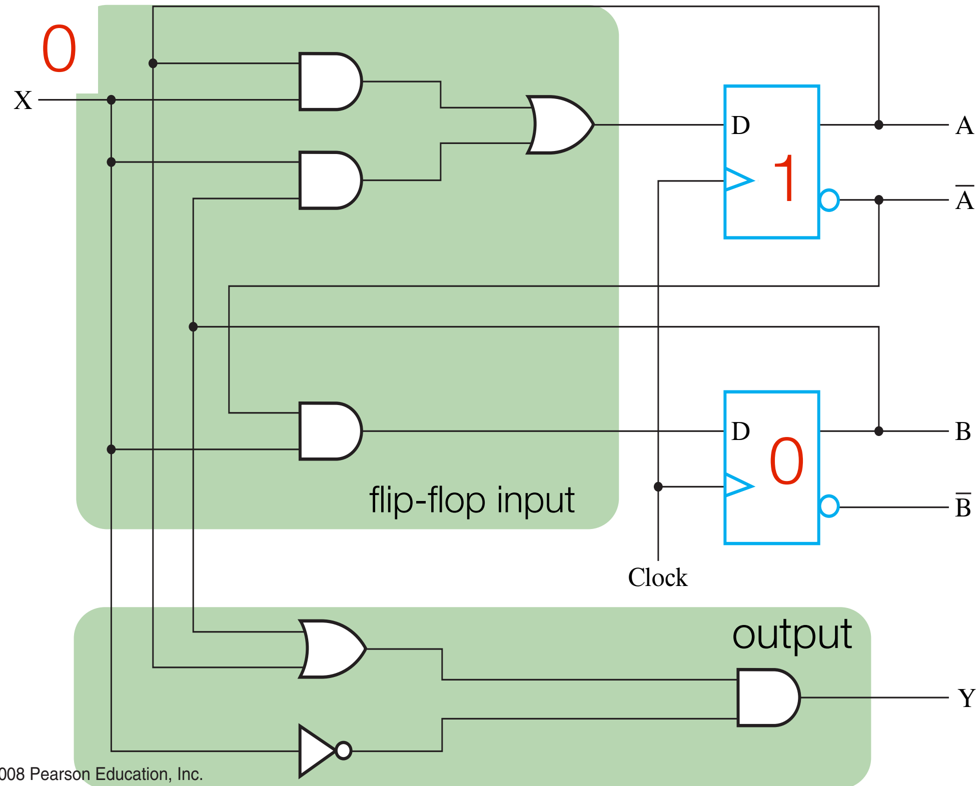


TABLE 5-1
State Table for Circuit of Figure 5-15

Present State		Input	Next State		Output
A(t)	B(t)	X(t)	A(t+1)	B(t+1)	Y(t)
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	0
1	1	0	0	0	1
1	1	1	1	0	0

← Time t → Time t+1 Time t

Sequential circuit (schematic)



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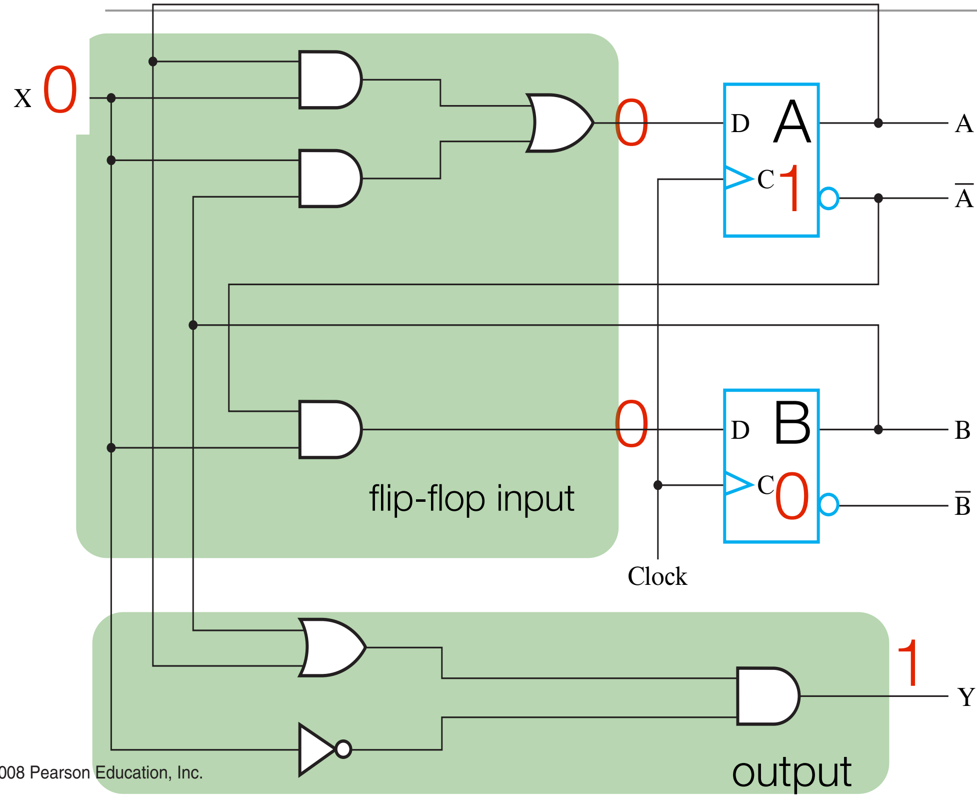
0	1	0	0	0	1
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	0
1	1	0	0	0	1
1	1	1	1	0	0
⟵ Time t ⟶			Time t+1		Time t

e.g., suppose A(t)=1, B(t)=0, X(t)=0

Sequential circuit (schematic)



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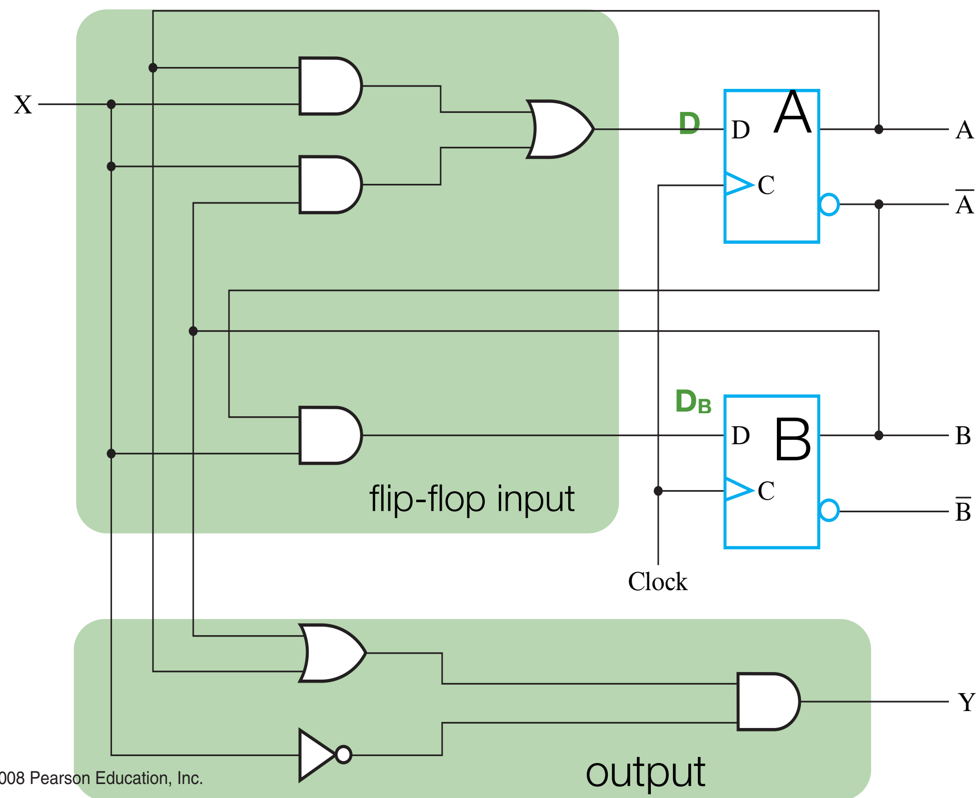
0	1	0	0	0	1
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	0
1	1	0	0	0	1
1	1	1	1	0	0

⟵ Time t ⟶ Time t+1 Time t

e.g., suppose A(t)=1, B(t)=0, X(t)=0

Then Y(t) = 1, A(t+1)=0, B(t+1)=0

Sequential circuit (schematic)



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1	1	0	0	0	1
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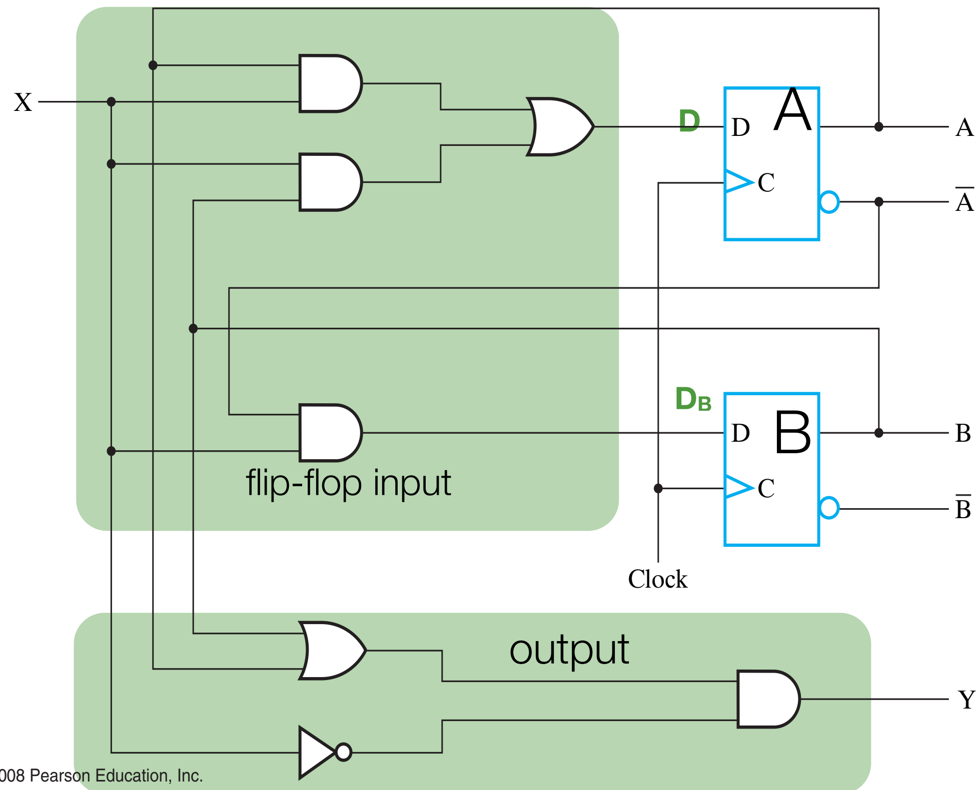
$A(t+1) = A(t)X(t) + X(t)B(t)$ $[A = AX+XB]$

$B(t+1) = \overline{A}(t)X(t)$ $[B = \overline{A}X]$

$Y(t) = (A(t) + B(t))\overline{X}(t)$ $[Y = (A+B)\overline{X}]$

At time t, next cycle's (t+1) FF's are set, but combinational output uses current (cycle t) FF values

Sequential circuit (schematic)



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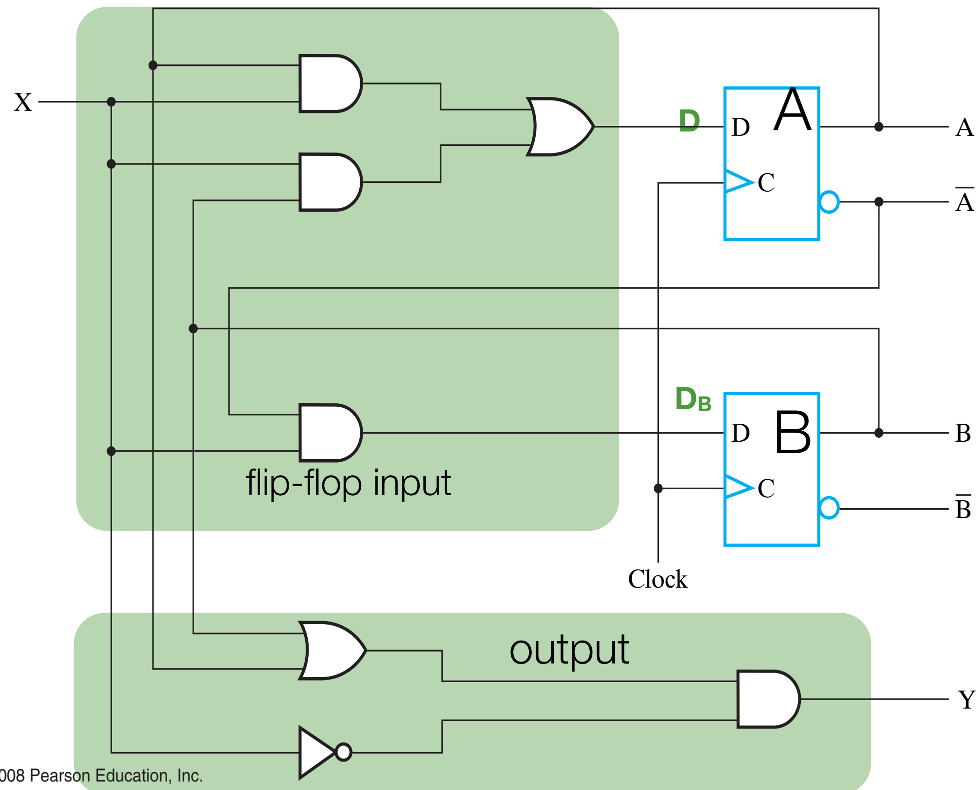
Example

t	A(t)	B(t)	X(t)	Y(t)
0	0	0	0	0
1	0	0	1	0
2	0	1	1	0
3	1	1	0	1
4	0	0	1	0
5	0	1	0	1
6	0	0	0	0
7	0	0	1	0
8	0	1	1	0
9	1	1	0	1

Showing behavior over time - not a truth table!

Sequential circuit (schematic)



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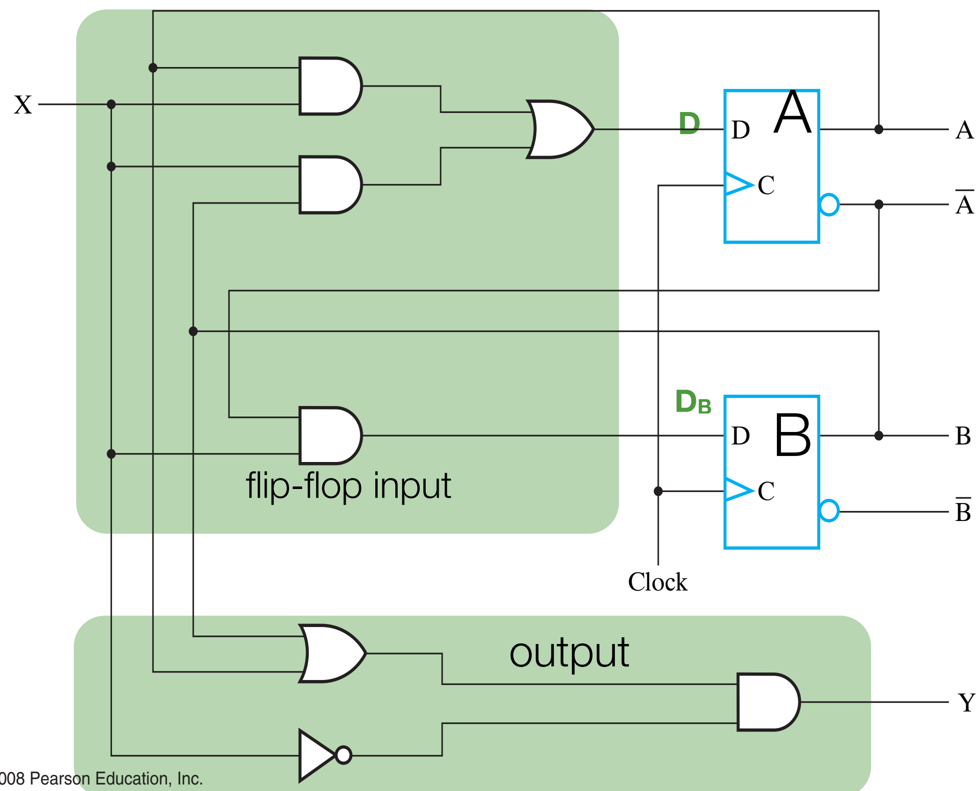
6	0	0	0	0
7	0	0	1	0
8	0	1	1	0
9	1	1	0	1

Green: current clock cycle
Red: current values to be used as function parameters
Blue: values being determined

Sequential circuit (schematic)



Example

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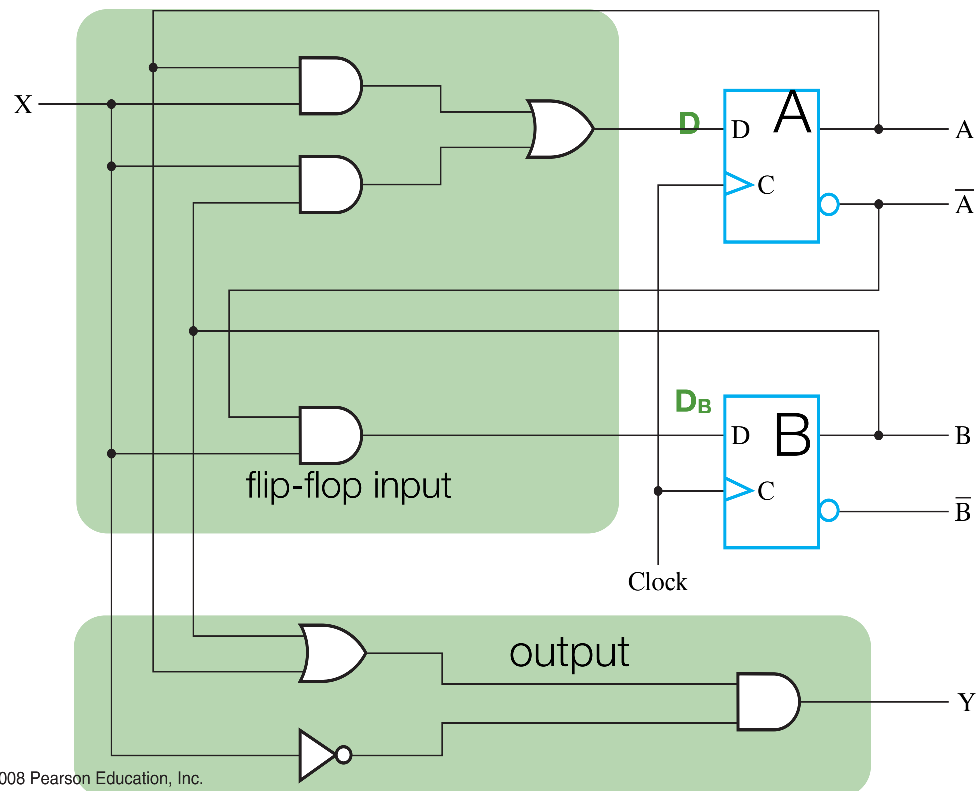

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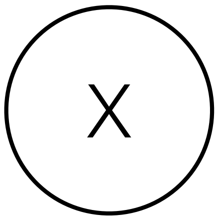
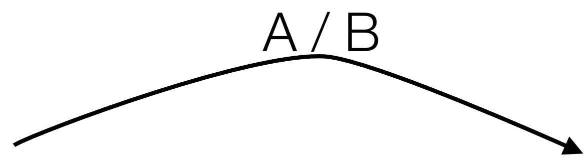
State Machines

State Machines

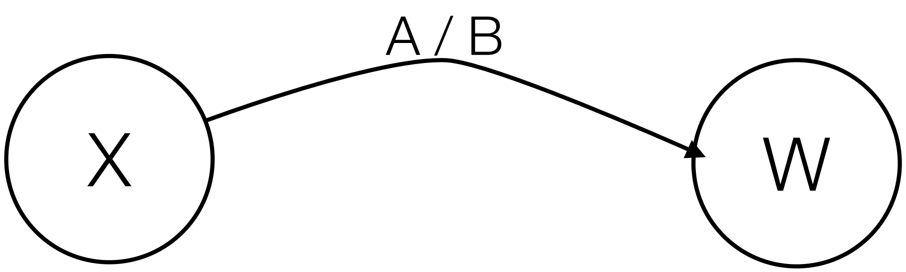
- We study the Mealy form (other form is Moore)



State with label “X”



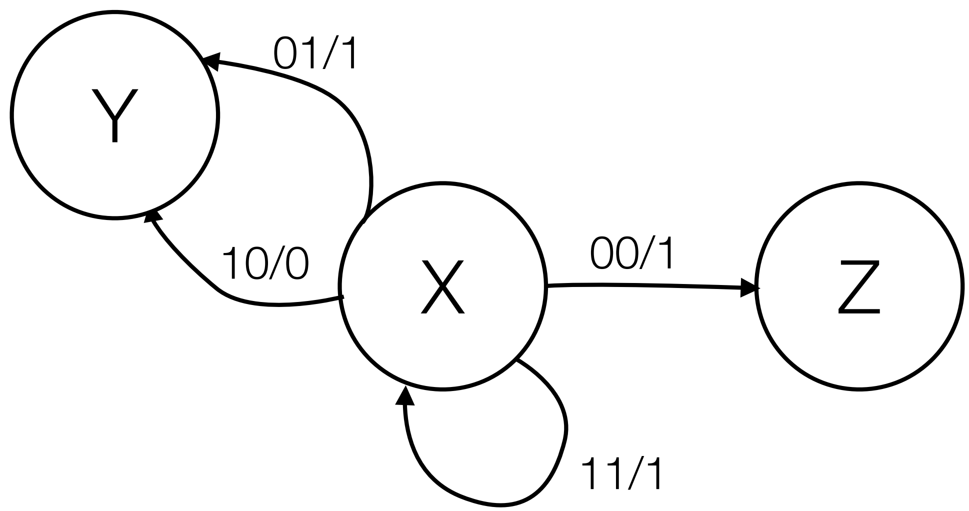
Transition with input “A”, outputs “B”



Time t: in state X, get input A, output B

Time t+1: in state W

Example: state X with 2 inputs, 1 output



Example State Machine

- 1 input, 1 output
- Let X = # of 1's input so far, Y = # of 0's input so far (incl current input).
- Output 1 whenever X-Y (excluding time t input) = 0 (mod 3)

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- Pseudocode:

```
Int X=0; int Y=0; bool j;
While (1==1){
    Input j;
    Int output_val = (X-Y)%3;
    If (j==1){
        X = X+1;
    }
    Else Y=Y+1;
    If output_val ==0{
        Output 1;
    }
    Else Output 0;
}
```

Over time, X and Y might grow very large, e.g., after 1,000,000 iterations, X+Y=1,000,000

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    Else Y=Y+1;
    If output_val ==0{
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    }
    Else Output 0;
}
```

- Reduced State Pseudocode:

```
Int D=0; bool j;
While (1==1){
    Input j;
    Int D_T = D; // temp D
    If (j==1){
        If (D_T==2){
            D_T=0;
        }
        Else D_T=D_T + 1;
    }
    Elsif (D_T==0){
        D_T=2;
    }
    Else D_T=D_T - 1;
    If (D==0){
        Output 1;
    }
    Else Output 0;
    D=D_T;
}
```

Just have variable D that keeps track of (X-Y) mod 3

NOTE: D only changed at very end of iteration

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- 1 input, 1 output
- Let X = # of 1's input so far, Y = # of 0's input so far (incl current input).
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      }
      Else D_T=D_T - 1;
      If (D==0){
            Output 1;
      }
      Else Output 0;
      D=D_T;
}
```

Just have variable D that keeps track of (X-Y) mod 3

NOTE: D only changed at very end of iteration

D could be implemented using 2 bits (takes values 0,1,2)

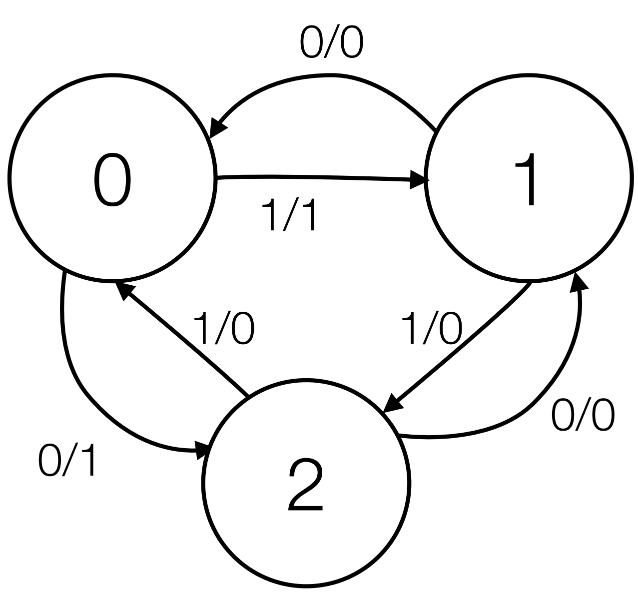
Also note: Output value depends on D, not D_T, i.e., not affected by current input

Example State Machine

- 1 input, 1 output
- Let D = # of 1's input so far - # of 0's input so far (incl current input) (mod 3)
- Output 1 whenever D (excluding time t input) = 0

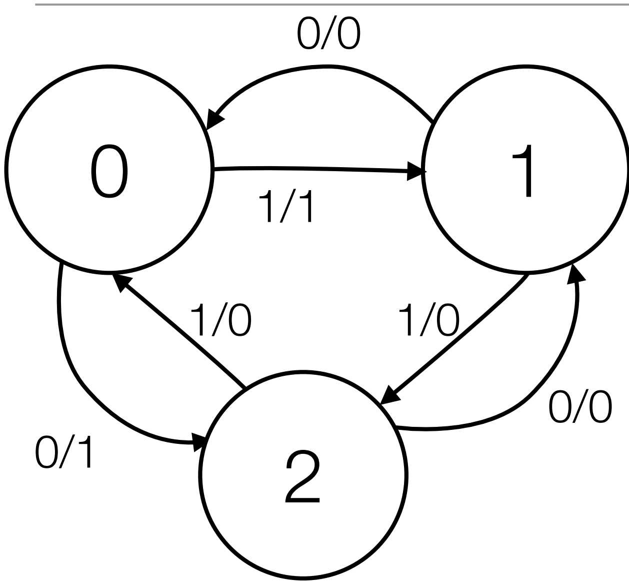
```
Int D=0; bool j;
While (1==1){
      Input j;
      Int D_T = D; // temp D
      If (j==1){
            If (D_T==2){
                  D_T=0;
            }
            Else D_T=D_T + 1;
      }
      Elsif (D_T==0){
            D_T=2;
      }
      Else D_T=D_T - 1;
      If (D==0){
            Output 1;
      }
      Else Output 0;
      D=D_T;
   }
```

"From" State label is value of D mod 3 not including current input.
"To" State is the value including current input



We want to output 1 when we start in state 0 (before considering the input)

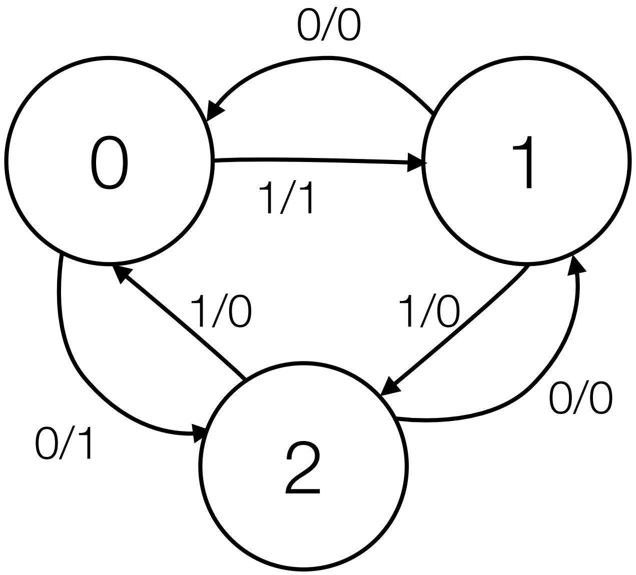
Example of the "X-Y = 0 mod 3"



t	D(t)	In	D(t+1)
0	0	0	2
1	2	0	1
2	1	1	2
3	2	0	1
4	1	1	2
5	2	1	0
6	0	0	2
7	2	1	0
8	0	0	2

Some arbitrarily chosen input sequence

Example of the “X-Y = 0 mod 3”



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3	2	0	0	1
4	1	1	0	2
5	2	1	0	0
6	0	0	1	2
7	2	1	0	0
8	0	0	1	2

Out(t) = 1 when and only when D(t) = 0!

Slight Tweak to problem

- 1 input, 1 output
- Let X = # of 1’s input so far, Y = # of 0’s input so far (incl current input).
- Output 1 whenever X-Y (including time t input) = 0 (mod 3)
  - (Previously we excluded current input)

Slight Tweak to problem

- 1 input, 1 output
- Let X = # of 1's input so far, Y = # of 0's input so far (incl current input).
- Output 1 whenever X-Y (including time t input) = 0 (mod 3)
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- Tweak to pseudocode

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            }
            Else D_T=D_T + 1;
      }
      Elsif (D_T==0){
            D_T=2;
      }
      Else D_T=D_T - 1;
      ~~If (D==0){~~      If (D_T=0){
            Output 1;
      }
      Else Output 0;
      D=D_T;
}
```

Slight Tweak to problem

- 1 input, 1 output
- Let X = # of 1's input so far, Y = # of 0's input so far (incl current input).
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    ~~If (D==0){~~      If (D_T=0){
        Output 1;
    }
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}
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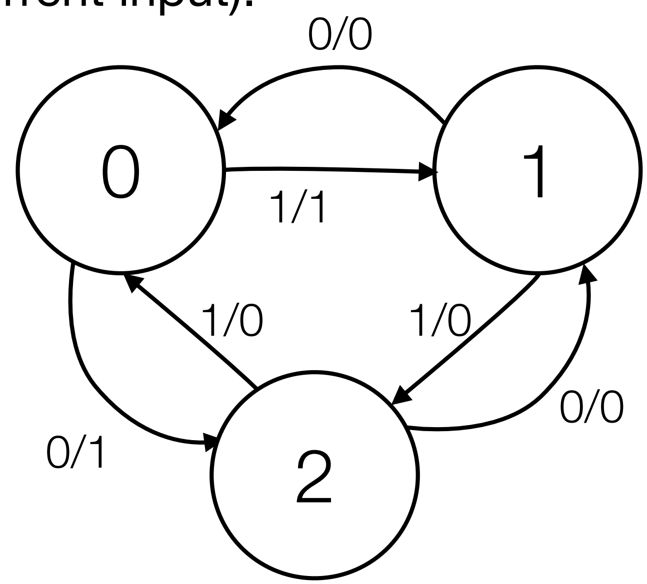
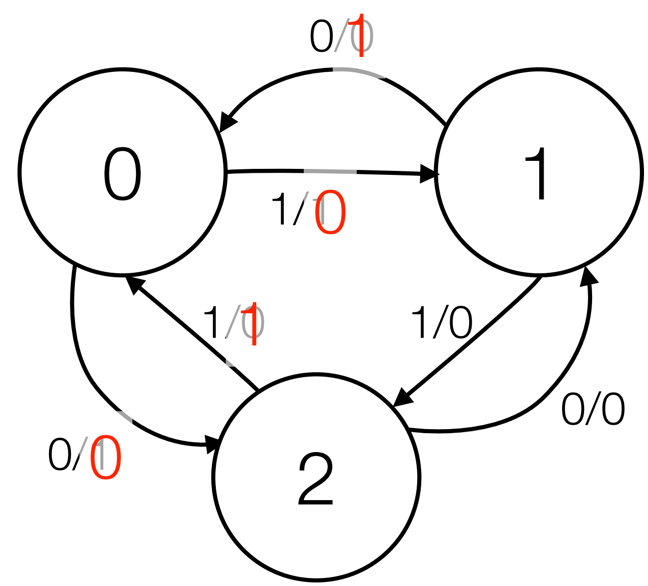
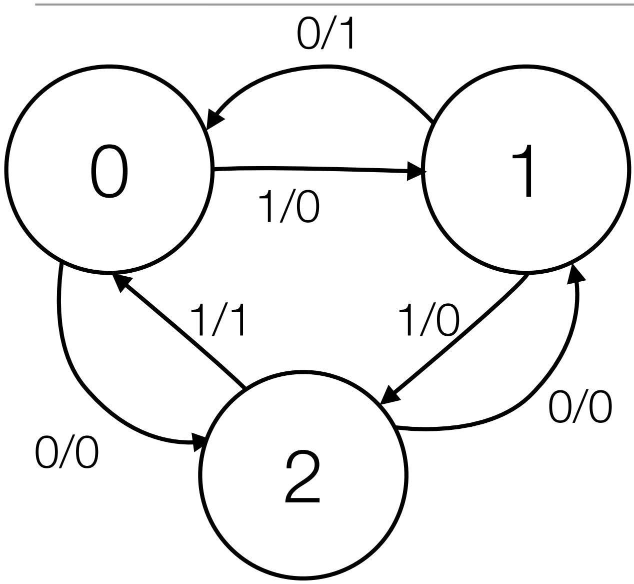
Original State Machine



Revised: the output @ cycle t depends on the input @ cycle t



Example of the alternate "X-Y = 0 mod 3"



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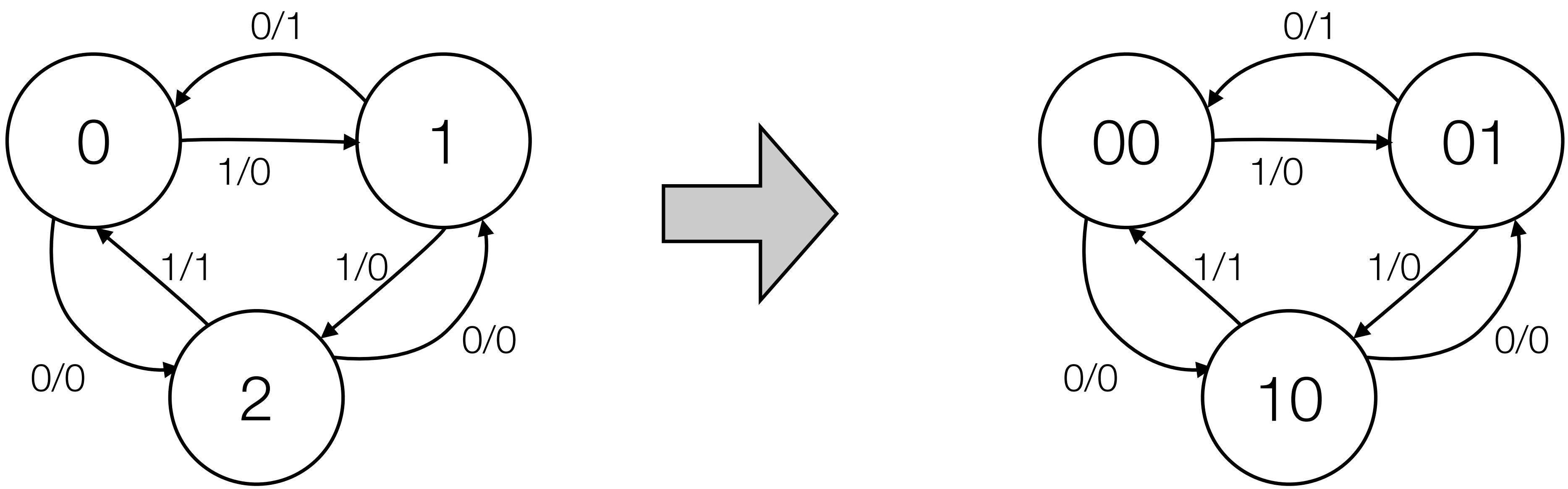
0	0	0	2	0
1	0	1	0	1
2	1	1	2	0
3	2	0	1	0
4	1	1	2	0
5	2	1	0	1
6	0	0	2	0
7	2	1	0	1
8	0	0	2	0

As constructed: Out(t) = 1 when D(t+1) = 0 mod 3

Using 2nd example (input @ time t affects output)

- 1 input, 1 output
- Let X = # of 1's input so far, Y = # of 0's input so far.
- Output 1 whenever X-Y (including time t input) = 0 (mod 3)

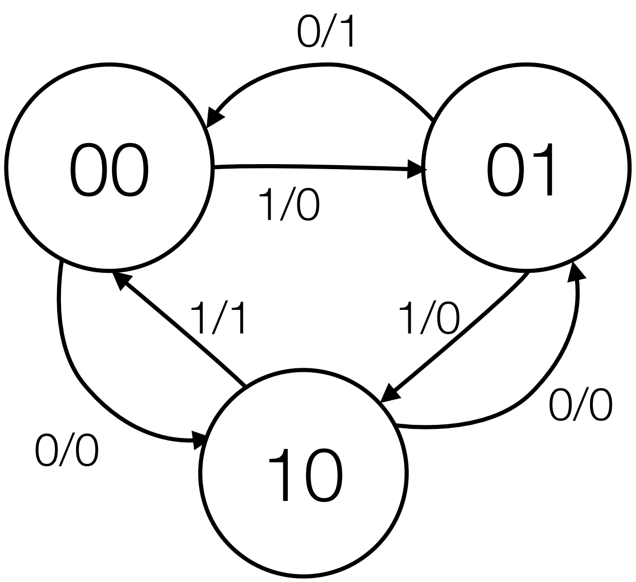


State label is current value of X-Y mod 3

Binary Labeling

From State Machine to Sequential Circuit

- Flip-Flop state represents current "state" of the State Machine
  - 1 FF per bit in the state representation



Let FF A store the high bit, B the low bit

A_{cur}	B_{cur}	In	A_{next}	B_{next}	Out
0	0	0	1	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	1	0	0
1	0	0	0	1	0
1	0	1	0	0	1
1	1	0	X	X	X
1	1	1	X	X	X

Design with D Flip-Flops

- D FF's are easy: we input the value to the FF that we want it set to

A_{cur}	B_{cur}	In	A_{next}	B_{next}	Out
0	0	0	1	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	1	0	0
1	0	0	0	1	0
1	0	1	0	0	1
1	1	0	X	X	X
1	1	1	X	X	X

A_{cur}	B_{cur}	In	A_{next}	D_A	B_{next}	D_B	Out
0	0	0	1	1	0	0	0
0	0	1	0	0	1	1	0
0	1	0	0	0	0	0	1
0	1	1	1	1	0	0	0
1	0	0	0	0	1	1	0
1	0	1	0	0	0	0	1
1	1	0	X	X	X	X	X
1	1	1	X	X	X	X	X

New columns indicate what values feed into D FF (mimic "next" values for that FF)

Design with D Flip-Flops Cont'd

- Build K-Maps, get equations for output and FF input vals (in terms of inputs and previous F vals)

A_{cur}	B_{cur}	In	A_{next}	D_A	B_{next}	D_B	Out
0	0	0	1	1	0	0	0
0	0	1	0	0	1	1	0
0	1	0	0	0	0	0	1
0	1	1	1	1	0	0	0
1	0	0	0	0	1	1	0
1	0	1	0	0	0	0	1
1	1	0	X	X	X	X	X
1	1	1	X	X	X	X	X

D_A:

In

	1	0	1	0
A_{cur}	0	0	X	X

B_{cur}

D_B:

In

	0	1	0	0
A_{cur}	1	0	X	X

B_{cur}

Out:

In

	0	0	0	1
A_{cur}	0	1	X	X

B_{cur}

$$D_A = \bar{A}\bar{B}\bar{I} + BI$$

$$D_B = A\bar{I} + \bar{A}\bar{B}I$$

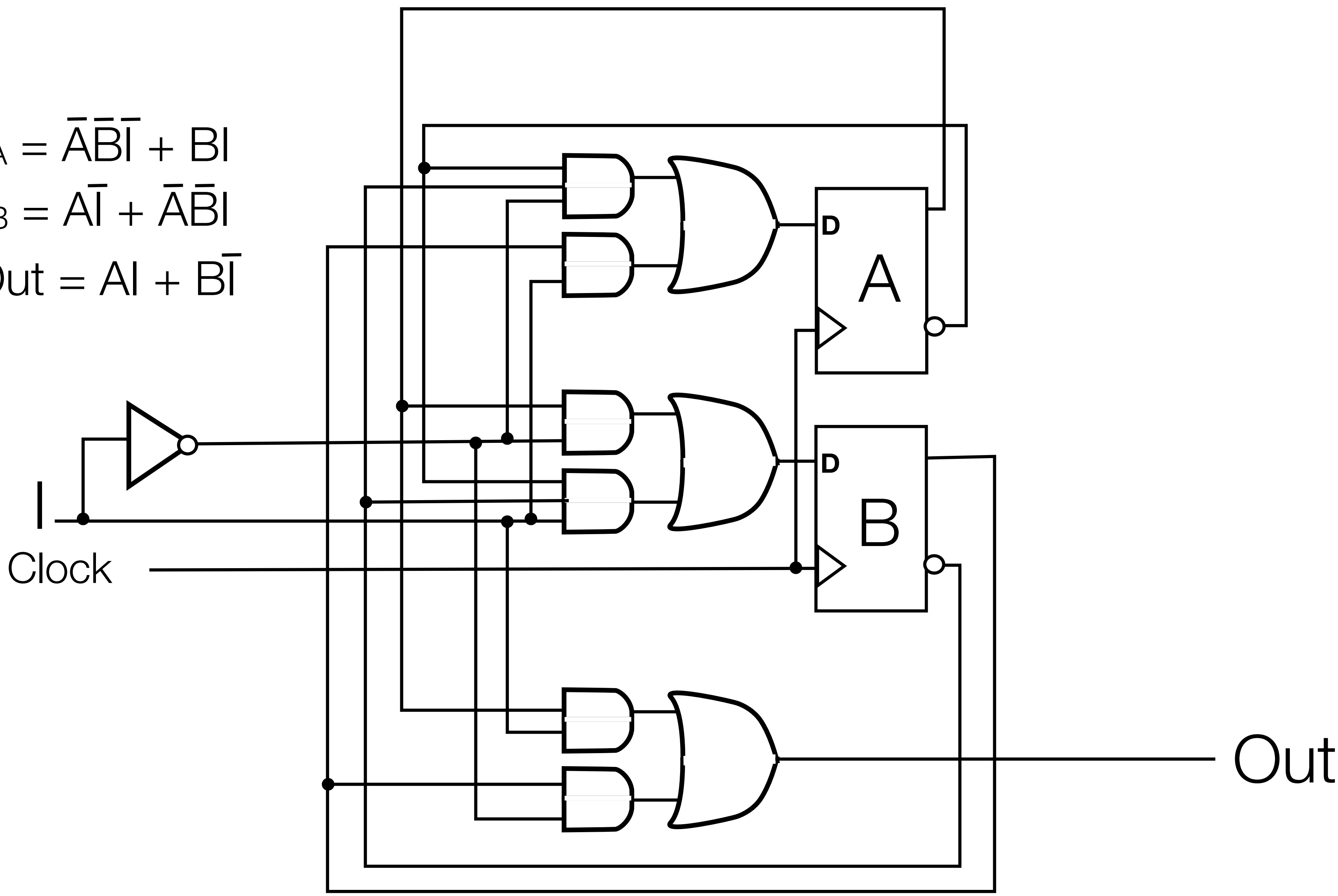
$$Out = AI + B\bar{I}$$

Design with D FF's cont'd

$D_A = \bar{A}\bar{B}\bar{I} + BI$

$D_B = A\bar{I} + \bar{A}\bar{B}I$

$Out = AI + B\bar{I}$



Design with T Flip Flops

- Value fed into T FF is 0 if FF should maintain value, 1 if it should flop

A_{cur}	B_{cur}	In	A_{next}	B_{next}	Out
0	0	0	1	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	1	0	0
1	0	0	0	1	0
1	0	1	0	0	1
1	1	0	X	X	X
1	1	1	X	X	X

A_{cur}	B_{cur}	In	A_{next}	T_A	B_{next}	T_B	Out
0	0	0	1	1	0		0
0	0	1	0	0	1		0
0	1	0	0	0	0		1
0	1	1	1	1	0		0
1	0	0	0	1	1		0
1	0	1	0	1	0		1
1	1	0	X	X	X		X
1	1	1	X	X	X		X

T_A:

In

	1	0	1	0
A_{cur}	1	1	X	X

B_{cur}

$$T_A = A + BI + \bar{B}\bar{I}$$

Design with T Flip Flops

- Value fed into T FF is 0 if FF should maintain value, 1 if it should flop

A_{cur}	B_{cur}	In	A_{next}	B_{next}	Out
0	0	0	1	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	1	0	0
1	0	0	0	1	0
1	0	1	0	0	1
1	1	0	X	X	X
1	1	1	X	X	X

A_{cur}	B_{cur}	In	A_{next}	T_A	B_{next}	T_B	Out
0	0	0	1	1	0	0	0
0	0	1	0	0	1	1	0
0	1	0	0	0	0	1	1
0	1	1	1	1	0	1	0
1	0	0	0	1	1	1	0
1	0	1	0	1	0	0	1
1	1	0	X	X	X	X	X
1	1	1	X	X	X	X	X

T_B:

In

	0	1	1	1
A_{cur}	1	0	X	X

B_{cur}

$$T_B = B + \bar{A}I + A\bar{I}$$

Design with JK Flip-Flops

- Note: to change A_{cur} to the correct A_{next} value, two possible input pairs can be fed into the J,K inputs of a JK Flip-Flop

J	K	Q(t+1)
0	0	Q(t)
0	1	0
1	0	1
1	1	$\overline{Q(t)}$

A_{cur}	A_{next}	J,K
0	0	0,0 or 0,1 (0,X)
0	1	1,0 or 1,1 (1,X)
1	0	0,1 or 1,1 (X,1)
1	1	0,0 or 1,0 (X,0)

Design with JK Flip-Flop

A_{cur}	A_{next}	J,K
0	0	0,0 or 0,1 (0,X)
0	1	1,0 or 1,1 (1,X)
1	0	0,1 or 1,1 (X,1)
1	1	0,0 or 1,0 (X,0)

A_{cur}	B_{cur}	In	A_{next}	B_{next}	Out
0	0	0	1	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	1	0	0
1	0	0	0	1	0
1	0	1	0	0	1
1	1	0	X	X	X
1	1	1	X	X	X

A_{cur}	B_{cur}	In	A_{next}	J_A	K_A	B_{next}	J_B	K_B	Out
0	0	0	1	1	X	0			0
0	0	1	0	0	X	1			0
0	1	0	0	0	X	0			1
0	1	1	1	1	X	0			0
1	0	0	0	X	1	1			0
1	0	1	0	X	1	0			1
1	1	0	X	X	X	X			X
1	1	1	X	X	X	X			X

Design with JK Flip-Flop

A_{cur}	A_{next}	J,K
0	0	0,0 or 0,1 (0,X)
0	1	1,0 or 1,1 (1,X)
1	0	0,1 or 1,1 (X,1)
1	1	0,0 or 1,0 (X,0)

A_{cur}	B_{cur}	In	A_{next}	B_{next}	Out
0	0	0	1	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	1	0	0
1	0	0	0	1	0
1	0	1	0	0	1
1	1	0	X	X	X
1	1	1	X	X	X

A_{cur}	B_{cur}	In	A_{next}	J_A	K_A	B_{next}	J_B	K_B	Out
0	0	0	1	1	X	0	0	X	0
0	0	1	0	0	X	1	1	X	0
0	1	0	0	0	X	0	X	1	1
0	1	1	1	1	X	0	X	1	0
1	0	0	0	X	1	1	1	X	0
1	0	1	0	X	1	0	0	X	1
1	1	0	X	X	X	X	X	X	X
1	1	1	X	X	X	X	X	X	X

Design with JK Flip-Flop

A_{cur}	B_{cur}	In	A_{next}	J_A	K_A	B_{next}	J_B	K_B	Out
0	0	0	1	1	X	0	0	X	0
0	0	1	0	0	X	1	1	X	0
0	1	0	0	0	X	0	X	1	1
0	1	1	1	1	X	0	X	1	0
1	0	0	0	X	1	1	1	X	0
1	0	1	0	X	1	0	0	X	1
1	1	0	X	X	X	X	X	X	X
1	1	1	X	X	X	X	X	X	X

J_A:

In

1	0	1	0
X	X	X	X

A_{cur}

B_{cur}

K_A:

In

X	X	X	X
1	1	X	X

A_{cur}

B_{cur}

J_B:

In

0	1	X	X
1	0	X	X

A_{cur}

B_{cur}

K_B:

In

X	X	1	1
X	X	X	X

A_{cur}

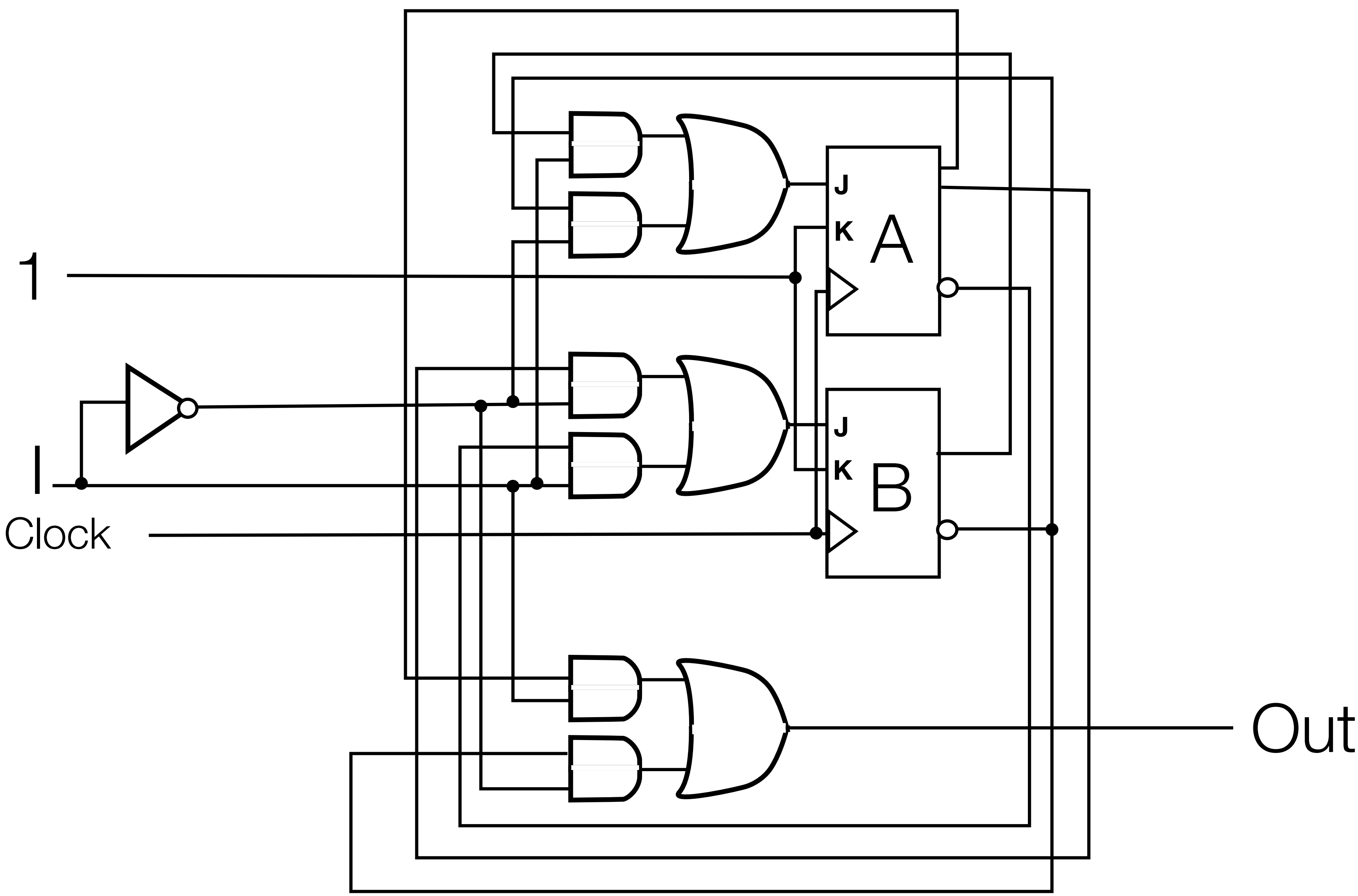
B_{cur}

$J_A = BI + \bar{B}\bar{I}$ $\quad K_A = 1$ $\quad J_B = \bar{A}I + A\bar{I}$ $\quad K_B = 1$

$B \oplus \bar{I}$ $\quad A \oplus I$

Design with JK FF's cont'd

$J_A = BI + \overline{B}\overline{I}$ $K_A = 1$ $J_B = \overline{A}I + A\overline{I}$ $K_B = 1$



2nd Example

Another Example: Message Generator

- Circuit's purpose is to generate a message.
- Each clock cycle, 1-bit input determines 1-bit output
    - When input is held fixed at 0: message is 011000000…
    - When input is held fixed at 1: message is 001111111…
    - Message restarts whenever current input differs from previous

Example

time	Input	Output
0	1	0
1	1	0
2	1	1
3	1	1
4	1	1
5	0	0
6	0	1
7	1	0
8	1	0
9	0	0
10	0	1
11	0	1
12	0	0
13	0	0

Figuring out state machine

- Q: What information needs to be retained?

- 0 input: message is 011000000…
- 1 input : message is 001111111…

Figuring out state machine

- Q: What information needs to be retained?
- A: 2 pieces of info
  - what previous input was (what msg was previously being delivered)
  - how far along we are in the message sequence

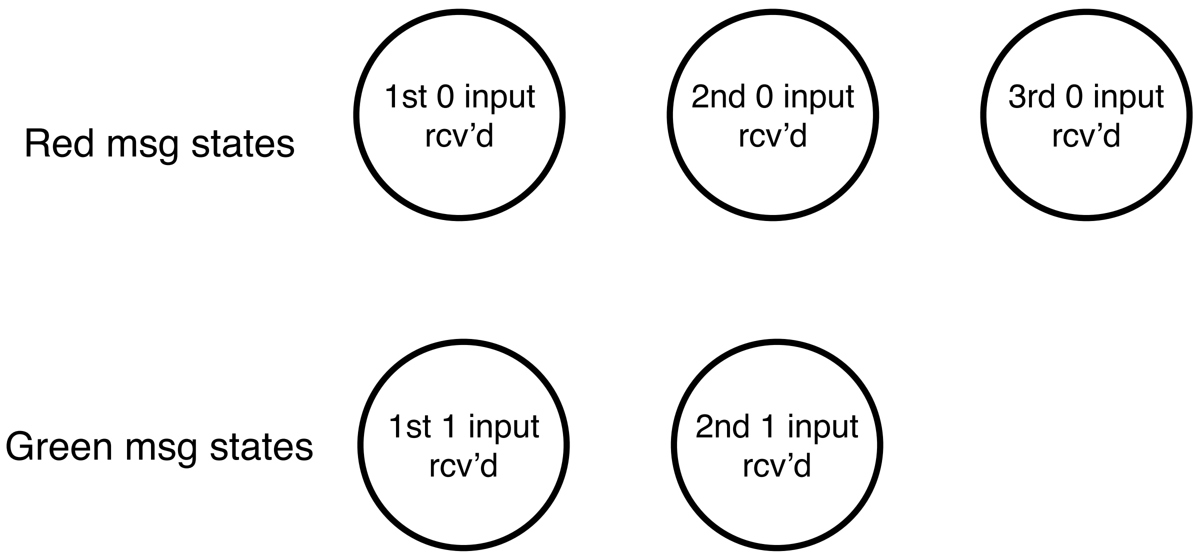
- 0 input: message is 011000000…
- 1 input : message is 001111111…

Figuring out state machine

- Q: What information needs to be retained?
- A: 2 pieces of info
  - what previous input was (what msg was previously being delivered)
  - how far along we are in the message sequence
- Note we only need to track where we are in the sequence to the point where it remains fixed.

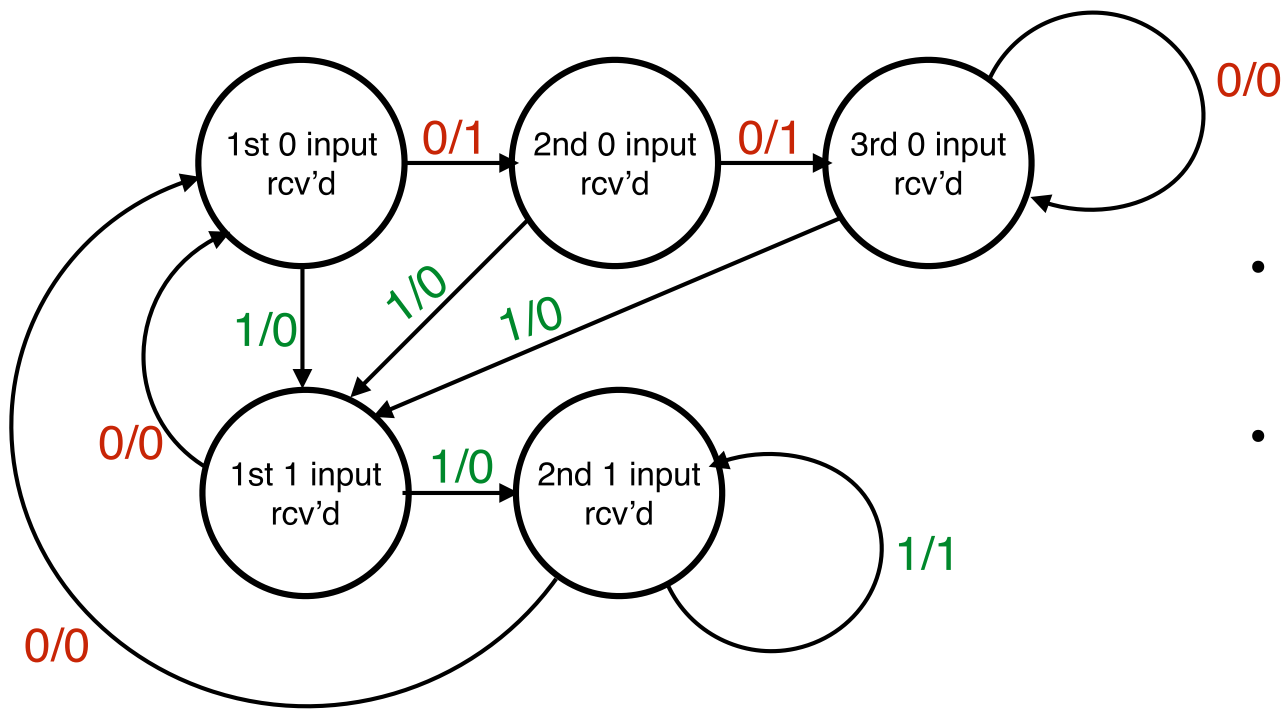
- 0 input: message is 011000000…
- 1 input : message is 001111111…

The States



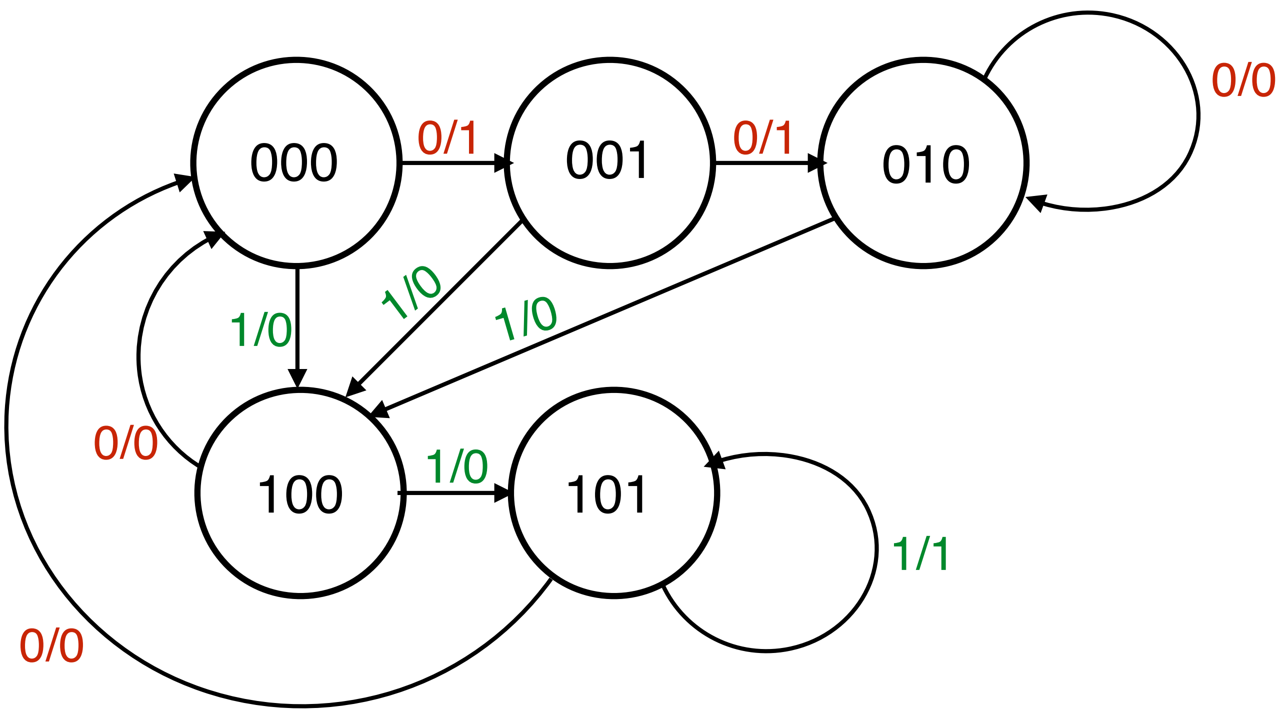
- 0 input: message is 011000000…
- 1 input : message is 001111111…

The States



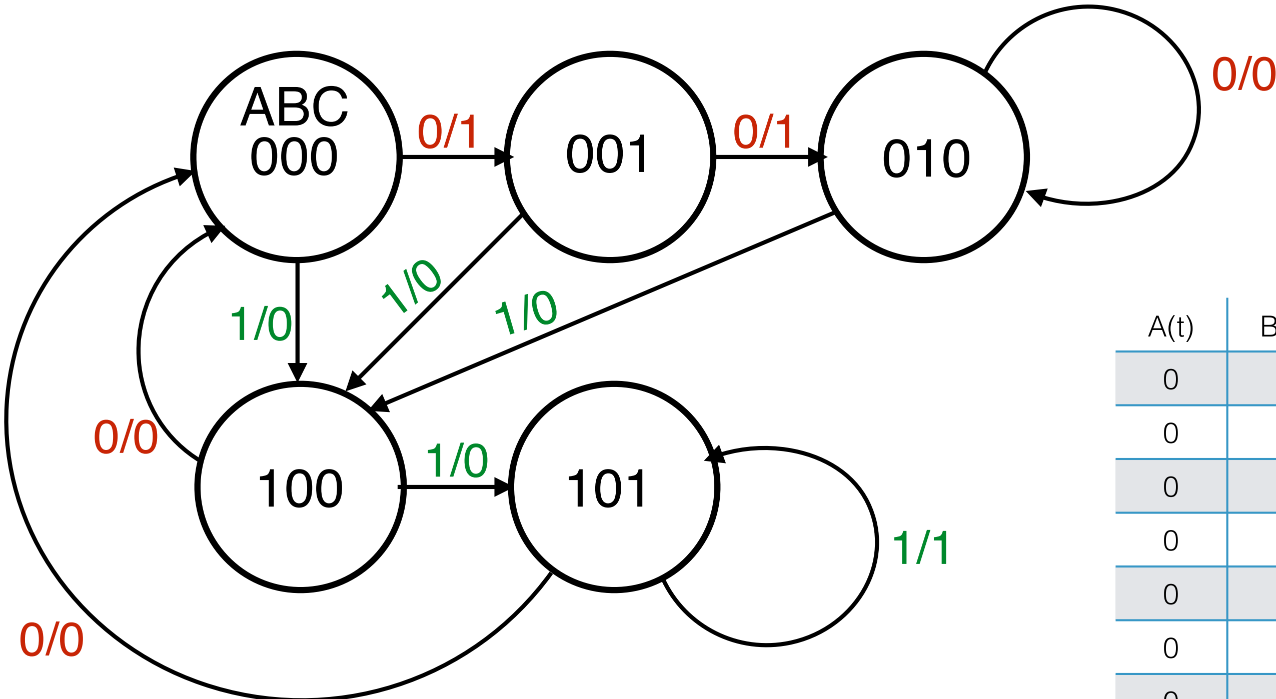
- 0 input: message is 011000000…
- 1 input : message is 001111111…

Number The States



- 5 states: each state needs a 3-bit labelling so that all states are unique
- Labelling can be arbitrary

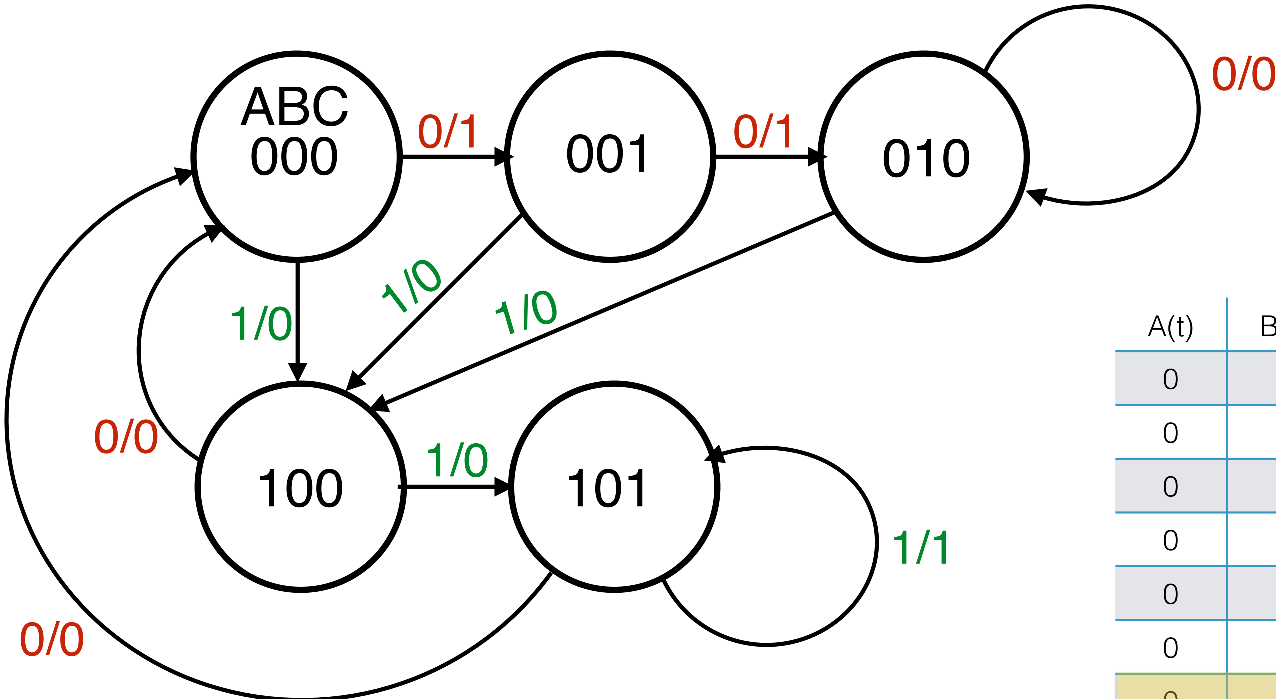
Convert State Machine to Table



Convert state machine to table form: consider all possible values of A(t),B(t),C(t), In(t)

A(t)	B(t)	C(t)	In(t)	Out(t)	A(t+1)	B(t+1)	C(t+1)
0	0	0	0				
0	0	0	1				
0	0	1	0				
0	0	1	1				
0	1	0	0				
0	1	0	1				
0	1	1	0				
0	1	1	1				
1	0	0	0				
1	0	0	1				
1	0	1	0				
1	0	1	1				
1	1	0	0				
1	1	0	1				
1	1	1	0				
1	1	1	1				

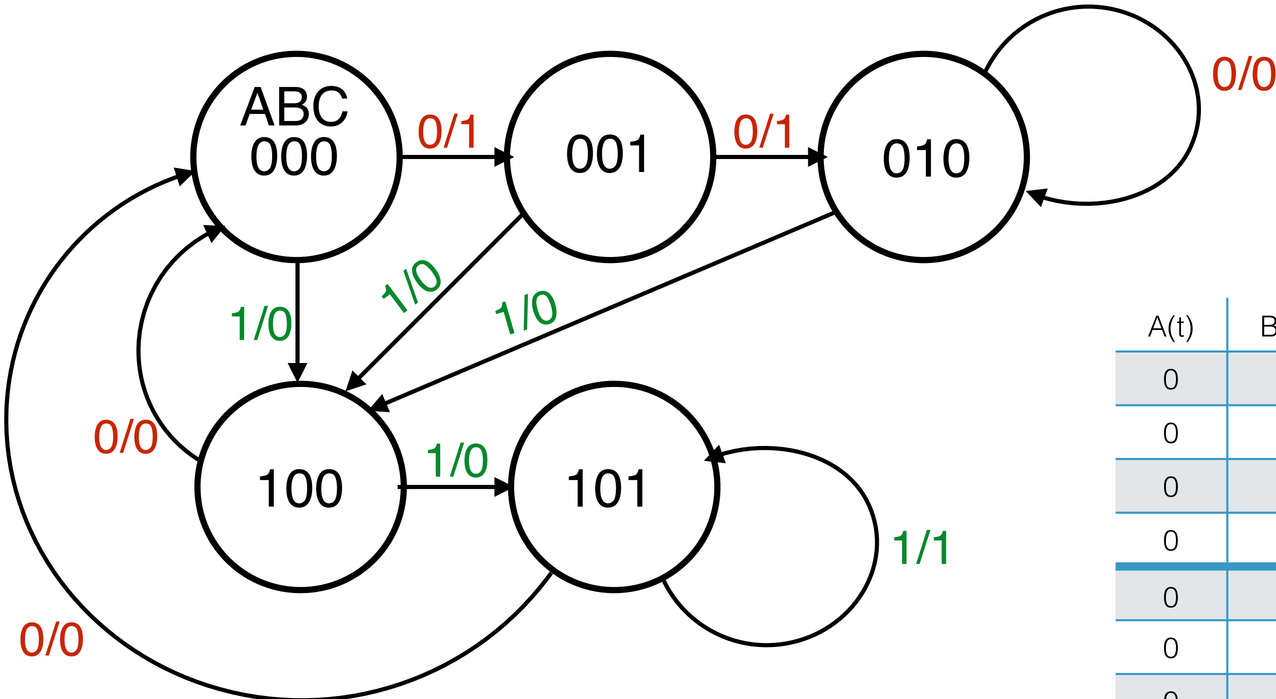
Don’t Cares for non-non-existent states



For table entries for states that don’t exist, can fill in with don’t-care conditions

A(t)	B(t)	C(t)	In(t)	Out(t)	A(t+1)	B(t+1)	C(t+1)
0	0	0	0				
0	0	0	1				
0	0	1	0				
0	0	1	1				
0	1	0	0				
0	1	0	1				
0	1	1	0	X	X	X	X
0	1	1	1	X	X	X	X
1	0	0	0				
1	0	0	1				
1	0	1	0				
1	0	1	1				
1	1	0	0	X	X	X	X
1	1	0	1	X	X	X	X
1	1	1	0	X	X	X	X
1	1	1	1	X	X	X	X

Determining Output



Out indicated on transitions: can already solve for output in terms of inputs via KMap

C(t)

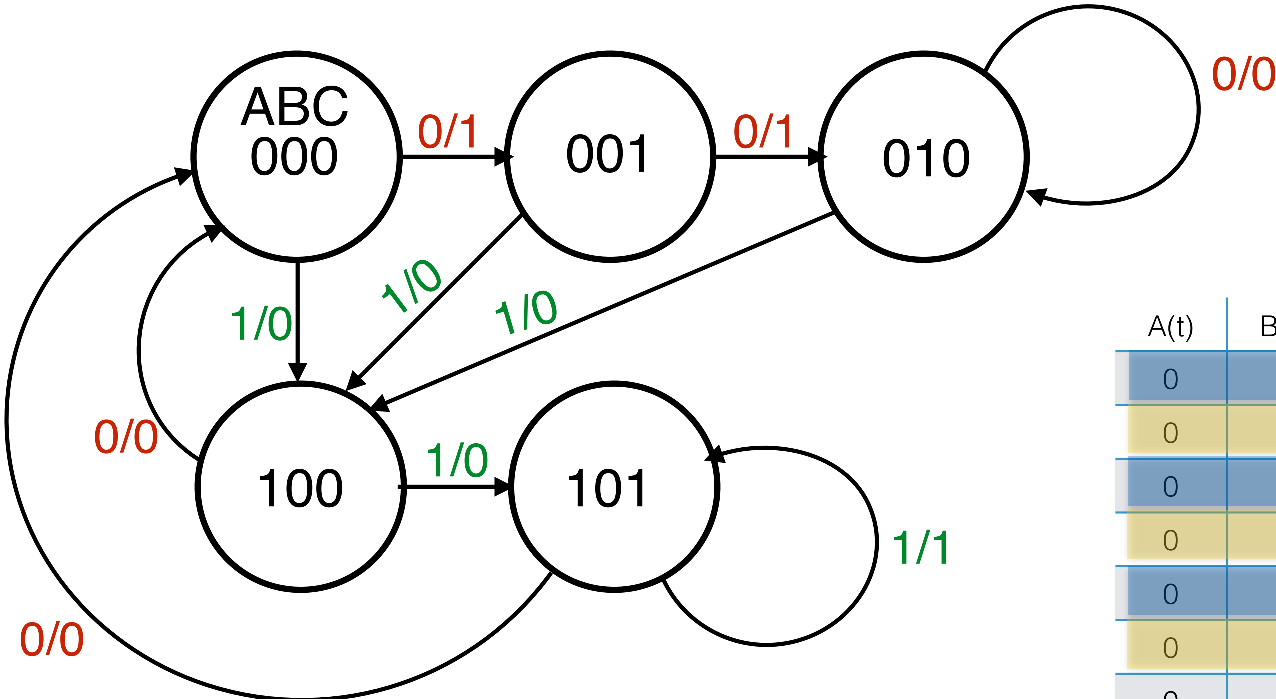
	1	0	0	1	
	0	0	X	X	B(t)
A(t)	X	X	X	X	B(t)
A(t)	0	0	1	0	

In(t)

$$Out(t) = ACI + \bar{A}\bar{B}\bar{I}$$

A(t)	B(t)	C(t)	In(t)	Out(t)	A(t+1)	B(t+1)	C(t+1)
0	0	0	0	1			
0	0	0	1	0			
0	0	1	0	1			
0	0	1	1	0			
0	1	0	0	0			
0	1	0	1	0			
0	1	1	0	X	X	X	X
0	1	1	1	X	X	X	X
1	0	0	0	0			
1	0	0	1	0			
1	0	1	0	0			
1	0	1	1	1			
1	1	0	0	X	X	X	X
1	1	0	1	X	X	X	X
1	1	1	0	X	X	X	X
1	1	1	1	X	X	X	X

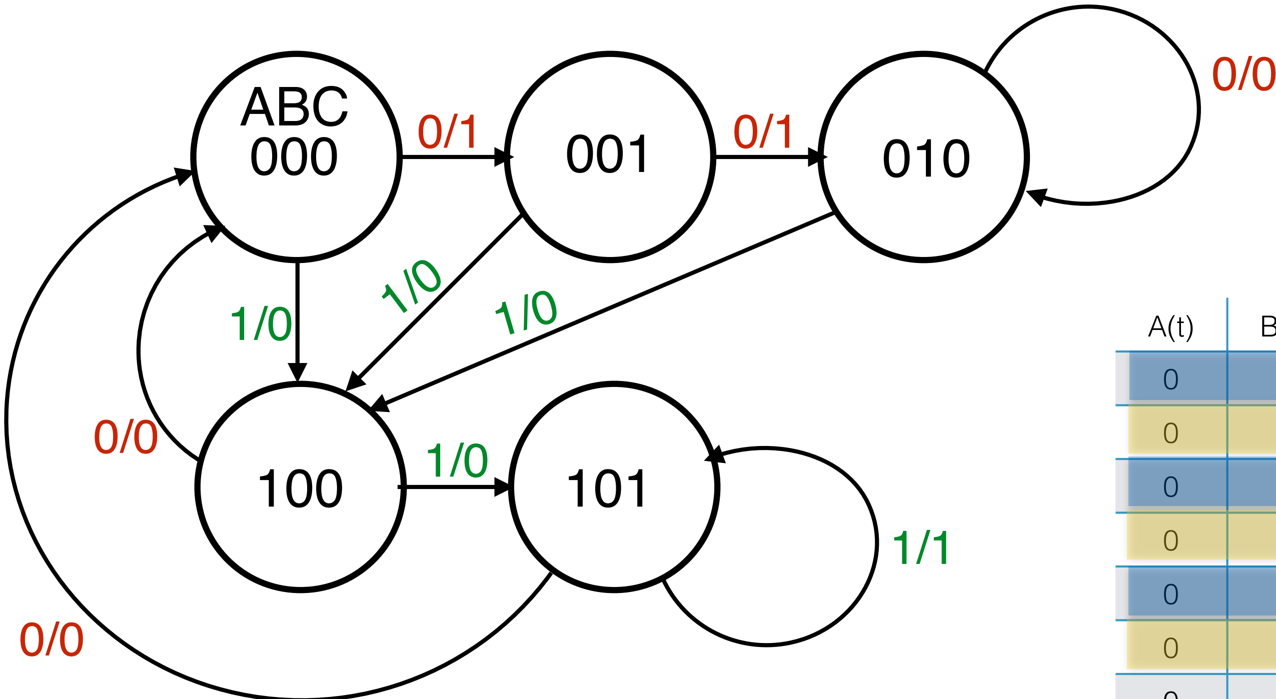
Enter in Next States



Can also determine next cycle's state (A(t+1), B(t+1), C(t+1)) based on current cycle's state (A(t), B(t), C(t)) and current input I(t)

A(t)	B(t)	C(t)	In(t)	Out(t)	A(t+1)	B(t+1)	C(t+1)
0	0	0	0	1	0	0	1
0	0	0	1	0	1	0	0
0	0	1	0	1	0	1	0
0	0	1	1	0	1	0	0
0	1	0	0	0	0	1	0
0	1	0	1	0	1	0	0
0	1	1	0	X	X	X	X
0	1	1	1	X	X	X	X
1	0	0	0	0	0	0	0
1	0	0	1	0	1	0	1
1	0	1	0	0	0	0	0
1	0	1	1	1	1	0	1
1	1	0	0	X	X	X	X
1	1	0	1	X	X	X	X
1	1	1	0	X	X	X	X
1	1	1	1	X	X	X	X

Which type of Flip-Flop to use?



A(t)	B(t)	C(t)	In(t)	Out(t)	A(t+1)	B(t+1)	C(t+1)
0	0	0	0	1	0	0	1
0	0	0	1	0	1	0	0
0	0	1	0	1	0	1	0
0	0	1	1	0	1	0	0
0	1	0	0	0	0	1	0
0	1	0	1	0	1	0	0
0	1	1	0	X	X	X	X
0	1	1	1	X	X	X	X
1	0	0	0	0	0	0	0
1	0	0	1	0	1	0	1
1	0	1	0	0	0	0	0
1	0	1	1	1	1	0	1
1	1	0	0	X	X	X	X
1	1	0	1	X	X	X	X
1	1	1	0	X	X	X	X
1	1	1	1	X	X	X	X

Now must decide which type of F.F. we are using before going further.

Let’s use JK Flip-Flops

JK Flip Flops: J& K columns

A(t)	B(t)	C(t)	In(t)	Out(t)	A(t+1)	$J_A(t)$	$K_A(t)$	B(t+1)	$J_B(t)$	$K_B(t)$	C(t+1)	$J_C(t)$	$K_C(t)$
0	0	0	0	1	0			0			1		
0	0	0	1	0	1			0			0		
0	0	1	0	1	0			1			0		
0	0	1	1	0	1			0			0		
0	1	0	0	0	0			1			0		
0	1	0	1	0	1			0			0		
0	1	1	0	X	X			X			X		
0	1	1	1	X	X			X			X		
1	0	0	0	0	0			0			0		
1	0	0	1	0	1			0			1		
1	0	1	0	0	0			0			0		
1	0	1	1	1	1			0			1		
1	1	0	0	X	X			X			X		
1	1	0	1	X	X			X			X		
1	1	1	0	X	X			X			X		
1	1	1	1	X	X			X			X		

A_{cur}	A_{next}	J,K
0	0	0,0 or 0,1 (0,X)
0	1	1,0 or 1,1 (1,X)
1	0	0,1 or 1,1 (X,1)
1	1	0,0 or 1,0 (X,0)

Add Columns to determine what J and K should be based on current state, current input, and how a particular FF's stored value should change

JK Flip Flops: J& K columns

A(t)	B(t)	C(t)	In(t)	Out(t)	A(t+1)	J_A(t)	K_A(t)	B(t+1)	J_B(t)	K_B(t)	C(t+1)	J_C(t)	K_C(t)
0	0	0	0	1	0			0			1		
0	0	0	1	0	1			0			0		
0	0	1	0	1	0			1			0		
0	0	1	1	0	1			0			0		
0	1	0	0	0	0			1			0		
0	1	0	1	0	1			0			0		
0	1	1	0	X	X			X			X		
0	1	1	1	X	X			X			X		
1	0	0	0	0	0			0			0		
1	0	0	1	0	1			0			1		
1	0	1	0	0	0			0			0		
1	0	1	1	1	1			0			1		
1	1	0	0	X	X			X			X		
1	1	0	1	X	X			X			X		
1	1	1	0	X	X			X			X		
1	1	1	1	X	X			X			X		

A_{cur}	A_{next}	J,K
0	0	0,0 or 0,1 (0,X)
0	1	1,0 or 1,1 (1,X)
1	0	0,1 or 1,1 (X,1)
1	1	0,0 or 1,0 (X,0)

When in state 100, and input is 0, go to state 000

JK Flip Flops: J& K columns

A(t)	B(t)	C(t)	In(t)	Out(t)	A(t+1)	$J_A(t)$	$K_A(t)$	B(t+1)	$J_B(t)$	$K_B(t)$	C(t+1)	$J_C(t)$	$K_C(t)$
0	0	0	0	1	0			0			1		
0	0	0	1	0	1			0			0		
0	0	1	0	1	0			1			0		
0	0	1	1	0	1			0			0		
0	1	0	0	0	0			1			0		
0	1	0	1	0	1			0			0		
0	1	1	0	X	X			X			X		
0	1	1	1	X	X			X			X		
1	0	0	0	0	0			0			0		
1	0	0	1	0	1			0			1		
1	0	1	0	0	0			0			0		
1	0	1	1	1	1			0			1		
1	1	0	0	X	X			X			X		
1	1	0	1	X	X			X			X		
1	1	1	0	X	X			X			X		
1	1	1	1	X	X			X			X		

A_{cur}	A_{next}	J,K
0	0	0,0 or 0,1 (0,X)
0	1	1,0 or 1,1 (1,X)
1	0	0,1 or 1,1 (X,1)
1	1	0,0 or 1,0 (X,0)

When in state 100, and input is 0, go to state 000

i.e., when A(t)=1, B(t)=0, C(t)=0, I(t)=0 then A(t+1)=0, B(t+1)=0, C(t+1)=0

JK Flip Flops: J& K columns

A(t)	B(t)	C(t)	In(t)	Out(t)	A(t+1)	$J_A(t)$	$K_A(t)$	B(t+1)	$J_B(t)$	$K_B(t)$	C(t+1)	$J_C(t)$	$K_C(t)$
0	0	0	0	1	0			0			1		
0	0	0	1	0	1			0			0		
0	0	1	0	1	0			1			0		
0	0	1	1	0	1			0			0		
0	1	0	0	0	0			1			0		
0	1	0	1	0	1			0			0		
0	1	1	0	X	X			X			X		
0	1	1	1	X	X			X			X		
1	0	0	0	0	0			0			0		
1	0	0	1	0	1			0			1		
1	0	1	0	0	0			0			0		
1	0	1	1	1	1			0			1		
1	1	0	0	X	X			X			X		
1	1	0	1	X	X			X			X		
1	1	1	0	X	X			X			X		
1	1	1	1	X	X			X			X		

A_{cur}	A_{next}	J,K
0	0	0,0 or 0,1 (0,X)
0	1	1,0 or 1,1 (1,X)
1	0	0,1 or 1,1 (X,1)
1	1	0,0 or 1,0 (X,0)

Focus just on A for now:

i.e., when A(t)=1, B(t)=0, C(t)=0, I(t)=0 then A(t+1)=0

JK Flip Flops: J& K columns

A(t)	B(t)	C(t)	In(t)	Out(t)	A(t+1)	J_A(t)	K_A(t)	B(t+1)	J_B(t)	K_B(t)	C(t+1)	J_C(t)	K_C(t)
0	0	0	0	1	0			0			1		
0	0	0	1	0	1			0			0		
0	0	1	0	1	0			1			0		
0	0	1	1	0	1			0			0		
0	1	0	0	0	0			1			0		
0	1	0	1	0	1			0			0		
0	1	1	0	X	X			X			X		
0	1	1	1	X	X			X			X		
1	0	0	0	0	0	X	1	0			0		
1	0	0	1	0	1			0			1		
1	0	1	0	0	0			0			0		
1	0	1	1	1	1			0			1		
1	1	0	0	X	X			X			X		
1	1	0	1	X	X			X			X		
1	1	1	0	X	X			X			X		
1	1	1	1	X	X			X			X		

A_{cur}	A_{next}	J,K
0	0	0,0 or 0,1 (0,X)
0	1	1,0 or 1,1 (1,X)
1	0	0,1 or 1,1 (X,1)
1	1	0,0 or 1,0 (X,0)

i.e., when A(t)=1, B(t)=0, C(t)=0, I(t)=0 then A(t+1)=0

so when A(t)=1, B(t)=0, C(t)=0, I(t)=0 then J_A(t)=X, K_A(t)=1

JK Flip Flops: J& K columns

A(t)	B(t)	C(t)	In(t)	Out(t)	A(t+1)	$J_A(t)$	$K_A(t)$	B(t+1)	$J_B(t)$	$K_B(t)$	C(t+1)	$J_C(t)$	$K_C(t)$
0	0	0	0	1	0			0			1		
0	0	0	1	0	1			0			0		
0	0	1	0	1	0			1			0		
0	0	1	1	0	1			0			0		
0	1	0	0	0	0			1			0		
0	1	0	1	0	1			0			0		
0	1	1	0	X	X			X			X		
0	1	1	1	X	X			X			X		
1	0	0	0	0	0	X	1	0	0	X	0	0	X
1	0	0	1	0	1			0			1		
1	0	1	0	0	0			0			0		
1	0	1	1	1	1			0			1		
1	1	0	0	X	X			X			X		
1	1	0	1	X	X			X			X		
1	1	1	0	X	X			X			X		
1	1	1	1	X	X			X			X		

A_{cur}	A_{next}	J,K
0	0	0,0 or 0,1 (0,X)
0	1	1,0 or 1,1 (1,X)
1	0	0,1 or 1,1 (X,1)
1	1	0,0 or 1,0 (X,0)

Same idea for B & C

JK Flip Flops: J& K columns

A(t)	B(t)	C(t)	In(t)	Out(t)	A(t+1)	J_A(t)	K_A(t)	B(t+1)	J_B(t)	K_B(t)	C(t+1)	J_C(t)	K_C(t)
0	0	0	0	1	0	0	X	0	0	X	1	1	X
0	0	0	1	0	1	1	X	0	0	X	0	0	X
0	0	1	0	1	0	0	X	1	1	X	0	X	1
0	0	1	1	0	1	1	X	0	0	X	0	X	1
0	1	0	0	0	0	0	X	1	X	0	0	0	X
0	1	0	1	0	1	1	X	0	X	1	0	0	X
0	1	1	0	X	X	X	X	X	X	X	X	X	X
0	1	1	1	X	X	X	X	X	X	X	X	X	X
1	0	0	0	0	0	X	1	0	0	X	0	0	X
1	0	0	1	0	1	X	0	0	0	X	1	1	X
1	0	1	0	0	0	X	1	0	0	X	0	X	1
1	0	1	1	1	1	X	0	0	0	X	1	X	0
1	1	0	0	X	X	X	X	X	X	X	X	X	X
1	1	0	1	X	X	X	X	X	X	X	X	X	X
1	1	1	0	X	X	X	X	X	X	X	X	X	X
1	1	1	1	X	X	X	X	X	X	X	X	X	X

A_{cur}	**A_{next}**	**J,K**
0	0	0,0 or 0,1 (0,X)
0	1	1,0 or 1,1 (1,X)
1	0	0,1 or 1,1 (X,1)
1	1	0,0 or 1,0 (X,0)

Same idea for all other rows

JK Flip Flops: J& K columns

A(t)	B(t)	C(t)	In(t)	Out(t)	A(t+1)	$J_A(t)$	$K_A(t)$	B(t+1)	$J_B(t)$	$K_B(t)$	C(t+1)	$J_C(t)$	$K_C(t)$
0	0	0	0	1	0	0	X	0	0	X	1	1	X
0	0	0	1	0	1	1	X	0	0	X	0	0	X
0	0	1	0	1	0	0	X	1	1	X	0	X	1
0	0	1	1	0	1	1	X	0	0	X	0	X	1
0	1	0	0	0	0	0	X	1	X	0	0	0	X
0	1	0	1	0	1	1	X	0	X	1	0	0	X
0	1	1	0	X	X	X	X	X	X	X	X	X	X
0	1	1	1	X	X	X	X	X	X	X	X	X	X
1	0	0	0	0	0	X	1	0	0	X	0	0	X
1	0	0	1	0	1	X	0	0	0	X	1	1	X
1	0	1	0	0	0	X	1	0	0	X	0	X	1
1	0	1	1	1	1	X	0	0	0	X	1	X	0
1	1	0	0	X	X	X	X	X	X	X	X	X	X
1	1	0	1	X	X	X	X	X	X	X	X	X	X
1	1	1	0	X	X	X	X	X	X	X	X	X	X
1	1	1	1	X	X	X	X	X	X	X	X	X	X

Just need these columns to get circuitry that feeds into the J,K inputs of each of the Flip-Flops

JK Flip Flops: J& K columns

A(t)	B(t)	C(t)	In(t)	$J_A(t)$	$K_A(t)$	$J_B(t)$	$K_B(t)$	$J_C(t)$	$K_C(t)$
0	0	0	0	0	X	0	X	1	X
0	0	0	1	1	X	0	X	0	X
0	0	1	0	0	X	1	X	X	1
0	0	1	1	1	X	0	X	X	1
0	1	0	0	0	X	X	0	0	X
0	1	0	1	1	X	X	1	0	X
0	1	1	0	X	X	X	X	X	X
0	1	1	1	X	X	X	X	X	X
1	0	0	0	X	1	0	X	0	X
1	0	0	1	X	0	0	X	1	X
1	0	1	0	X	1	0	X	X	1
1	0	1	1	X	0	0	X	X	0
1	1	0	0	X	X	X	X	X	X
1	1	0	1	X	X	X	X	X	X
1	1	1	0	X	X	X	X	X	X
1	1	1	1	X	X	X	X	X	X

J_A:

C(t)

0	1	1	0
0	1	X	X
X	X	X	X
X	X	X	X

B(t)

A(t)

In(t)

$$= In(t) = I$$

K_A:

C(t)

X	X	X	X
X	X	X	X
X	X	X	X
1	0	0	1

B(t)

A(t)

In(t)

$$= \overline{In(t)} = \bar{I}$$

Just need these columns to get circuitry that feeds into the J,K inputs of each of the Flip-Flops

JK Flip Flops: J& K columns

A(t)	B(t)	C(t)	In(t)	J_A(t)	K_A(t)	J_B(t)	K_B(t)	J_C(t)	K_C(t)
0	0	0	0	0	X	0	X	1	X
0	0	0	1	1	X	0	X	0	X
0	0	1	0	0	X	1	X	X	1
0	0	1	1	1	X	0	X	X	1
0	1	0	0	0	X	X	0	0	X
0	1	0	1	1	X	X	1	0	X
0	1	1	0	X	X	X	X	X	X
0	1	1	1	X	X	X	X	X	X
1	0	0	0	X	1	0	X	0	X
1	0	0	1	X	0	0	X	1	X
1	0	1	0	X	1	0	X	X	1
1	0	1	1	X	0	0	X	X	0
1	1	0	0	X	X	X	X	X	X
1	1	0	1	X	X	X	X	X	X
1	1	1	0	X	X	X	X	X	X
1	1	1	1	X	X	X	X	X	X

J_B:

C(t)

0	0	0	1
X	X	X	X
X	X	X	X
0	0	0	0

A(t), B(t), In(t)

$= \bar{A}C\bar{I}$

K_B:

C(t)

X	X	X	X
0	1	X	X
X	X	X	X
X	X	X	X

A(t), B(t), In(t)

$= I$

Just need these columns to get circuitry that feeds into the J,K inputs of each of the Flip-Flops

JK Flip Flops: J& K columns

A(t)	B(t)	C(t)	In(t)	$J_A(t)$	$K_A(t)$	$J_B(t)$	$K_B(t)$	$J_C(t)$	$K_C(t)$
0	0	0	0	0	X	0	X	1	X
0	0	0	1	1	X	0	X	0	X
0	0	1	0	0	X	1	X	X	1
0	0	1	1	1	X	0	X	X	1
0	1	0	0	0	X	X	0	0	X
0	1	0	1	1	X	X	1	0	X
0	1	1	0	X	X	X	X	X	X
0	1	1	1	X	X	X	X	X	X
1	0	0	0	X	1	0	X	0	X
1	0	0	1	X	0	0	X	1	X
1	0	1	0	X	1	0	X	X	1
1	0	1	1	X	0	0	X	X	0
1	1	0	0	X	X	X	X	X	X
1	1	0	1	X	X	X	X	X	X
1	1	1	0	X	X	X	X	X	X
1	1	1	1	X	X	X	X	X	X

J_C:

			$\overbrace{\quad\quad}^{C(t)}$	
	1	0	X	X
B(t)	0	0	X	X
A(t), B(t)	X	X	X	X
A(t)	0	1	X	X
		$\underbrace{\quad\quad}_{In(t)}$		

$$= IA + \bar{A}\bar{I}\bar{B}$$

K_C:

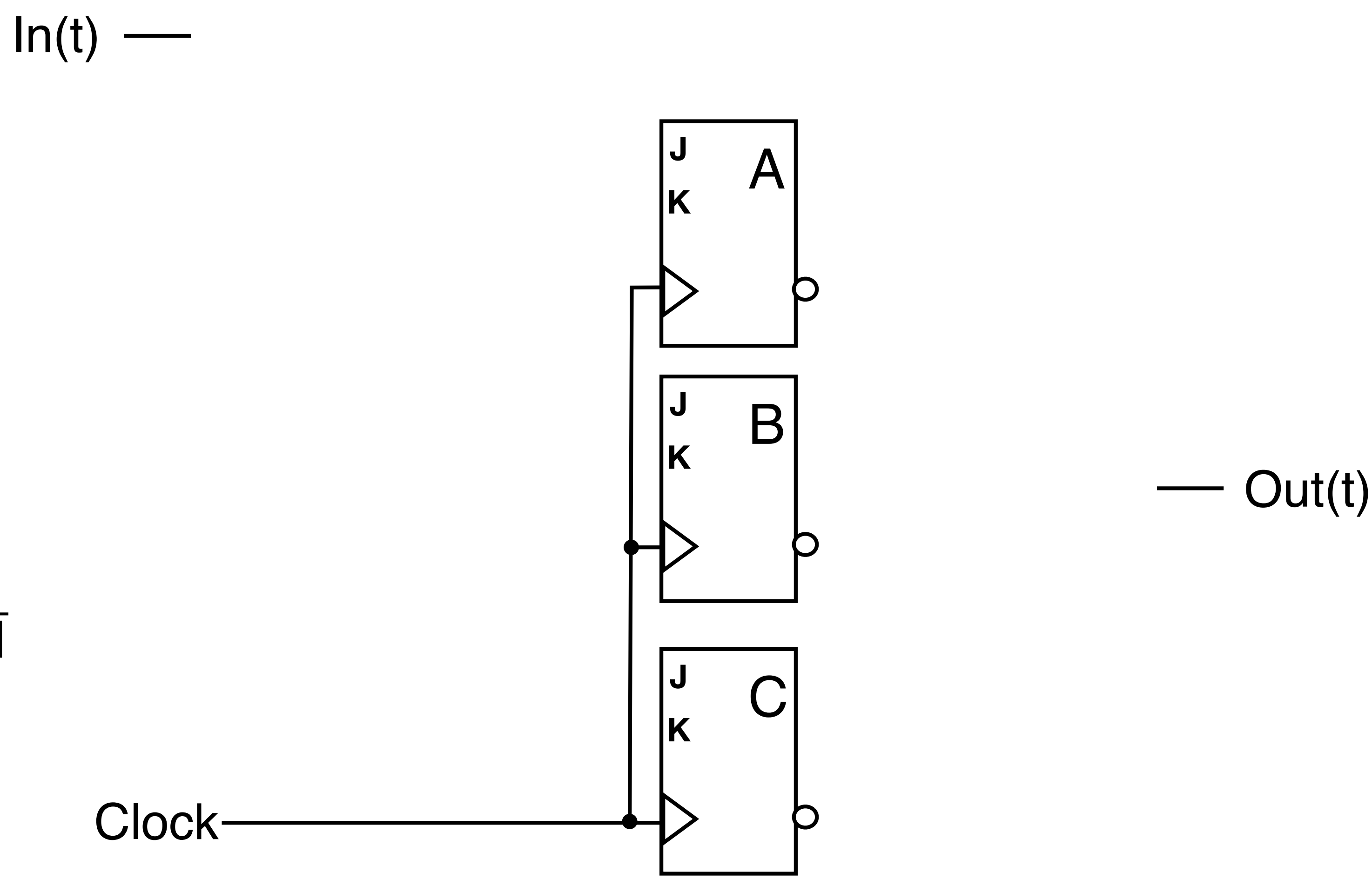
			$\overbrace{\quad\quad}^{C(t)}$	
	X	X	1	1
B(t)	X	X	X	X
A(t), B(t)	X	X	X	X
A(t)	X	X	0	1
		$\underbrace{\quad\quad}_{In(t)}$		

$$= \bar{A} + \bar{I}C$$

Just need these columns to get circuitry that feeds into the J,K inputs of each of the Flip-Flops

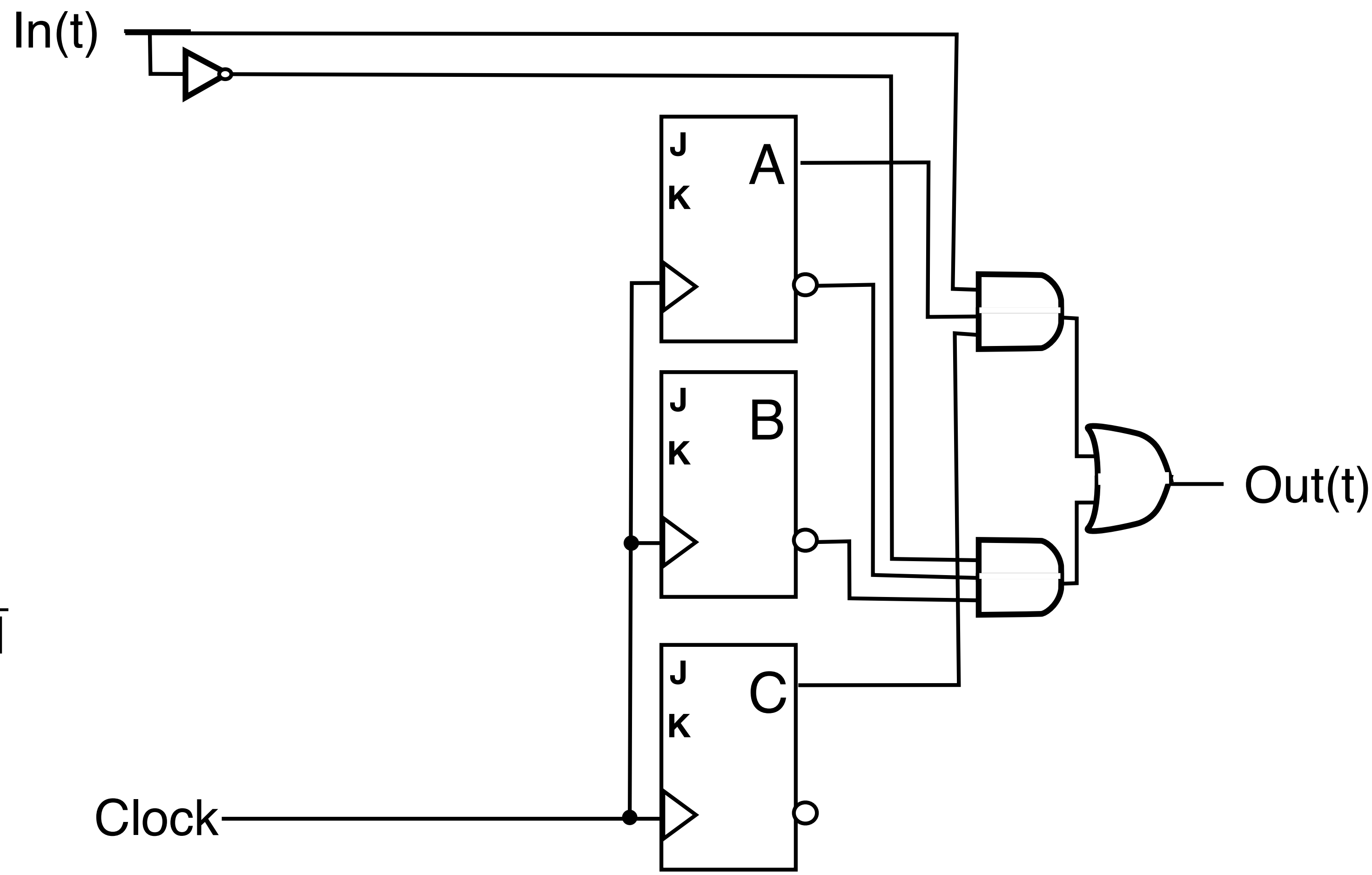
The Sequential Circuit

- $J_A = I$
- $K_A = \bar{I}$
- $J_B = \bar{A}C\bar{I}$
- $K_B = I$
- $J_C = IA$
- $K_C = \bar{A}$
- $Out = ACI + \bar{A}\bar{B}\bar{I}$



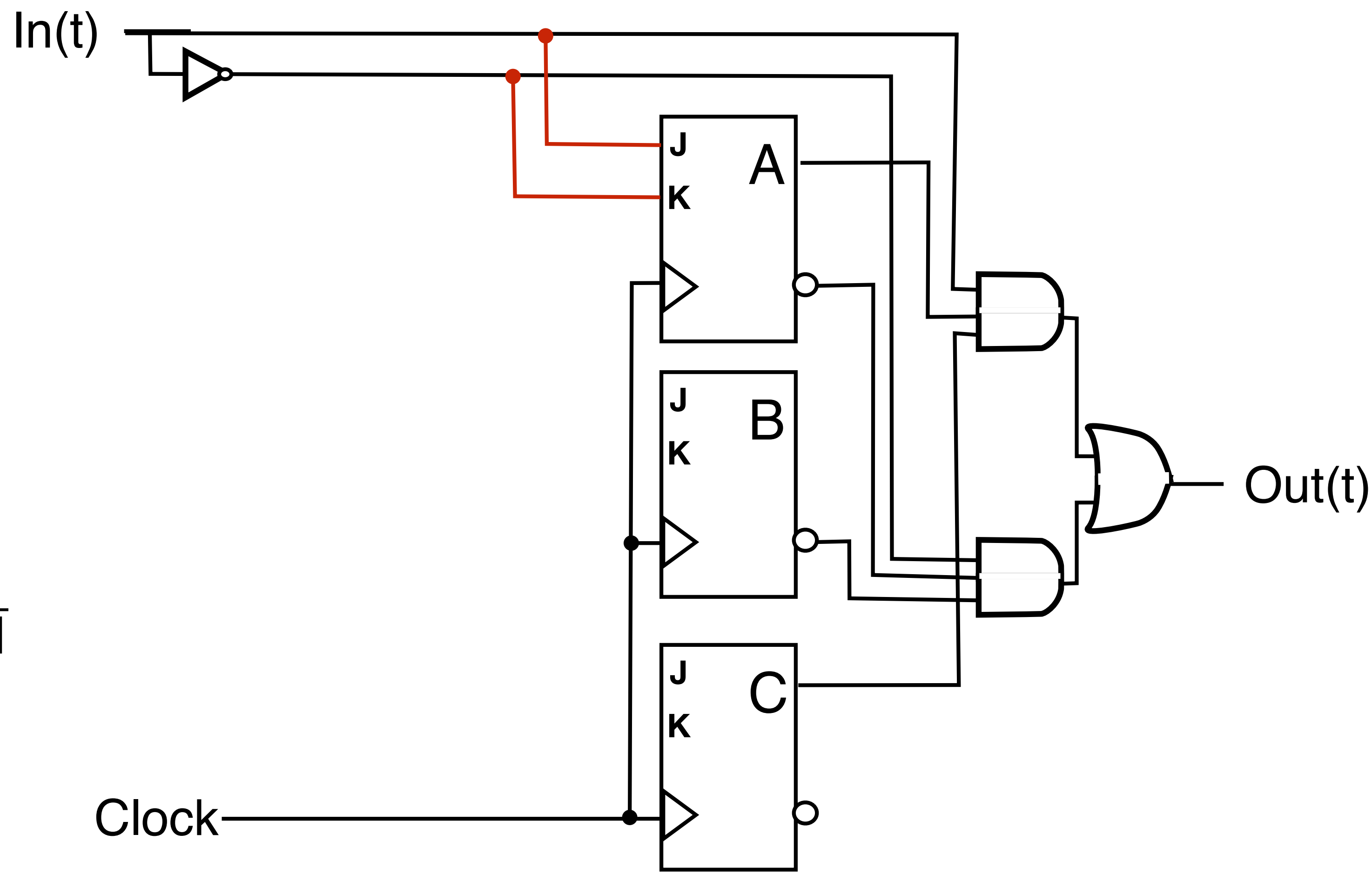
The Sequential Circuit

- $J_A = I$
- $K_A = \bar{I}$
- $J_B = \bar{A}C\bar{I}$
- $K_B = I$
- $J_C = IA$
- $K_C = \bar{A} + \bar{I}C$
- $Out = ACI + \bar{A}\bar{B}\bar{I}$



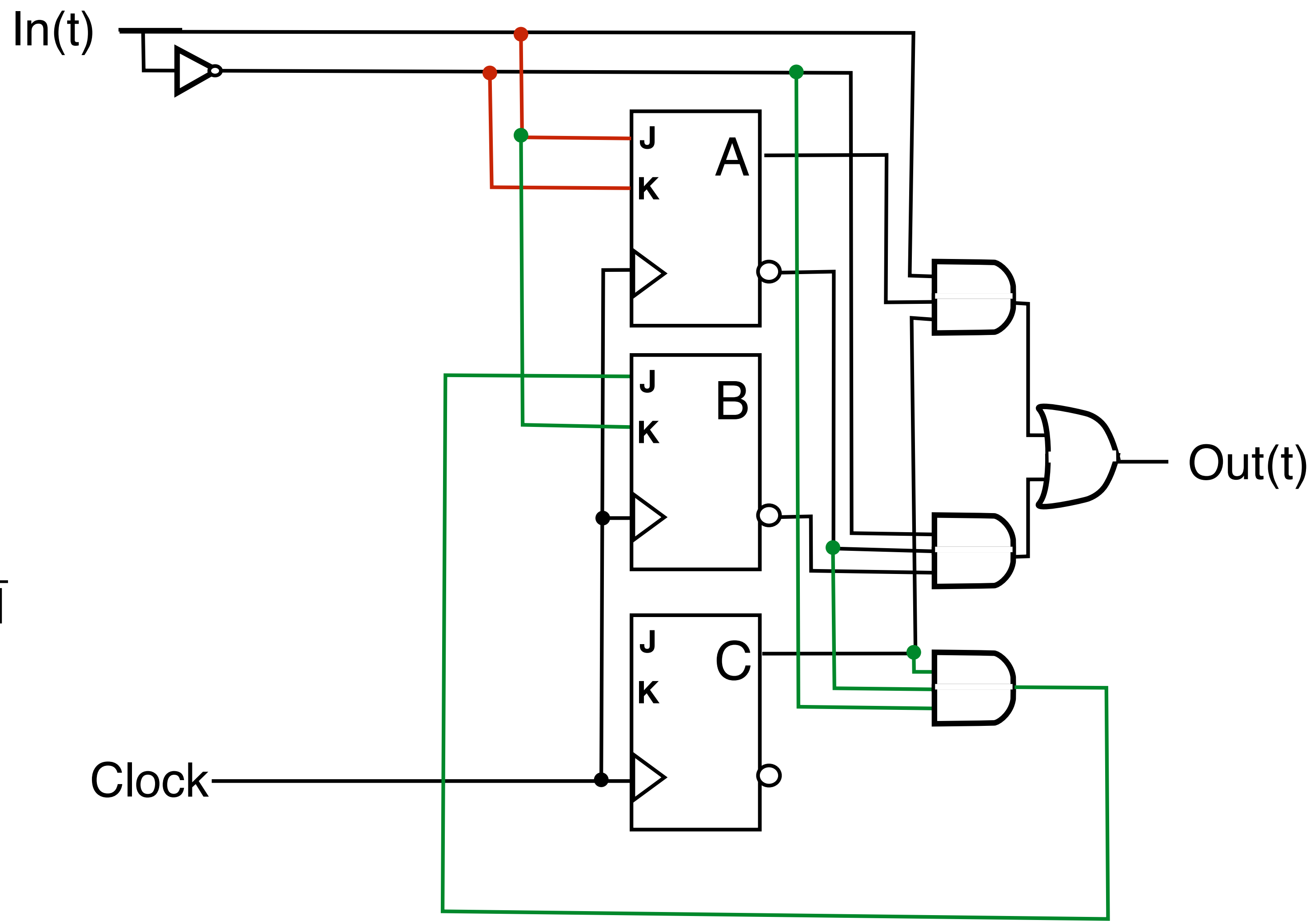
The Sequential Circuit

- $J_A = I$
- $K_A = \bar{I}$
- $J_B = \bar{A}C\bar{I}$
- $K_B = I$
- $J_C = IA$
- $K_C = \bar{A} + \bar{I}C$
- $Out = ACI + \bar{A}\bar{B}\bar{I}$



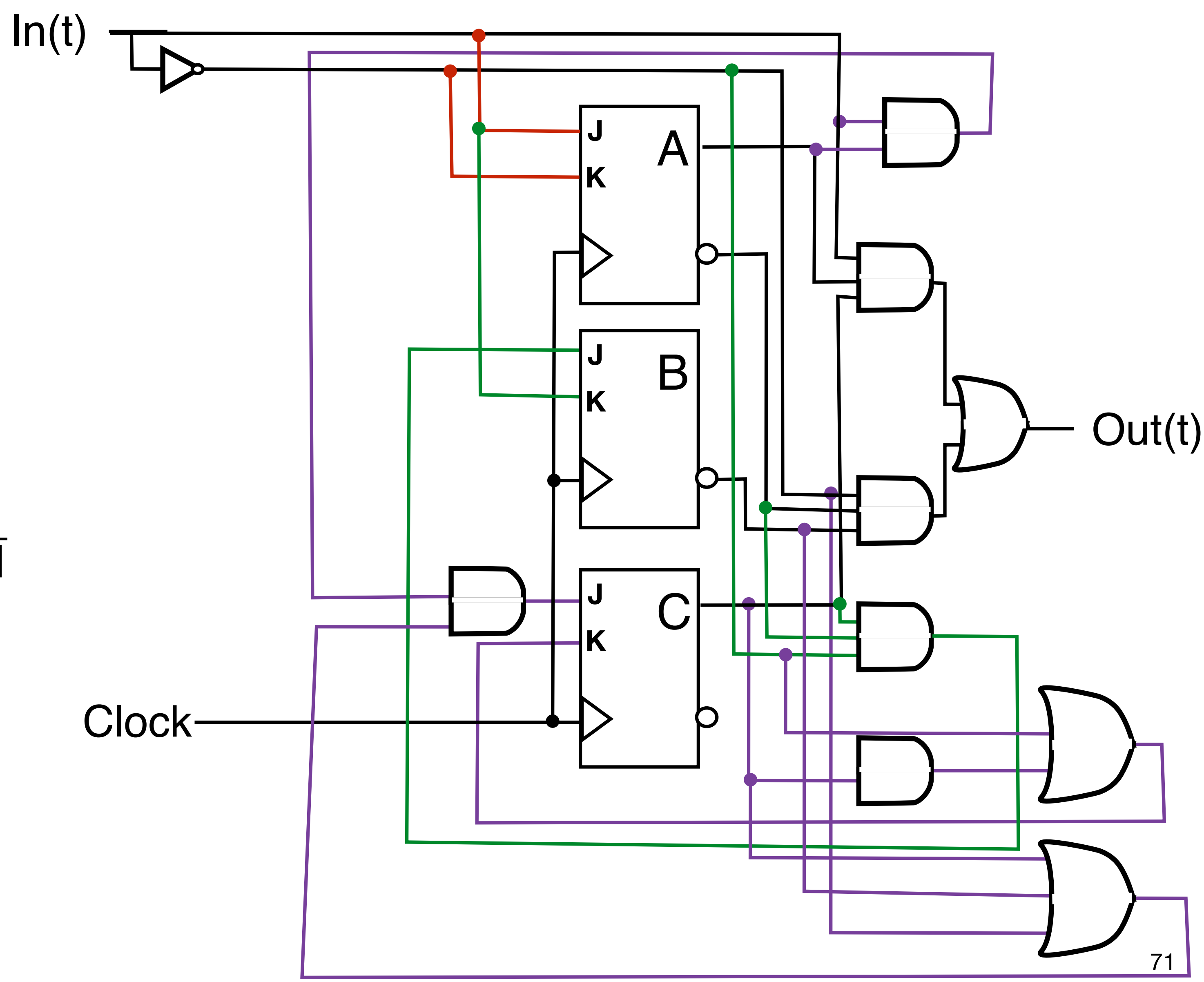
The Sequential Circuit

- $J_A = I$
- $K_A = \bar{I}$
- $J_B = \bar{A}C\bar{I}$
- $K_B = I$
- $J_C = IA$
- $K_C = \bar{A} + \bar{I}C$
- $Out = ACI + \bar{A}\bar{B}\bar{I}$



The Sequential Circuit

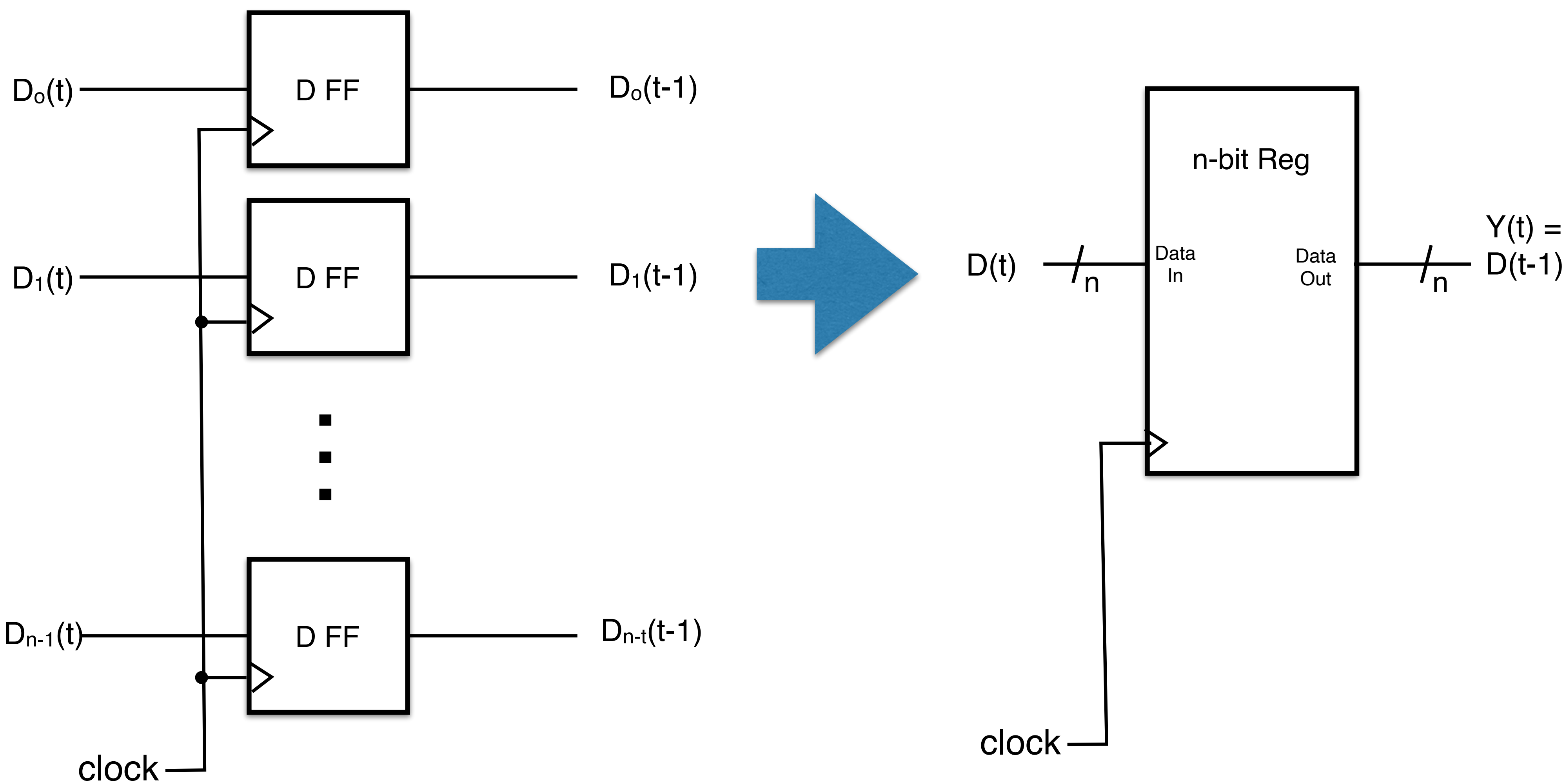
- $J_A = I$
- $K_A = \bar{I}$
- $J_B = \bar{A}C\bar{I}$
- $K_B = I$
- $J_C = IA$
- $K_C = \bar{A} + \bar{I}C$
- $Out = ACI + \bar{A}\bar{B}\bar{I}$



Registers

Registers

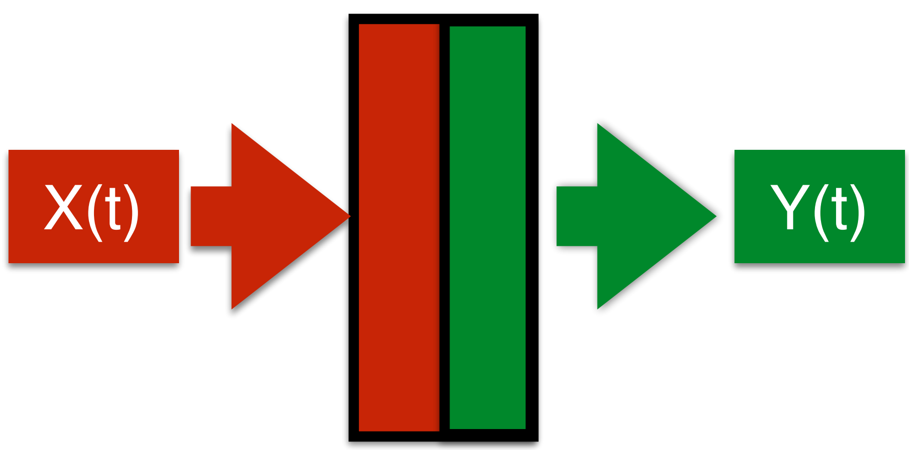
A flip-flop can store 1 bit. A register is a set of n flip flops that stores n bits in parallel.



Register Behavior

- Each clock cycle t, register "set" to an n-bit input value $X(t)=X_{n-1}(t)\ X_{n-2}(t)\ \ldots\ X_1(t)\ X_0(t)$
- Output
  - $Y(t) = Y_{n-1}(t)\ Y_{n-2}(t)\ \ldots\ Y_1(t)\ Y_0(t) = X_{n-1}(t-1)\ X_{n-2}(t-1)\ \ldots\ X_1(t-1)\ X_0(t-1) = X(t-1)$.
  - and of course, $Y(t+1) = X(t)$
- In other words, the output of a register is the input from the previous clock cycle

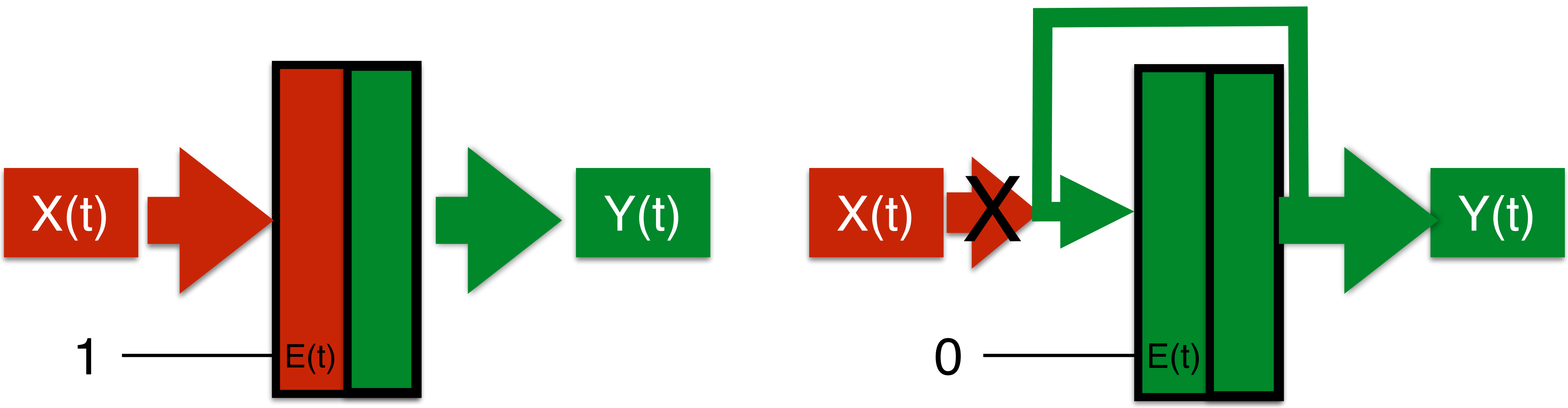
Snapshot of register at time t (tth clock cycle)



Register w/ Write Enable

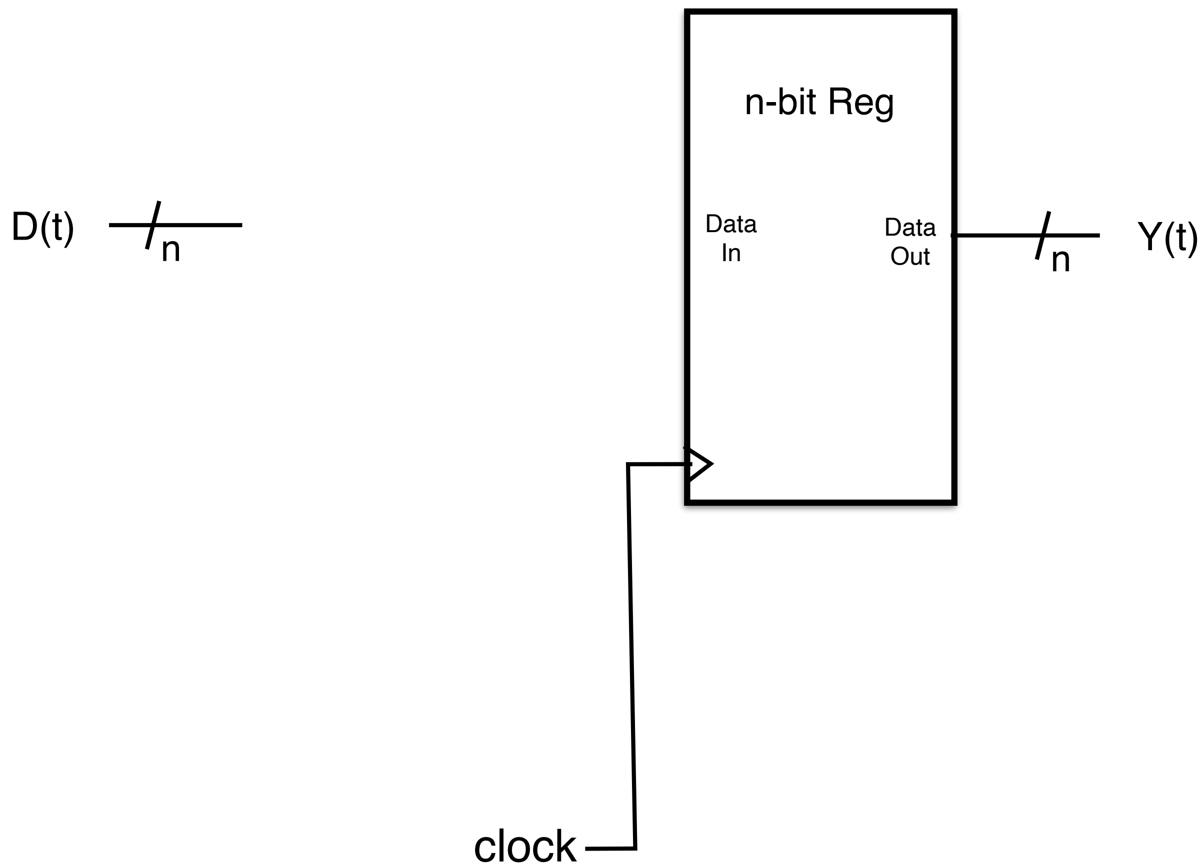
Register w/ Write Enable

- A register with an additional (1-bit) input signal, E(t)
  - E(t)=1: behave like register w/o enable, Y(t+1) = X(t) will be output next clock cycle
  - E(t)=0: hold value in register, ignore current input: Y(t+1) = Y(t)

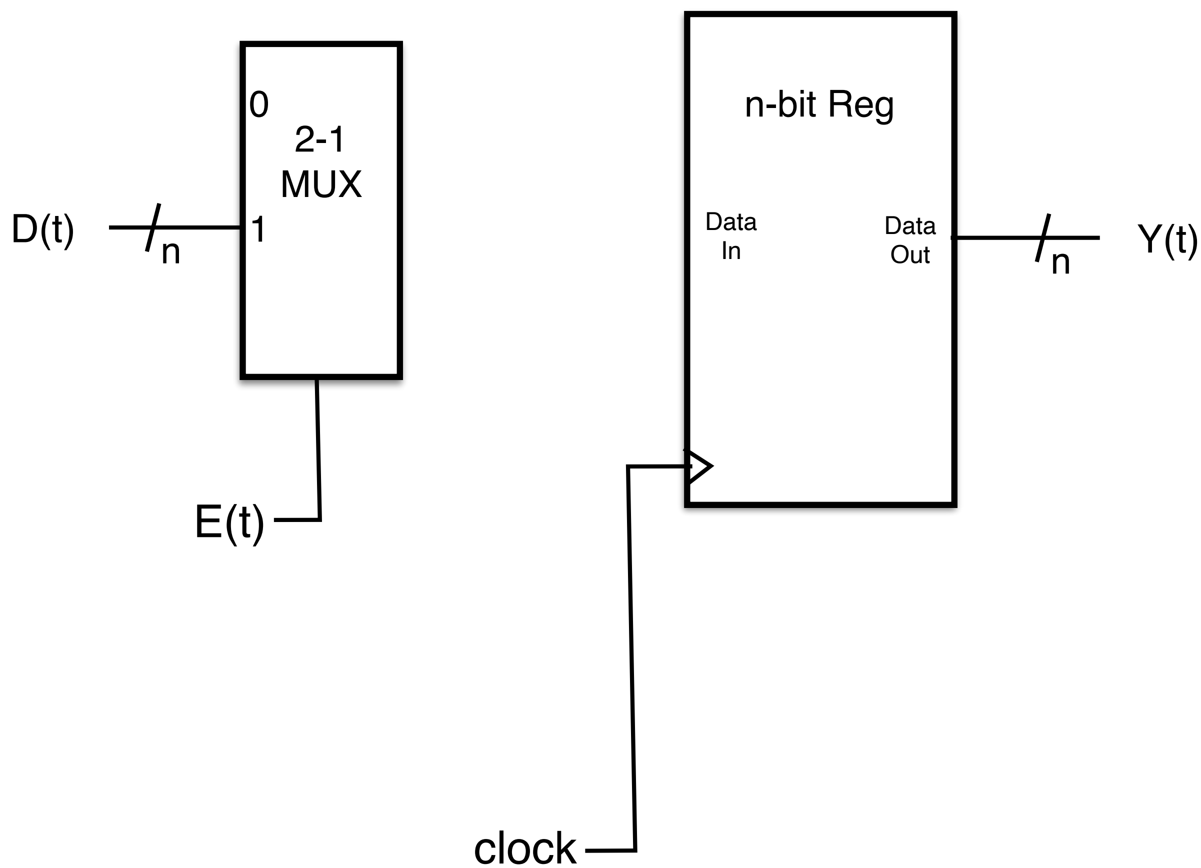


Note: When not enabled, the register value is not being “zero‘d“out!

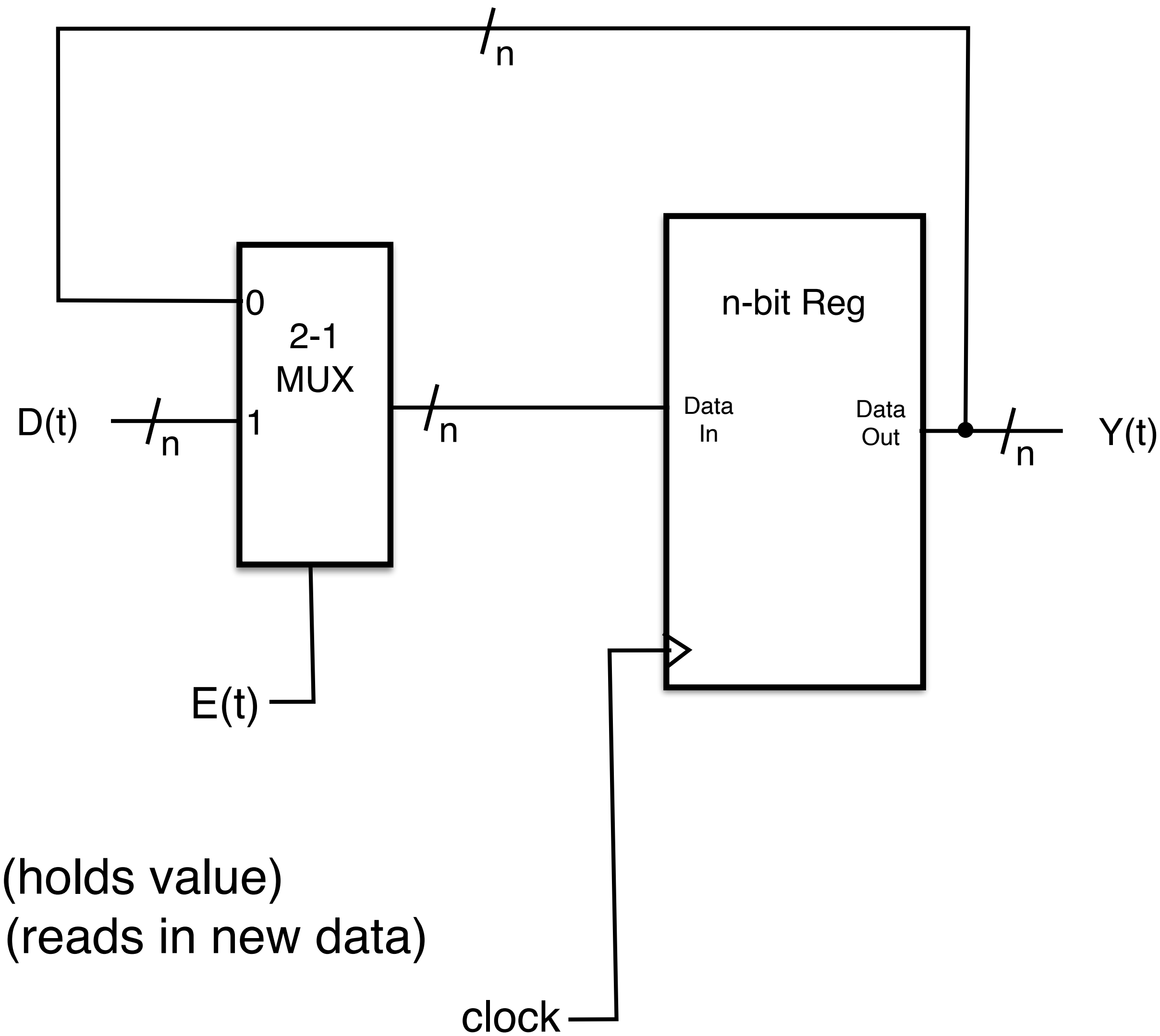
Building Register w/ Write Enable from Register



Building Register w/ Write Enable from Register



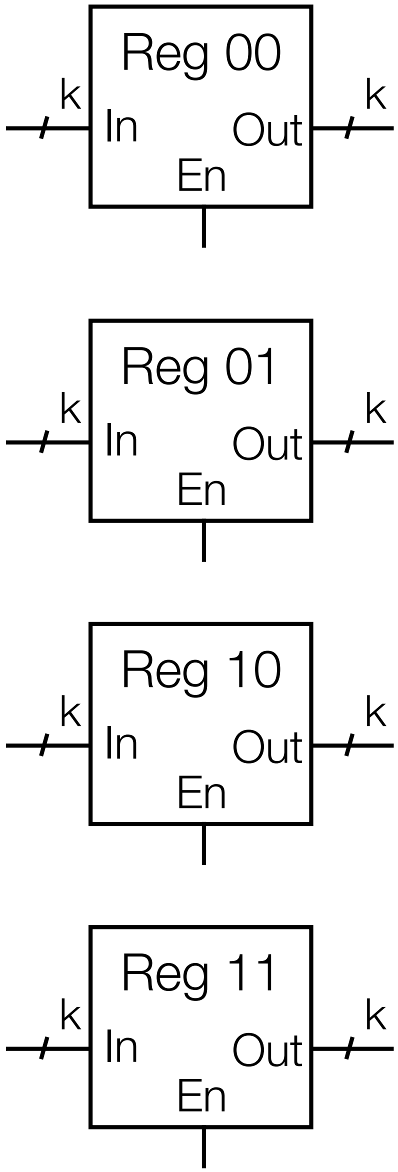
Building Register w/ Write Enable from Register

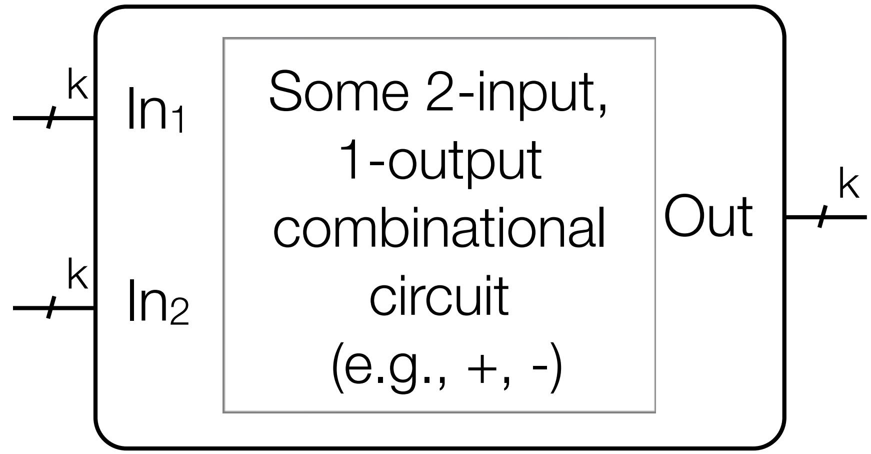


E(t)=0: Y(t+1) = Y(t)   (holds value)
E(t)=1: Y(t+1) = D(t)   (reads in new data)

Register Selection (using MUXs)

Register MUXing



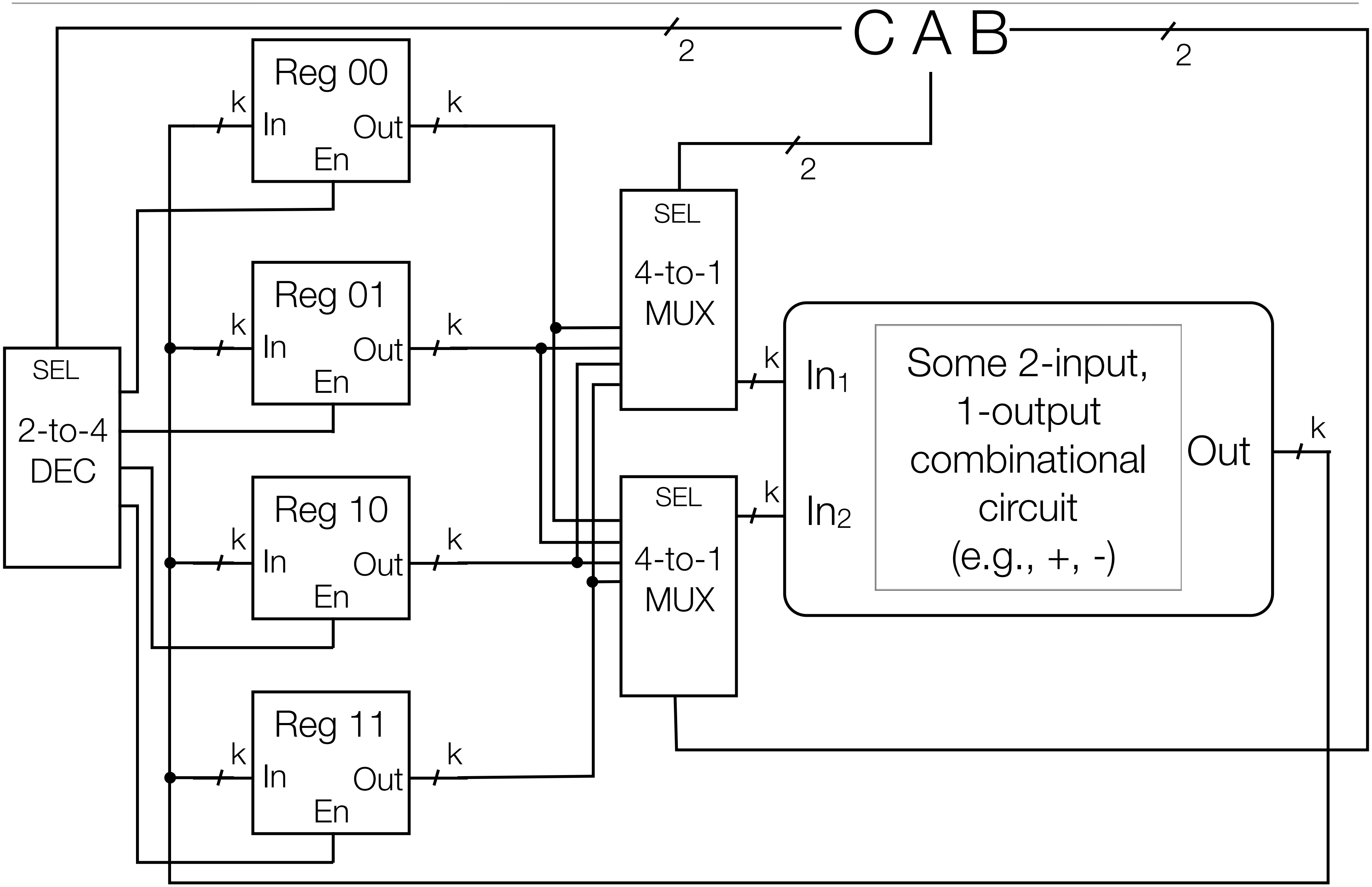


- Suppose have
  - 4 k-bit registers with Enable (for writing)
  - combinational logic to perform some operation OP
- Goal: Allow system to select registers
  - 2 registers selected for inputs (A & B)
  - 1 register for output (C)
  - At end of cycle: C = A OP B
- Note: A&B can be same register, C can also be same as A or B

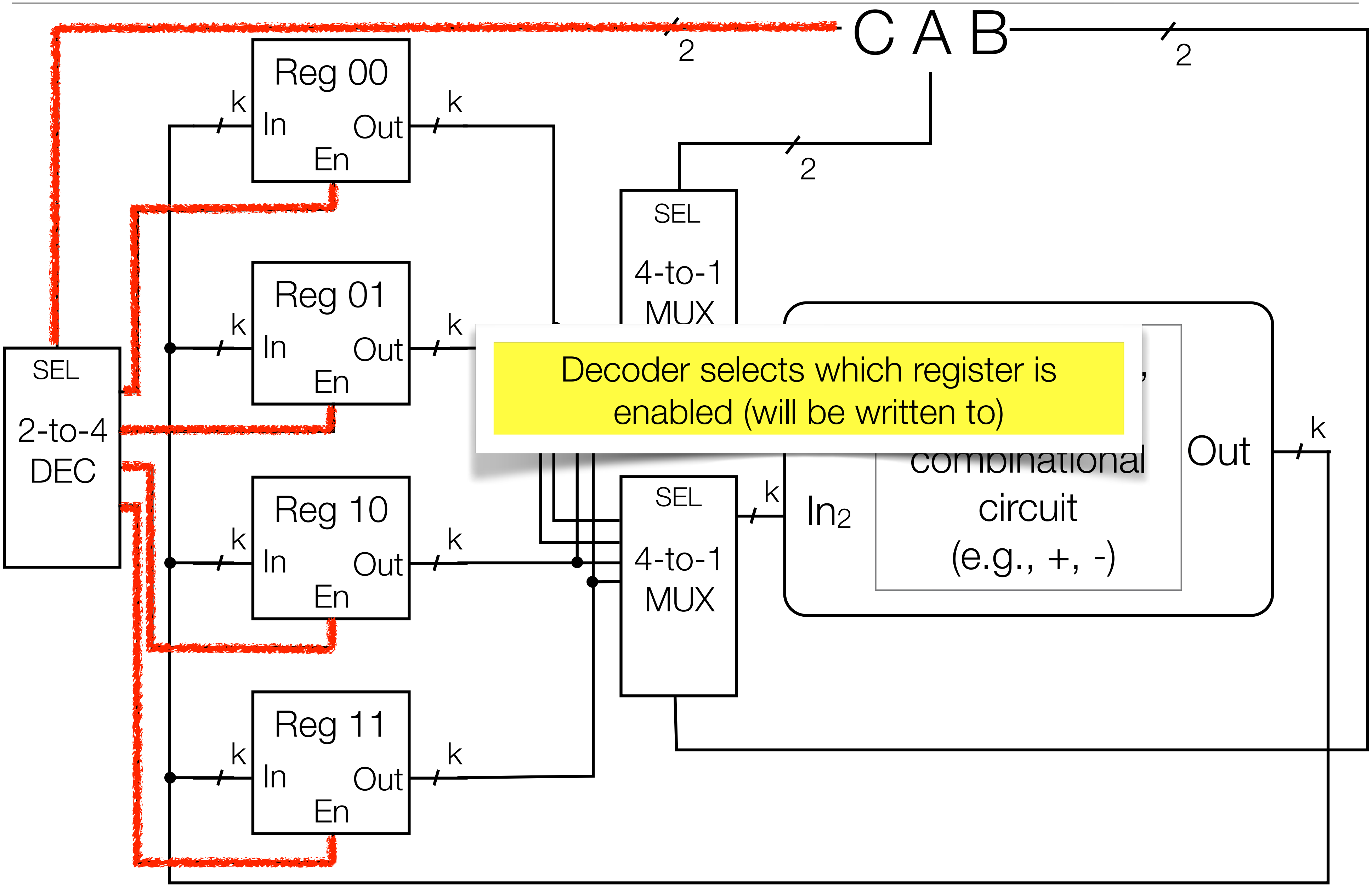
Register MUXing example

- e.g.,
  - Reg01 = Reg01 OP Reg10
  - Reg10 = Reg10 OP Reg10 (if OP were +, this would do Reg10 *= 2)
- OP selecting: be able to choose from different OPs, e.g.
  - Reg01 = Reg01 + Reg10
  - Reg10 = Reg01 * Reg00

Register MUXing

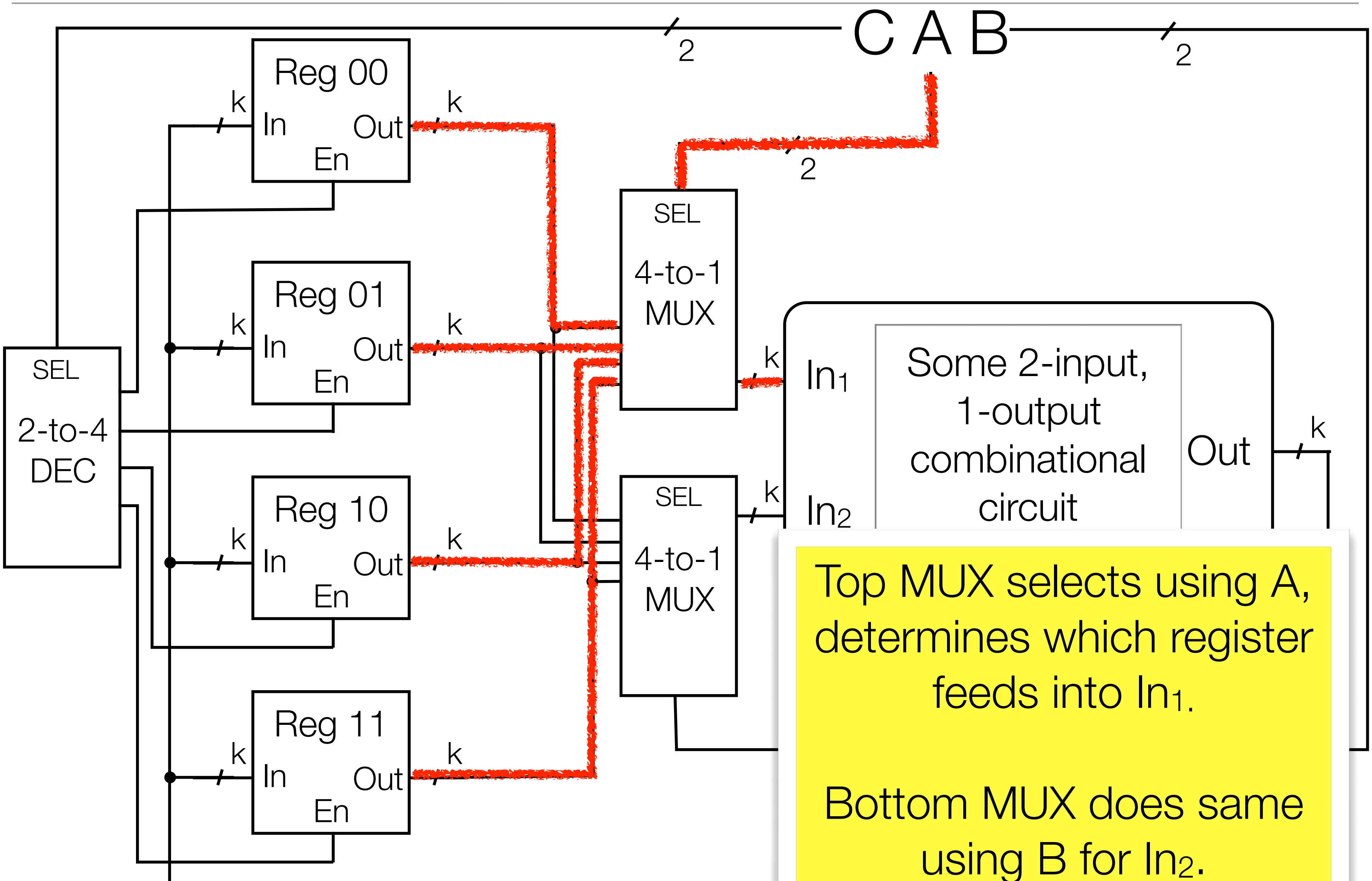


Register MUXing



Decoder selects which register is enabled (will be written to)

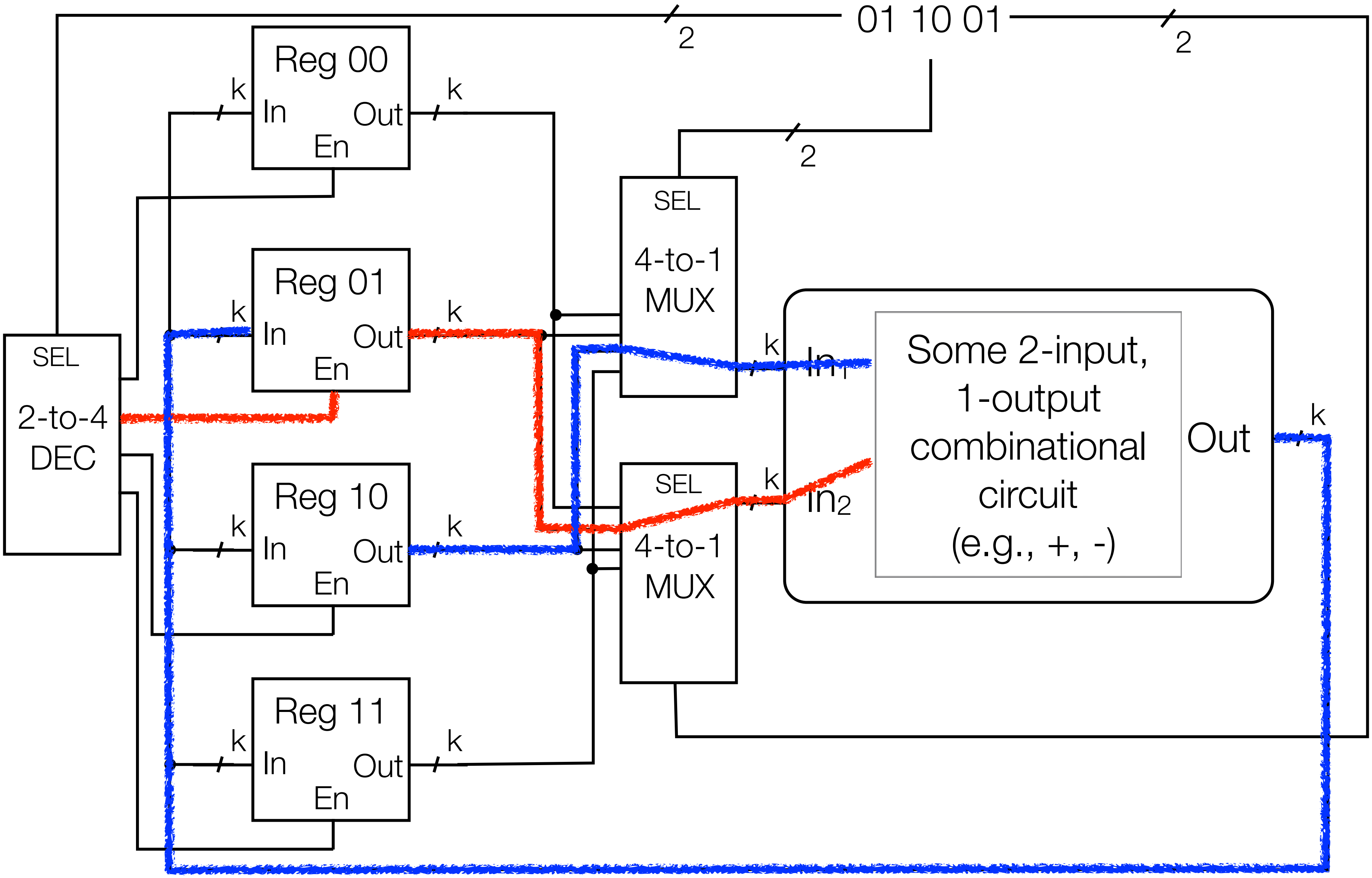
Register MUXing



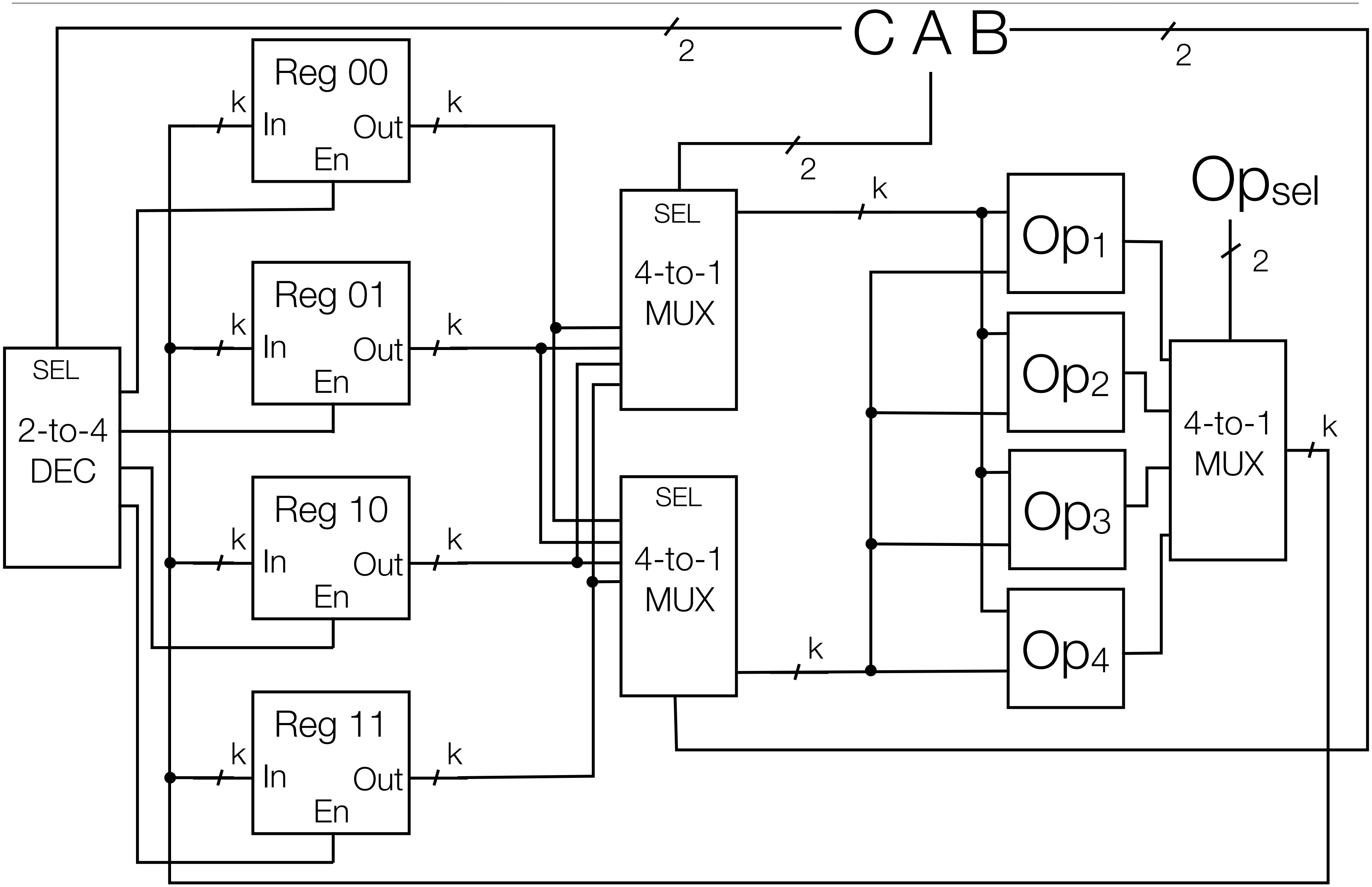
Top MUX selects using A, determines which register feeds into In_1.

Bottom MUX does same using B for In_2.

Example: Reg01 = Reg10 OP Reg01



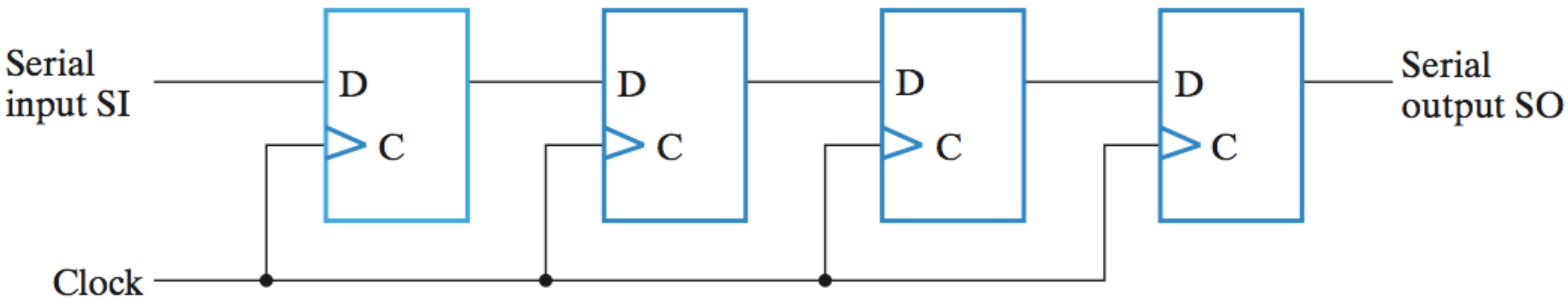
Register MUXing and OP MUXing



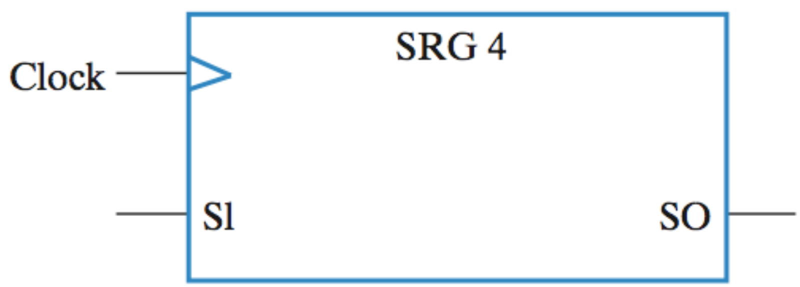
Specialized Registers

Shift register

A register capable of shifting bits laterally in one direction

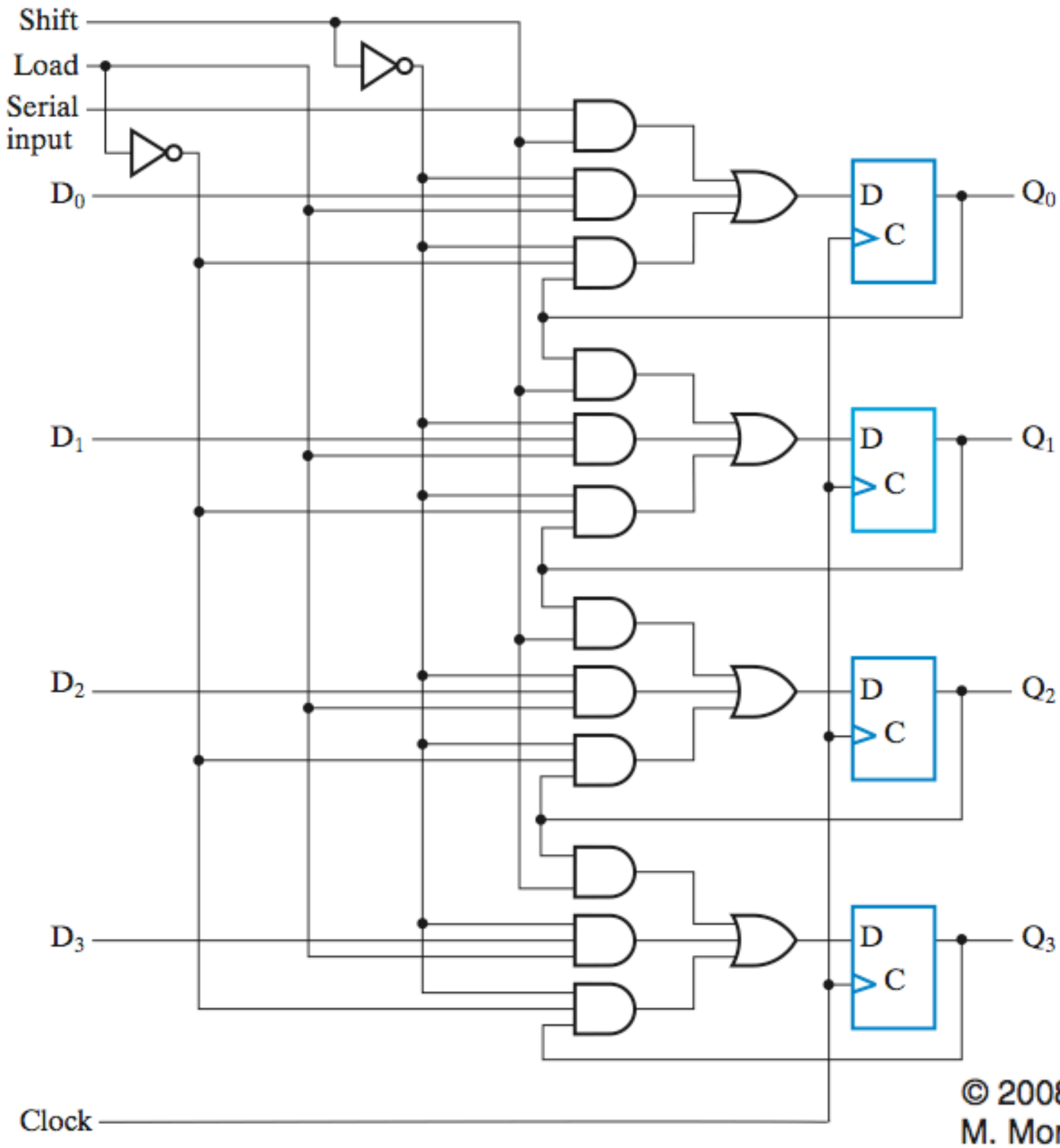


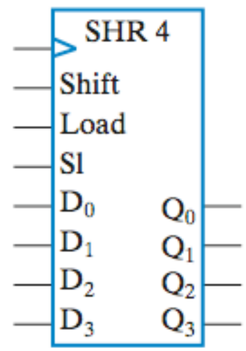
(a) Logic diagram



(b) Symbol

Shift register w. parallel load





(b) Symbol

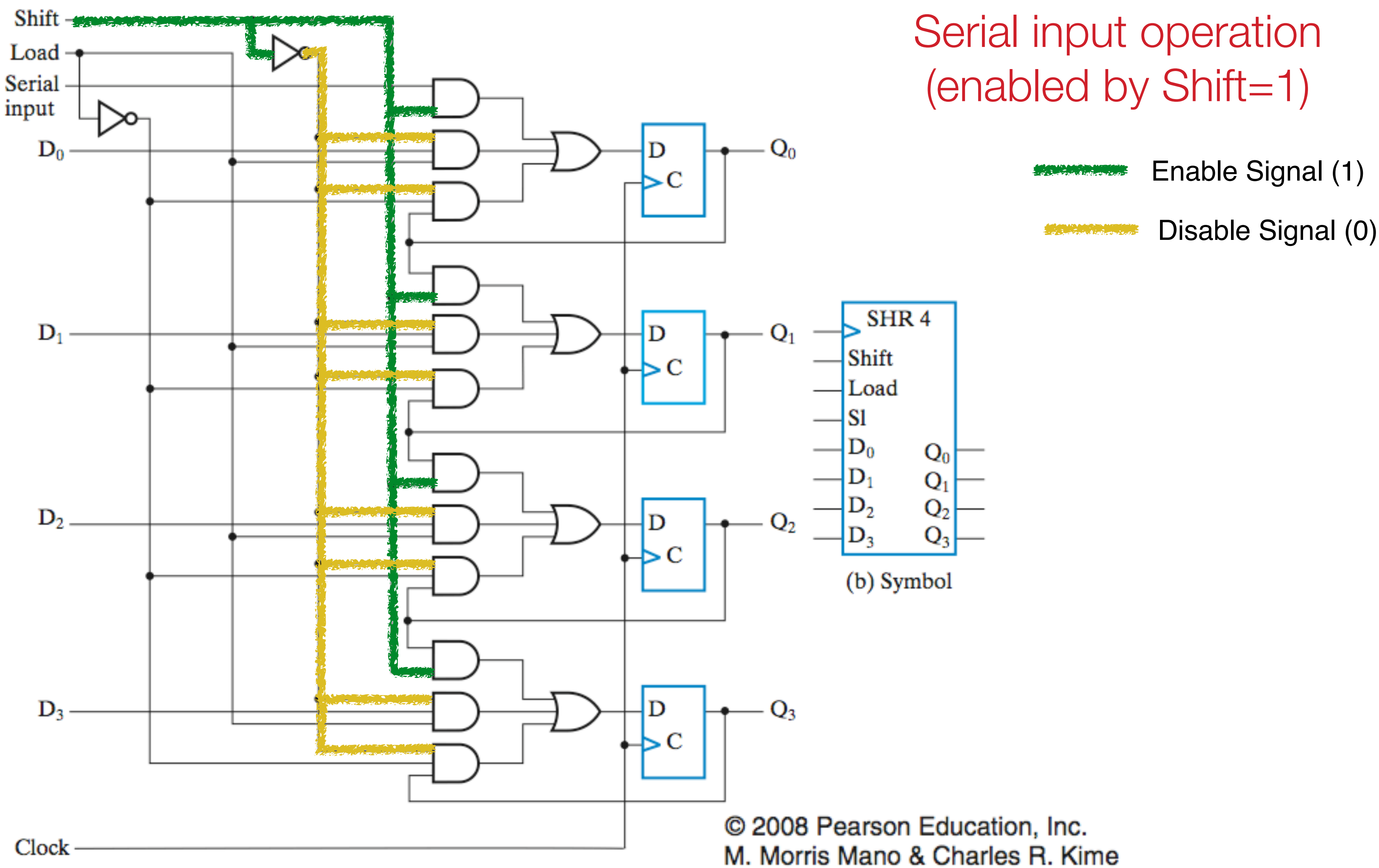
Three modes:

Shift (w/ serial input)

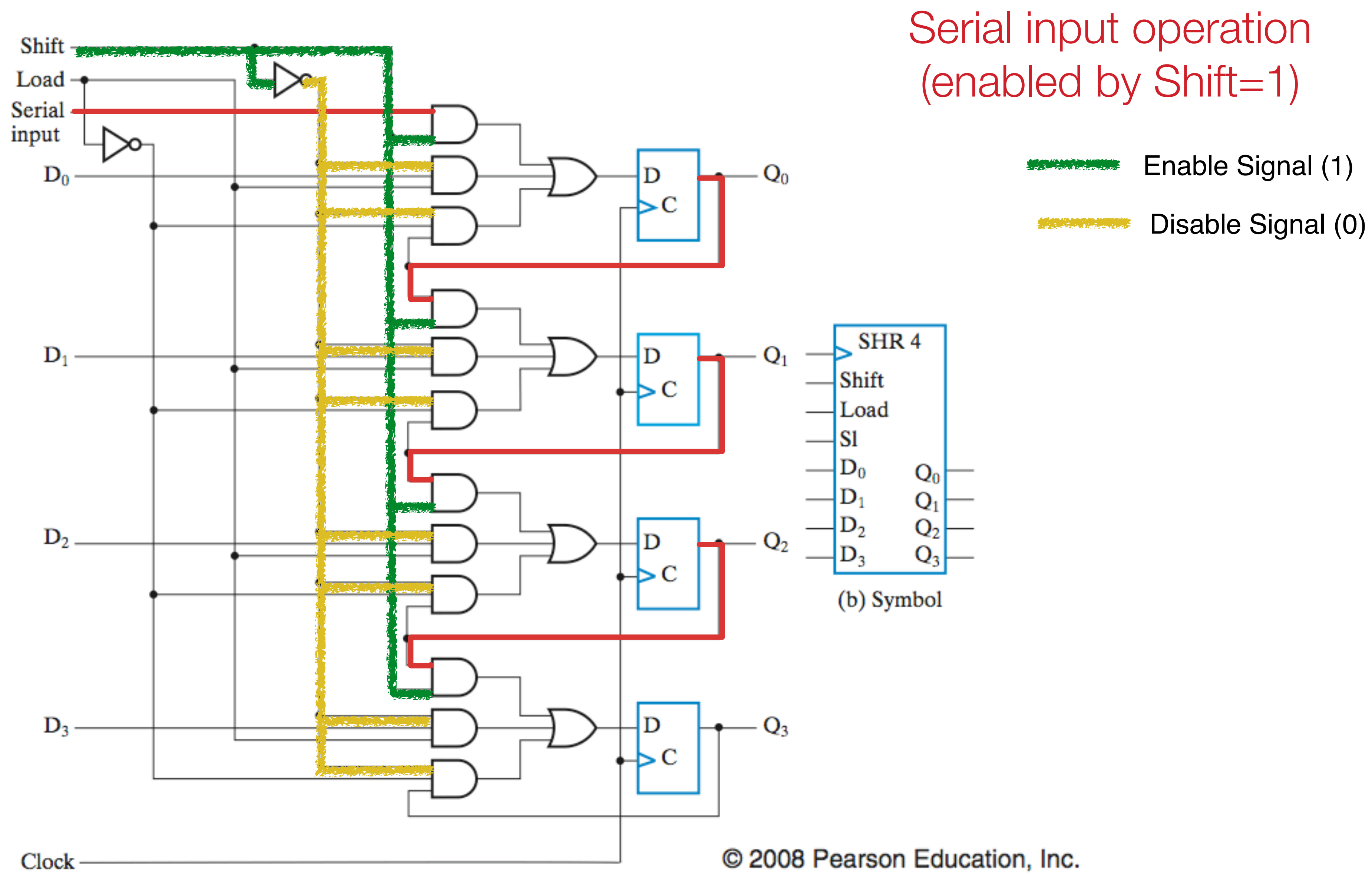
Parallel input

No input (do nothing)

Shift register w. parallel load



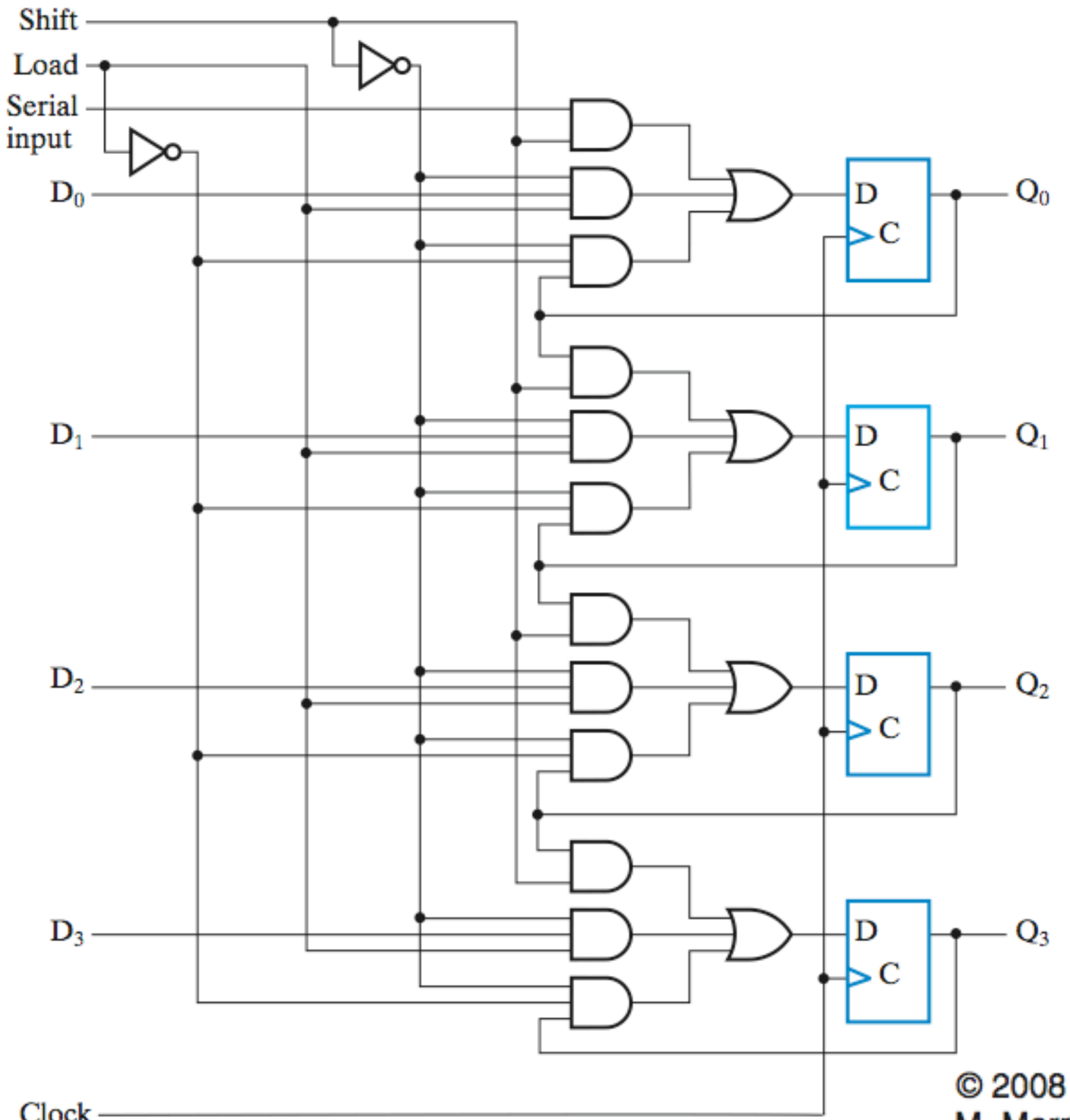
Shift register w. parallel load

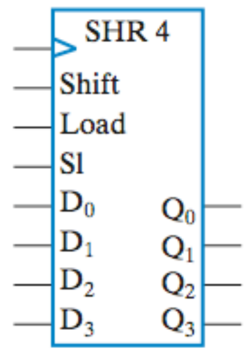


(b) Symbol

Shift register w. parallel load

Parallel load operation
(enabled by Load AND $\overline{\text{Shift}}$)

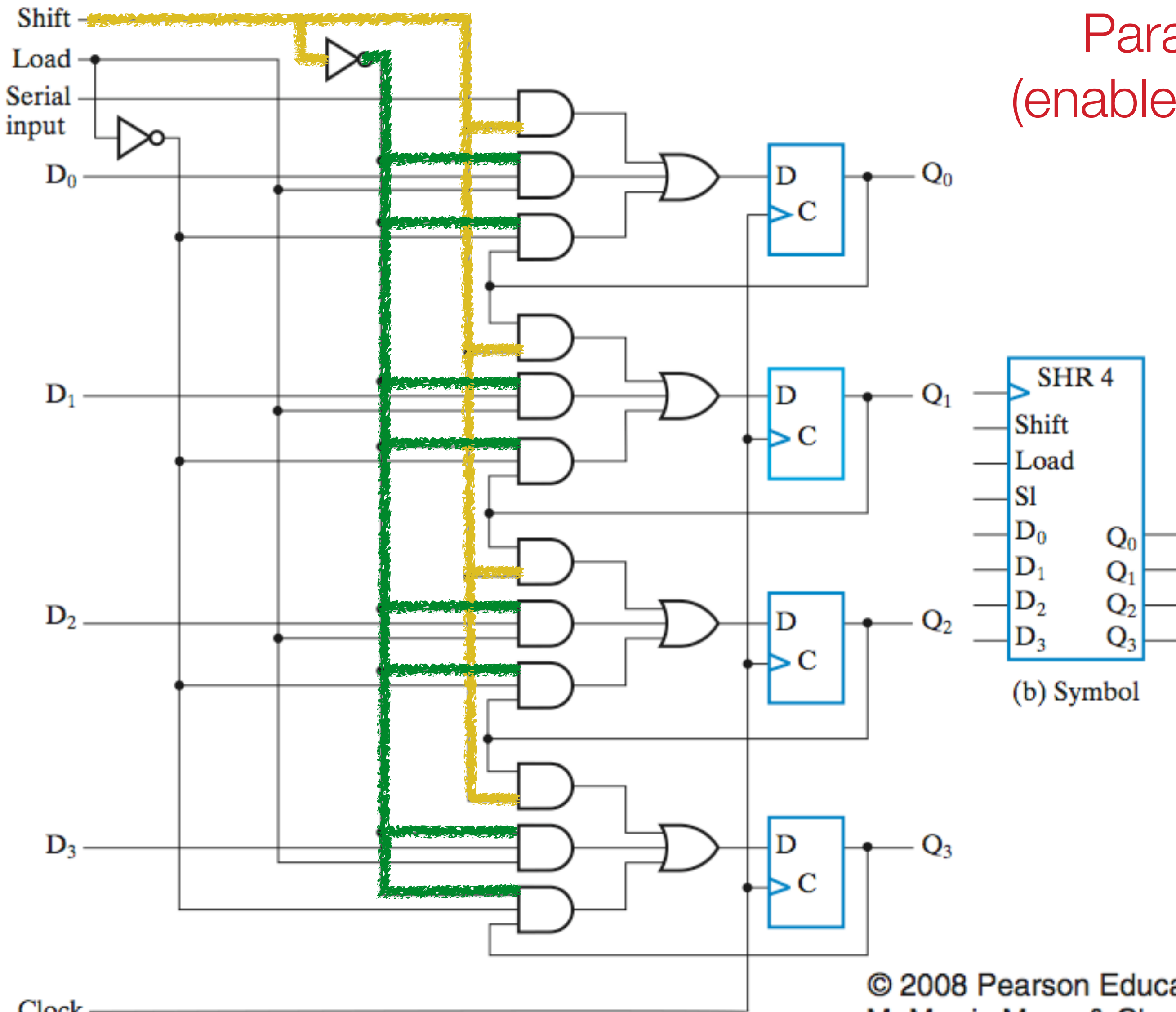




(b) Symbol

Shift register w. parallel load

Parallel load operation
(enabled by Load AND $\overline{\text{Shift}}$)



(b) Symbol

Enable Signal (1)

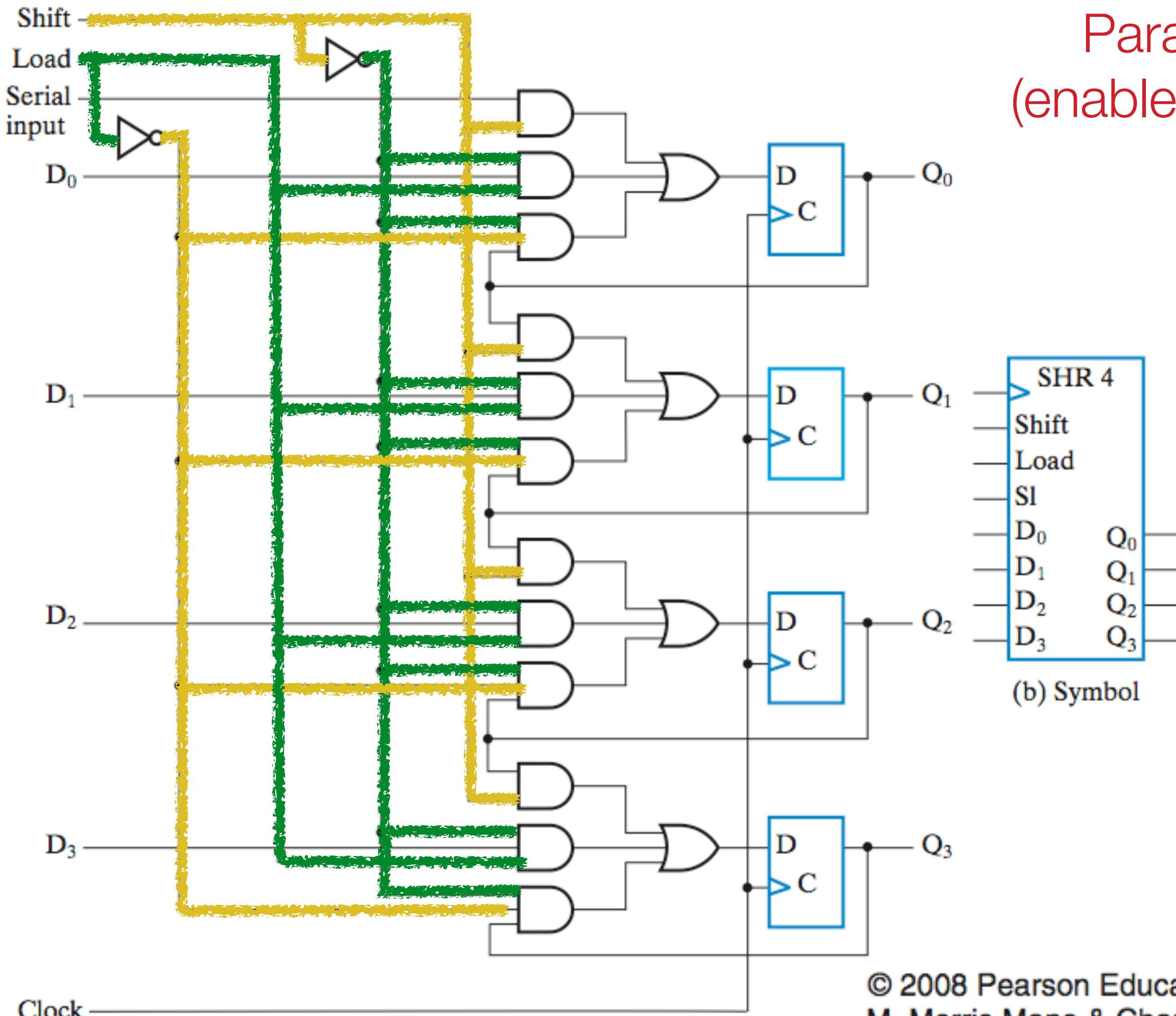
Disable Signal (0)

Shift register w. parallel load

Parallel load operation
(enabled by Load AND $\overline{\text{Shift}}$)

Enable Signal (1)

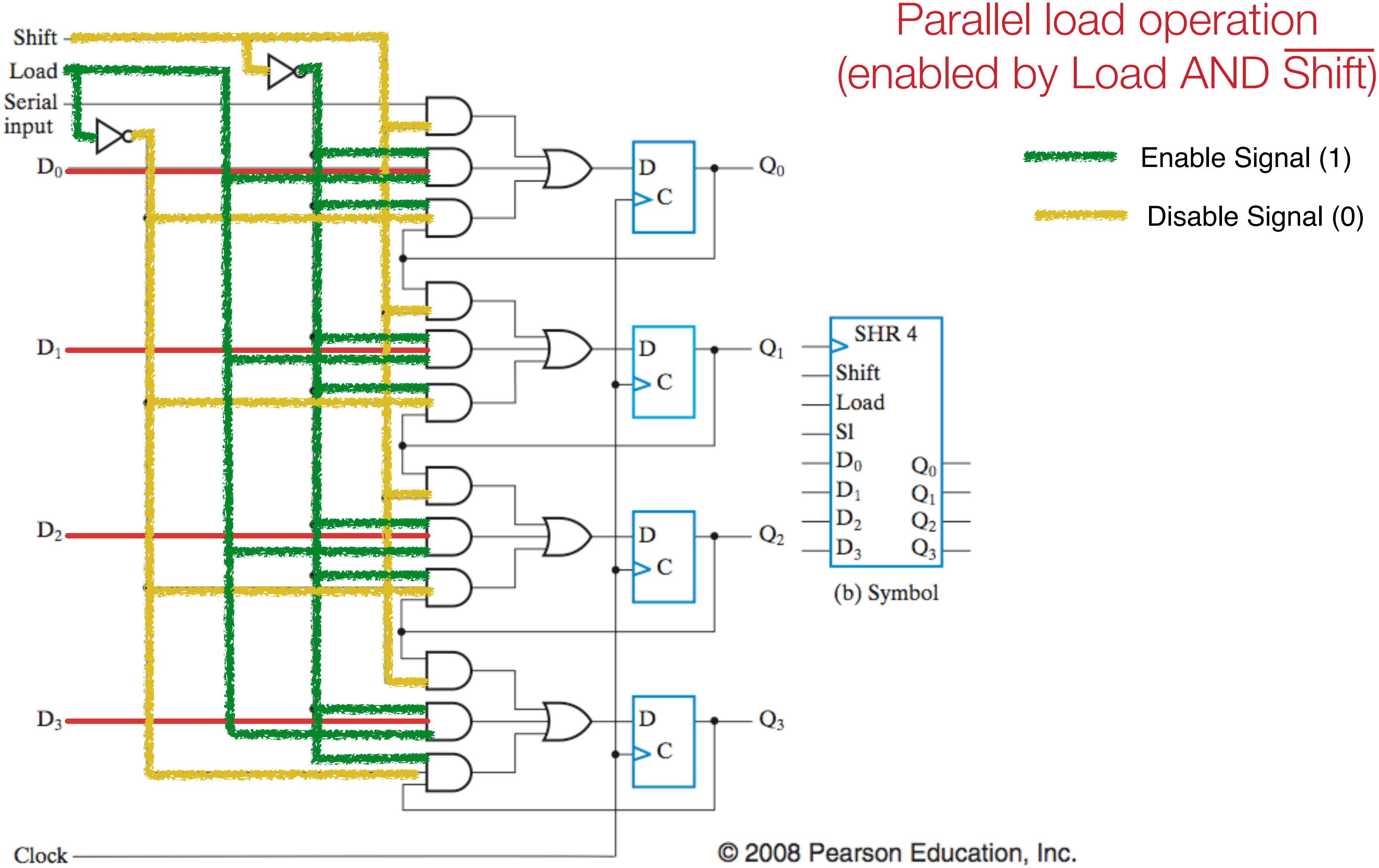
Disable Signal (0)



(b) Symbol

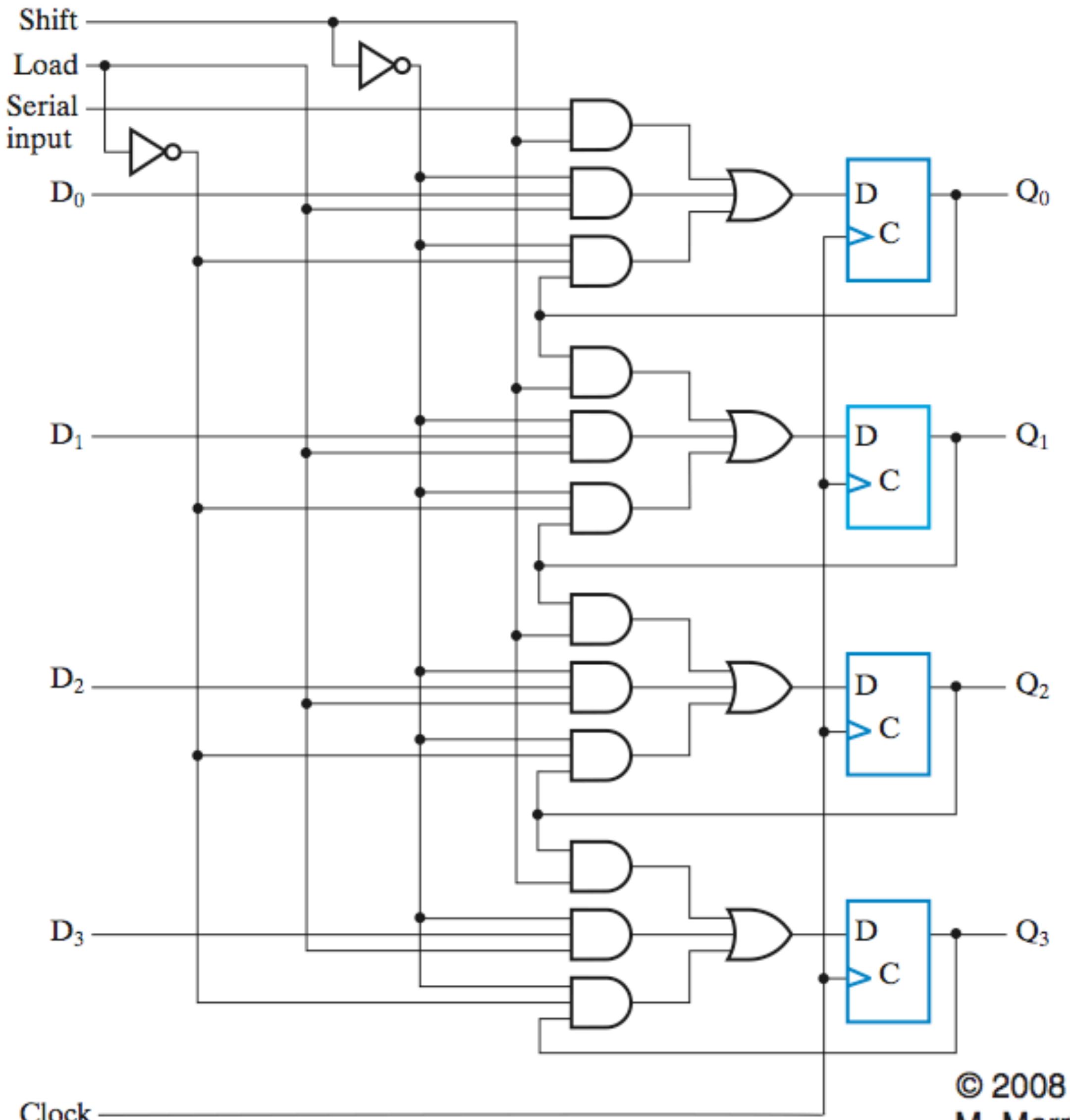
Shift register w. parallel load

Parallel load operation
(enabled by Load AND $\overline{\text{Shift}}$)

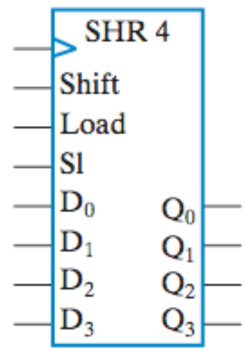


(b) Symbol

Shift register w. parallel load



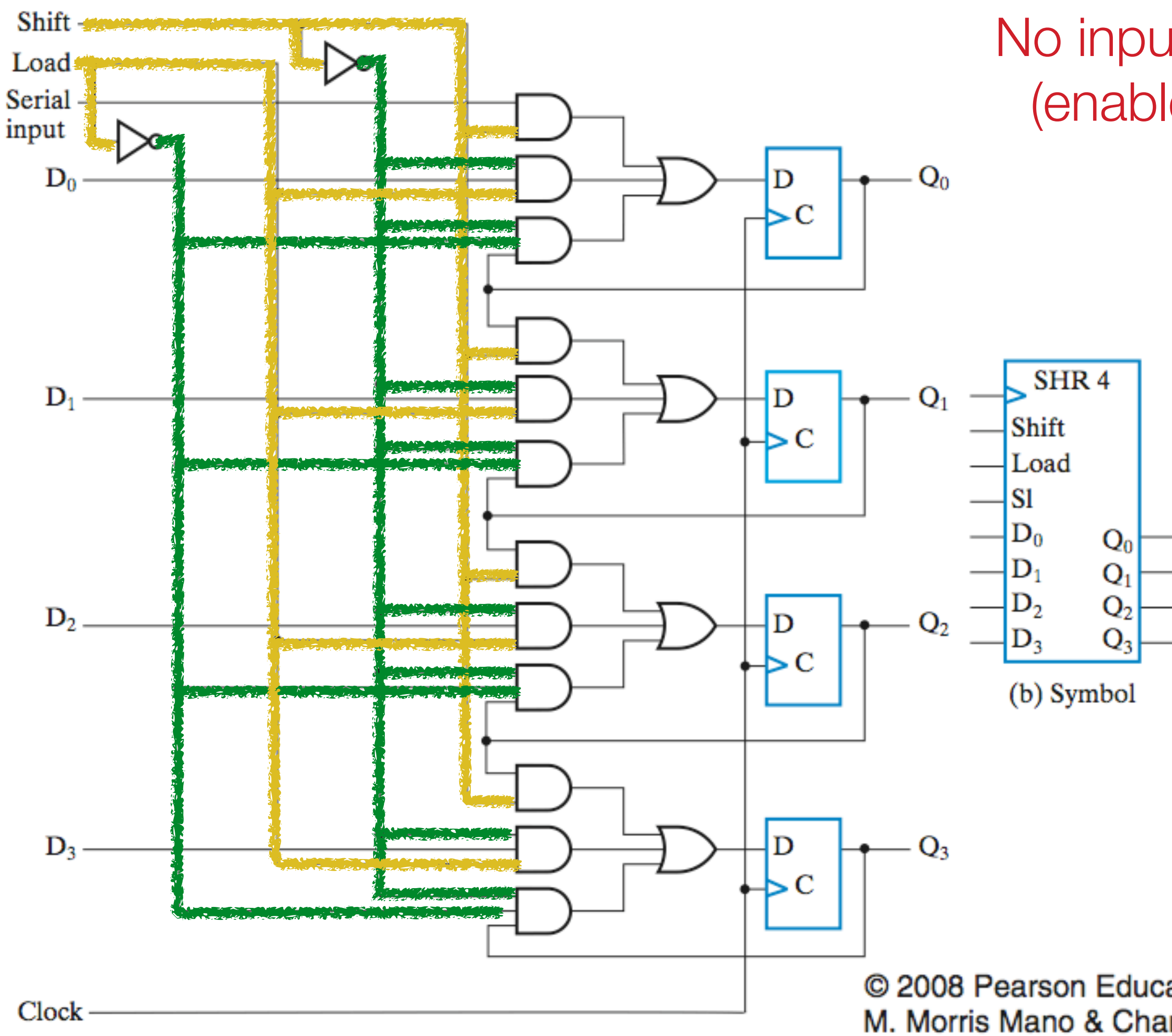
No input (Do nothing) operation
(enabled by $\overline{\text{Load}}$ AND $\overline{\text{Shift}}$)



(b) Symbol

Shift register w. parallel load

No input (Do nothing) operation
(enabled by $\overline{\text{Load}}$ AND $\overline{\text{Shift}}$)



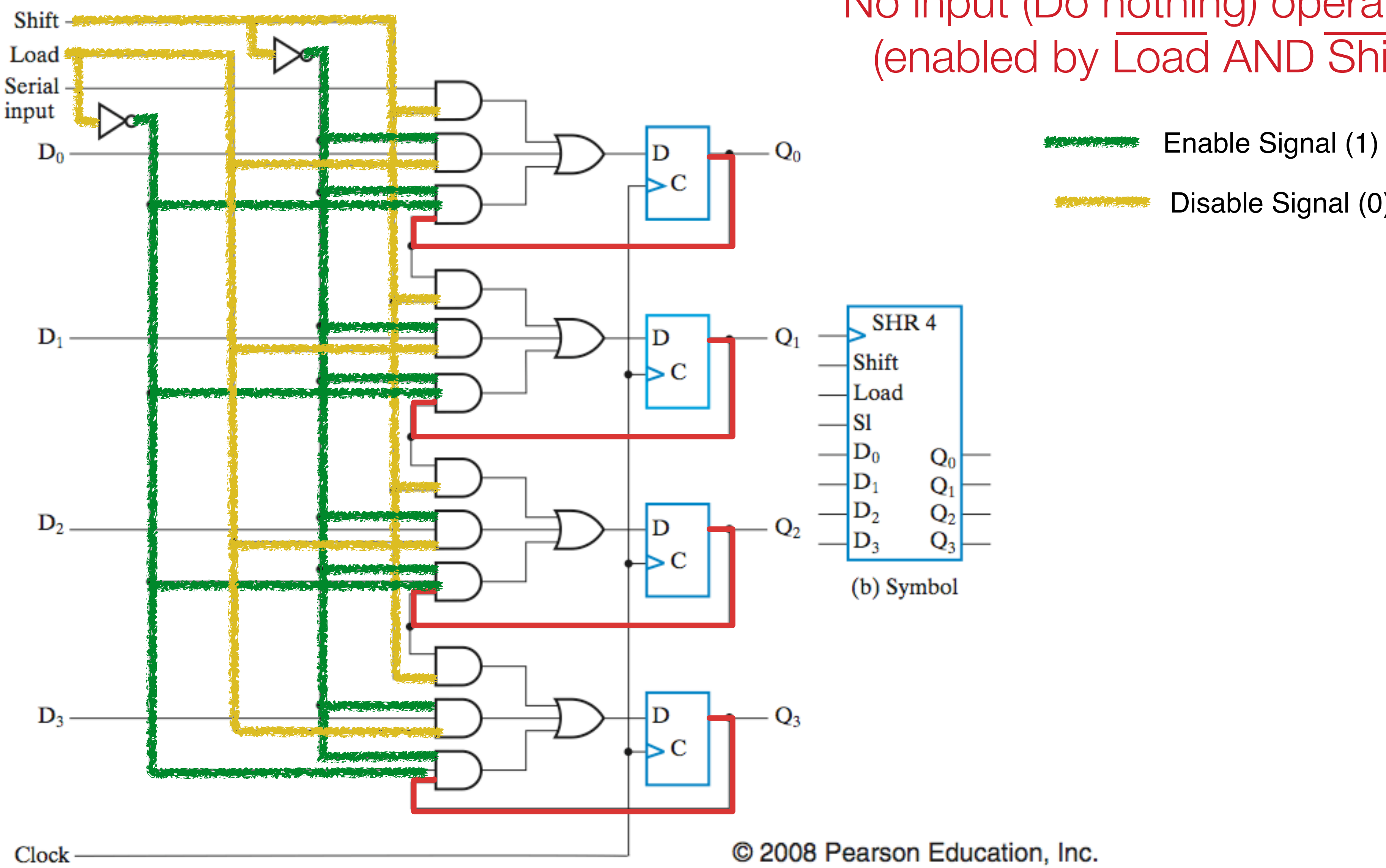
(b) Symbol

Enable Signal (1)

Disable Signal (0)

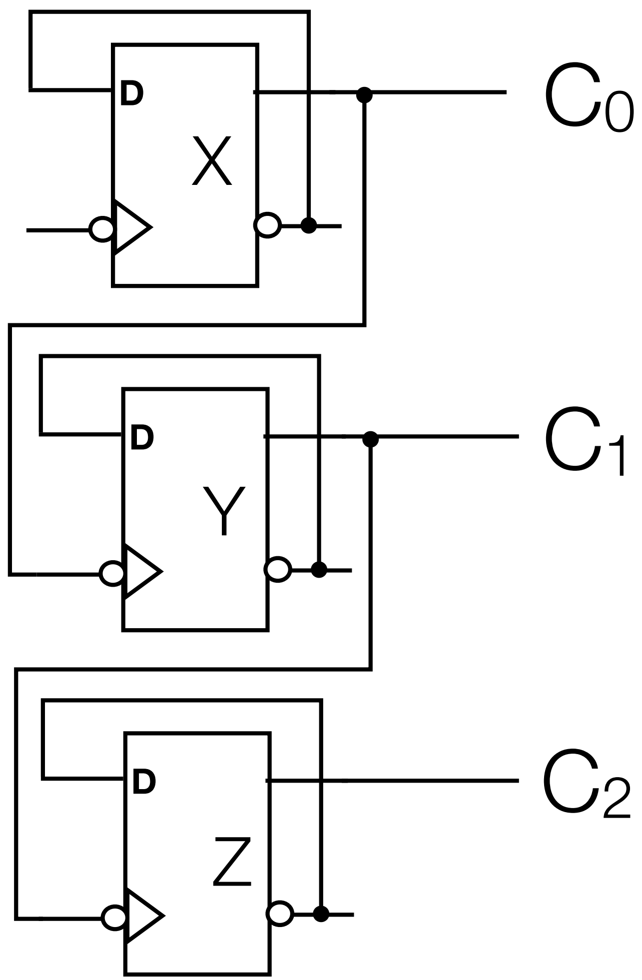
Shift register w. parallel load

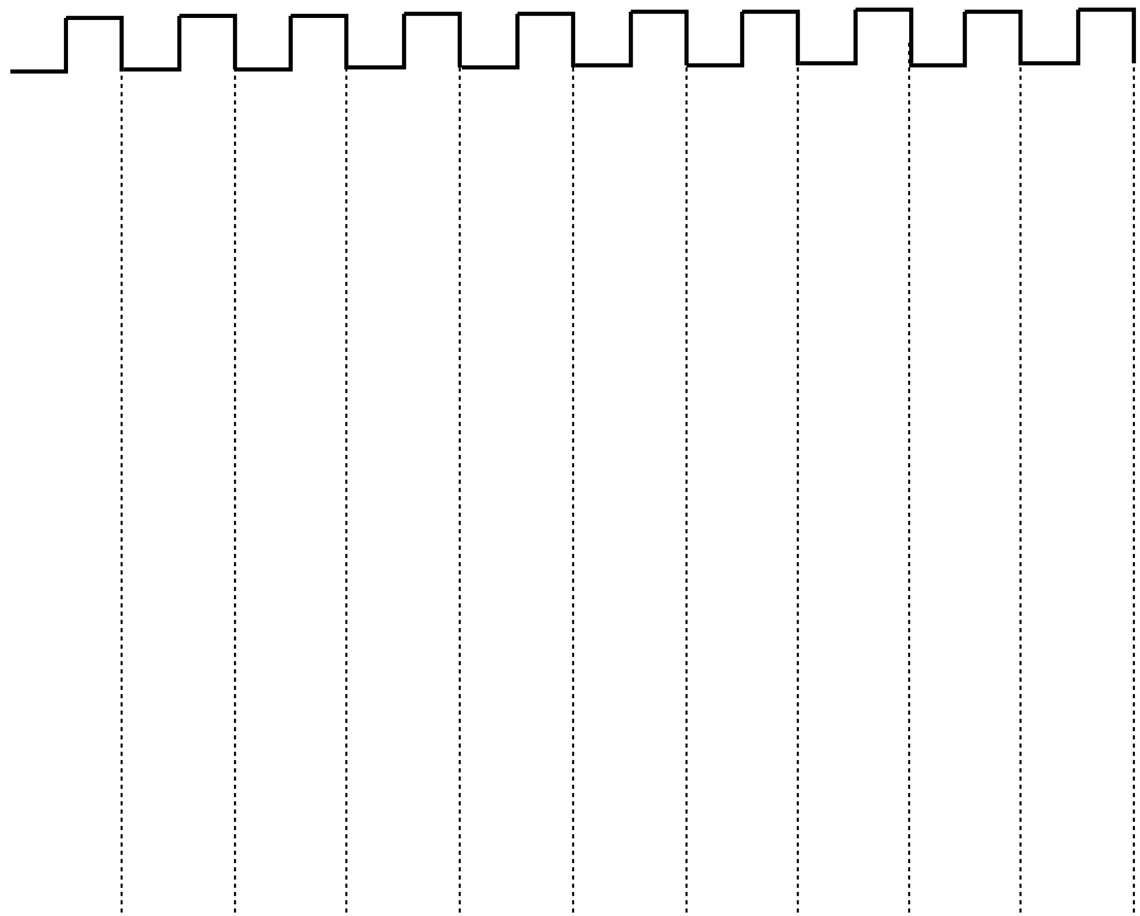
No input (Do nothing) operation
(enabled by $\overline{\text{Load}}$ AND $\overline{\text{Shift}}$)



(b) Symbol

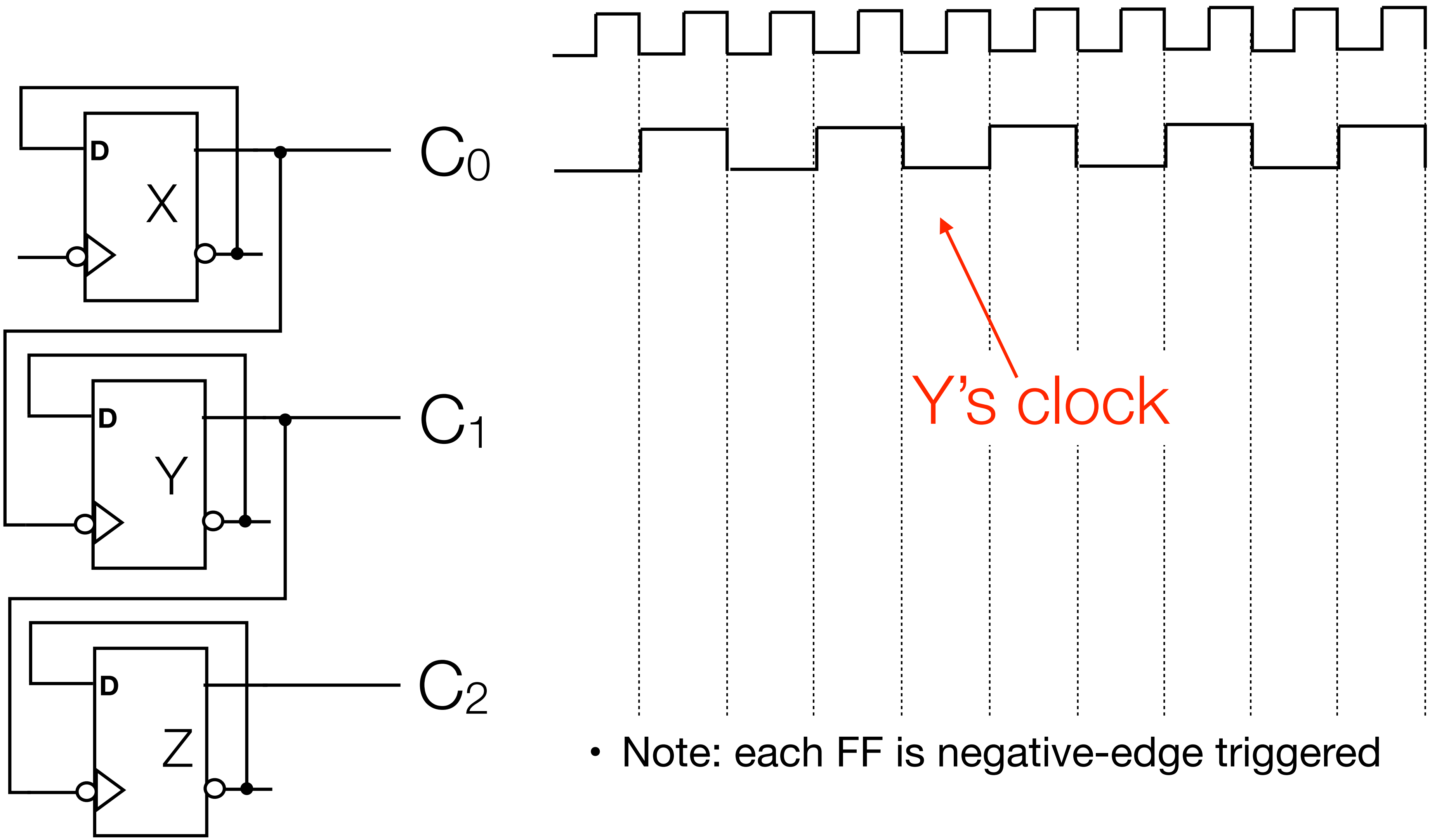
Ripple Counter





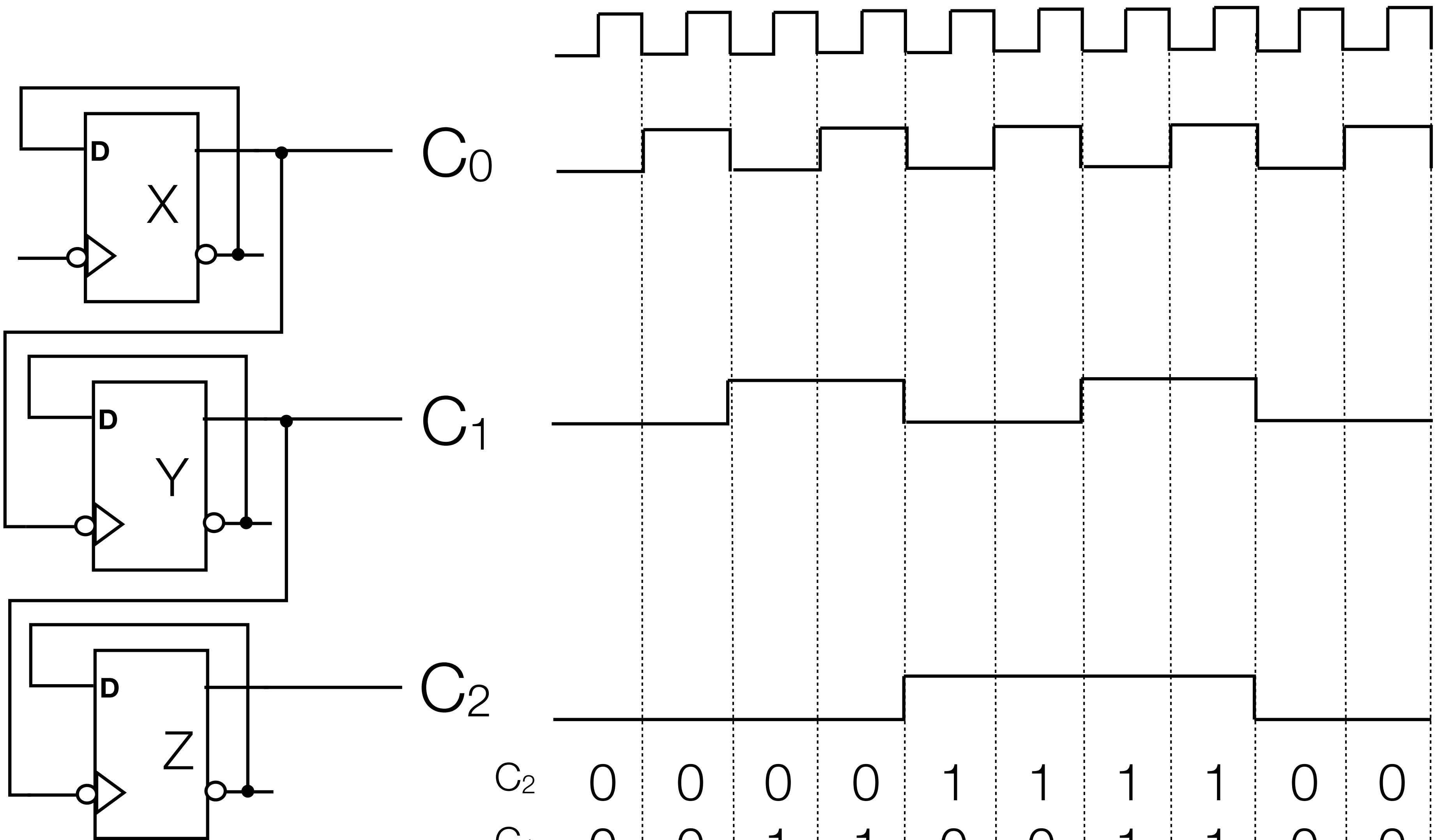
- Each flip-flop feeds Q into D: complements its value
- X's Q feeds into Y's clock, Y's Q feeds into Z's clock - why?

Ripple Counter



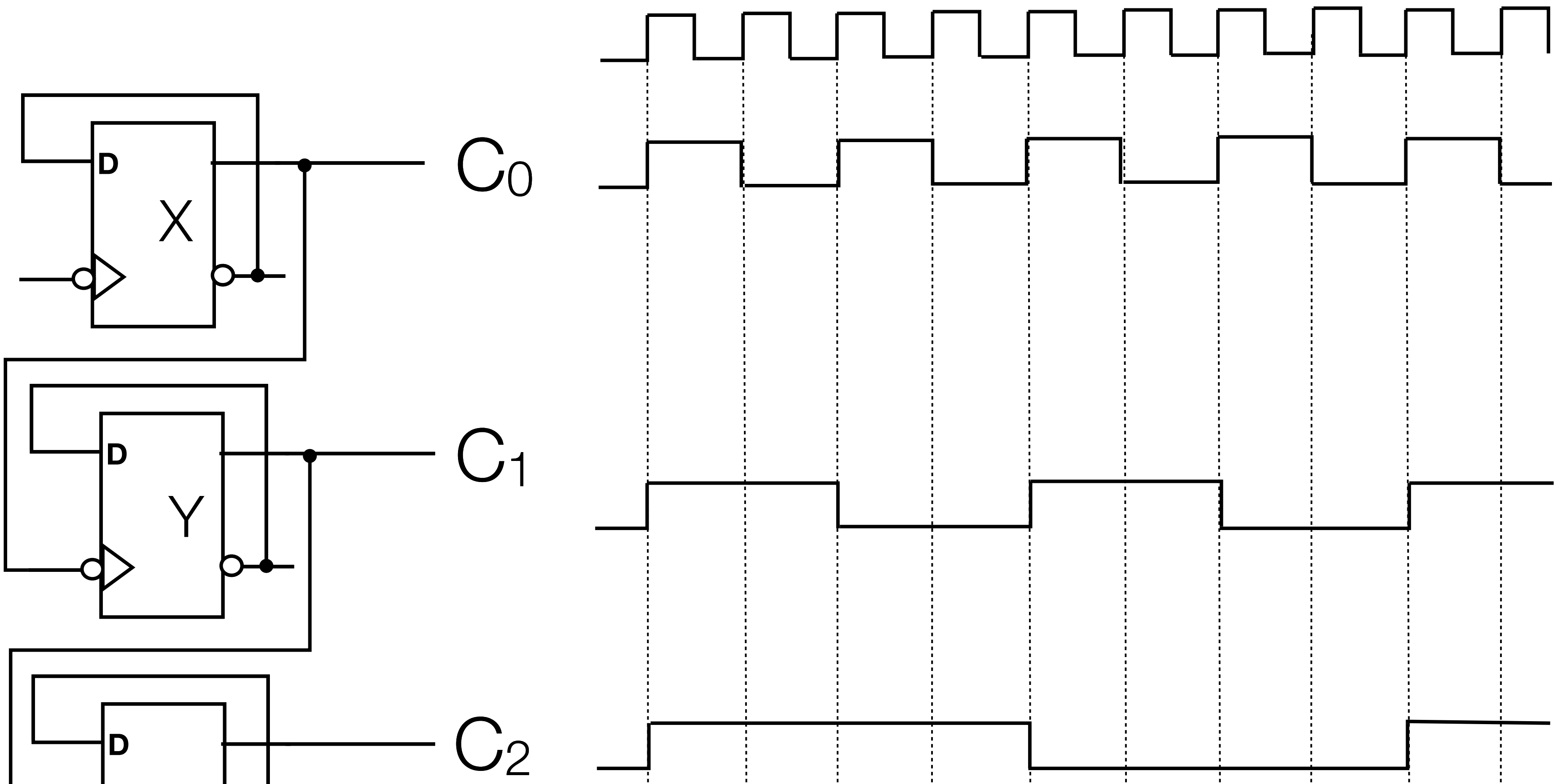
- Note: each FF is negative-edge triggered

Ripple Counter



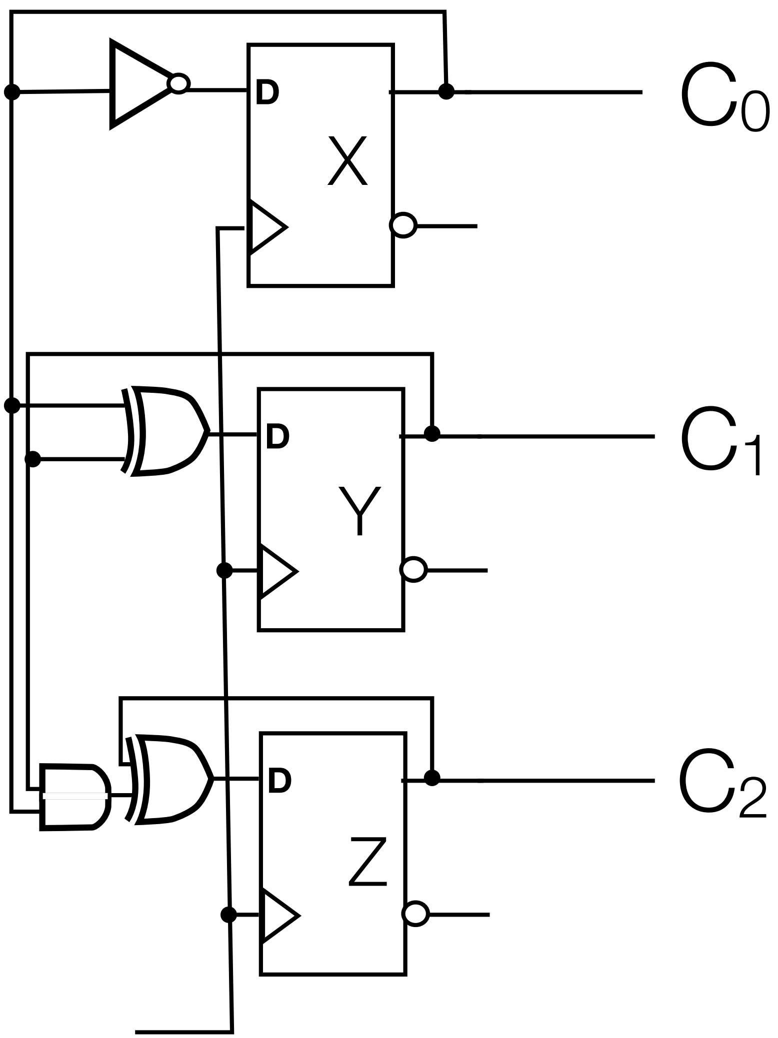
C_2	0	0	0	0	1	1	1	1	0	0
C_1	0	0	1	1	0	0	1	1	0	0
C_0	0	1	0	1	0	1	0	1	0	1

Ripple Counter with Positive Edge Triggering



C_2	0	1	1	1	1	0	0	0	0	1	1
C_1	0	1	1	0	0	1	1	0	0	1	1
C_0	0	1	0	1	0	1	0	1	0	1	0

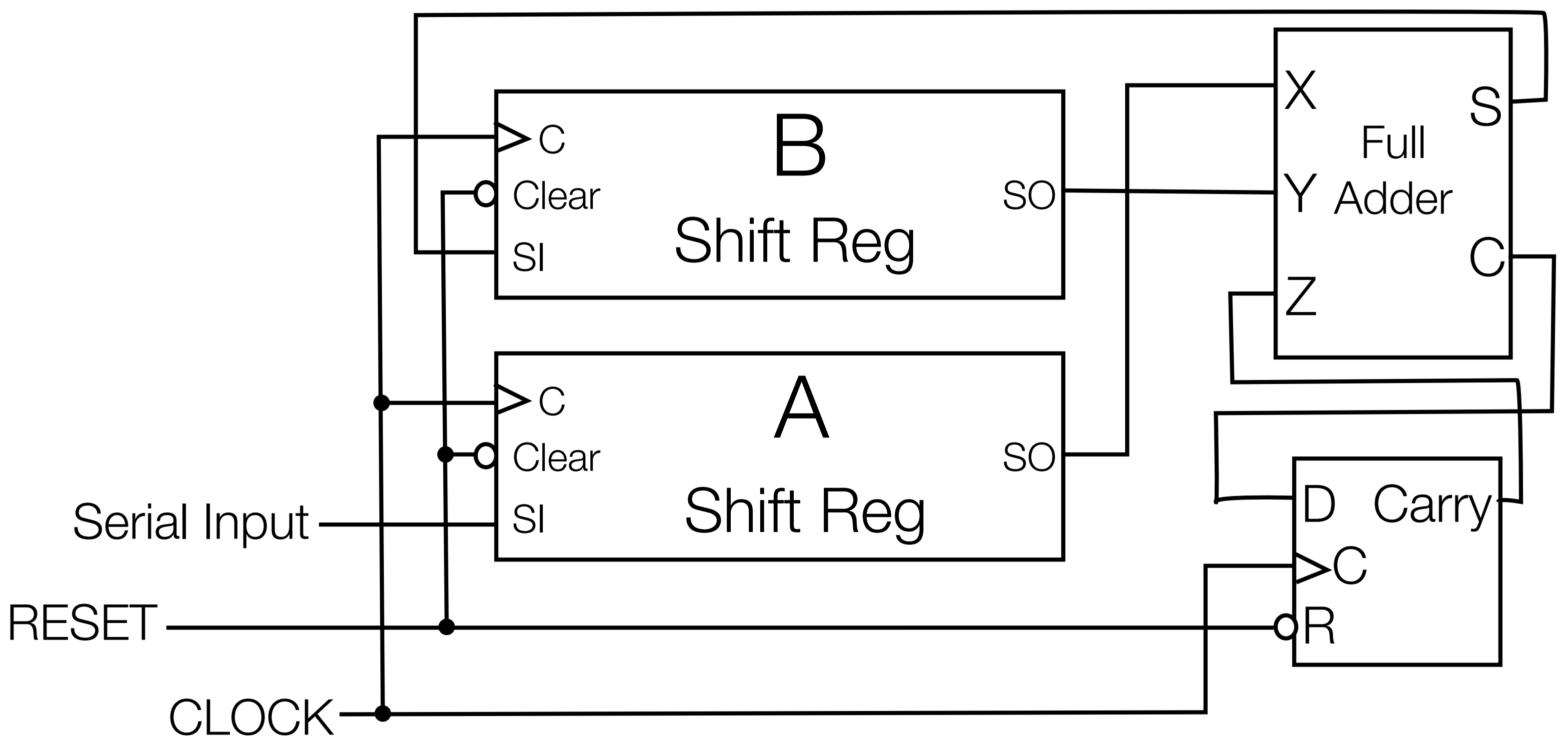
Serial Counter (counting upward)



C_2	0	0	0	0	1	1	1	1	0	0
C_1	0	0	1	1	0	0	1	1	0	0
C_0	0	1	0	1	0	1	0	1	0	1

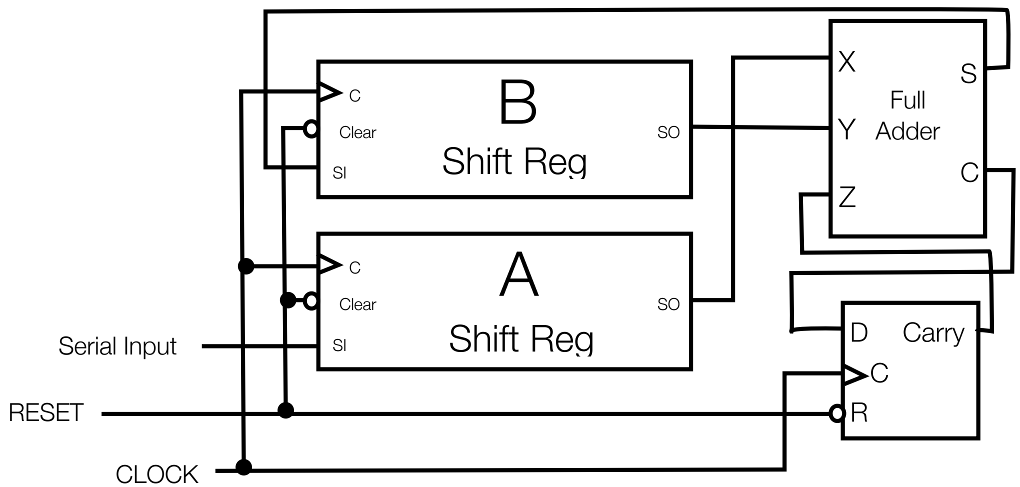
- Y(t+1) differs from Y(t) only when X(t)=1
- Z(t+1) differs from Z(t) when both X(t)=1 and Y(t)=1
- so Y changes when X (evaluates to TRUE), Z changes when XY (evaluates to TRUE)
- $X = \bar{X}$, $Y = Y \oplus X$, $Z = Z \oplus XY$
- for the jth bit (in a j+1 bit counter), $B_j = B_j \oplus B_{j-1}B_{j-2}...B_1B_0$

Serial Adder Circuit



- New Value pushed into A register
- Result pushed into B register
- If Shift Reg holds N bits, SUM starts calculating during Nth cycle (0 is initial cycle)
- Done 2N cycles later (3N total cycles)

Serial Adder Circuit Example: 0101 + 0011 = 1000



To time 7, just rolling values into registers by adding with 0's (B output)

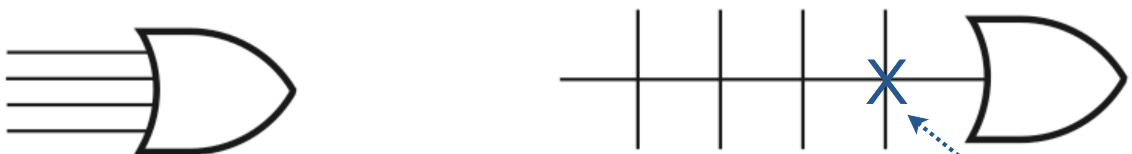
Adding starts time 8 with least significant bits

Time	In	A	B	S	C
0	1	"0000"	"0000"	0	0
1	0	"1000"	"0000"	0	0
2	1	"0100"	"0000"	0	0
3	0	"1010"	"0000"	0	0
4	1	"0101"	"0000"	1	0
5	1	"1010"	"1000"	0	0
6	0	"1101"	"0100"	1	0
7	0	"0110"	"1010"	0	0
8	X	"0011"	"0101"	0	1
9	X	"X001"	"0010"	0	1
10	X	"XX00"	"0001"	0	1
11	X	"XXX0"	"0000"	1	0
12	X	"XXXX"	"1000"	0	0

Programmable Logic Devices

Programmable Logic Devices

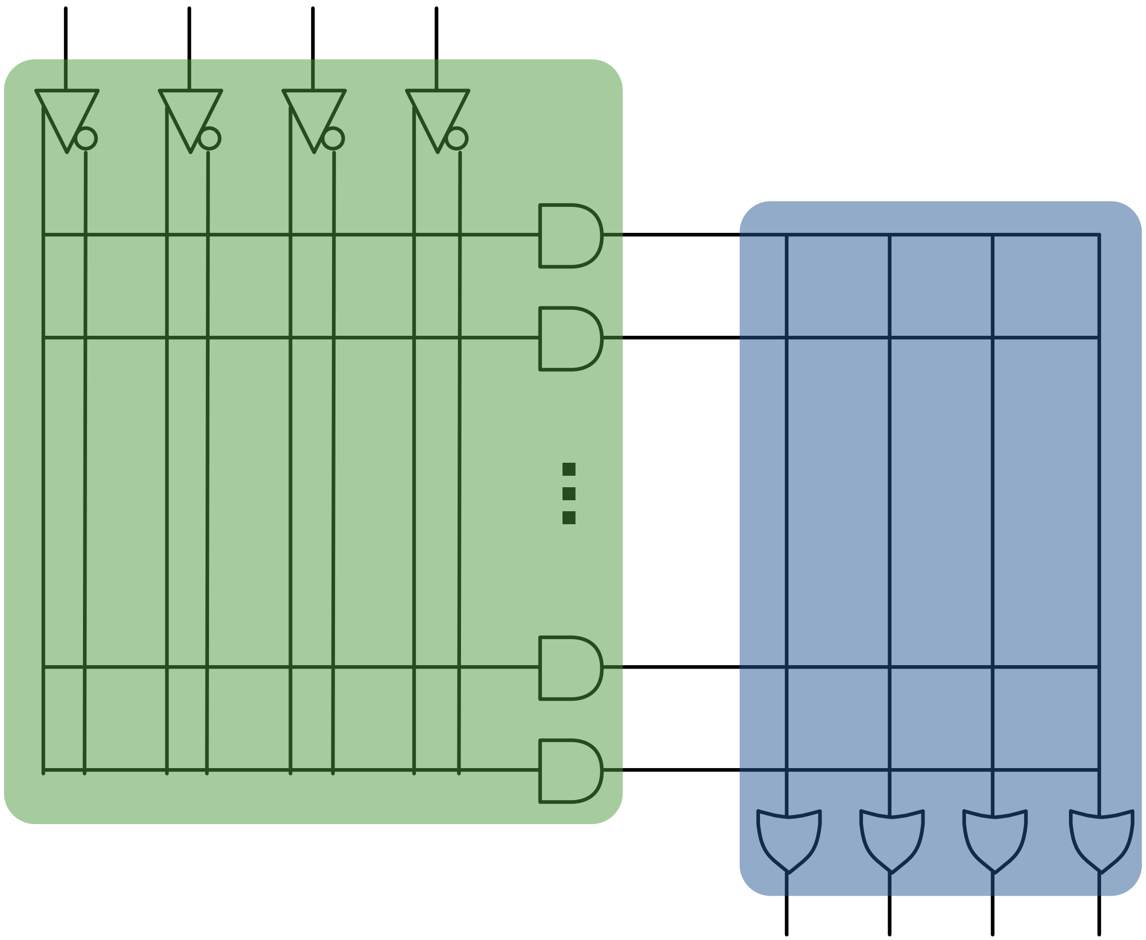
- Programmable logic devices (PLDs)
  - Structured like memories
  - Used to implement combinational logic



(a) Conventional symbol (b) Array logic symbol

  - “X” on array logic means wire connected to logic gate (e.g. above)
  - Connections can be either permanent (e.g., fuse, mask) or not (e.g., Flash)

General PLD architecture

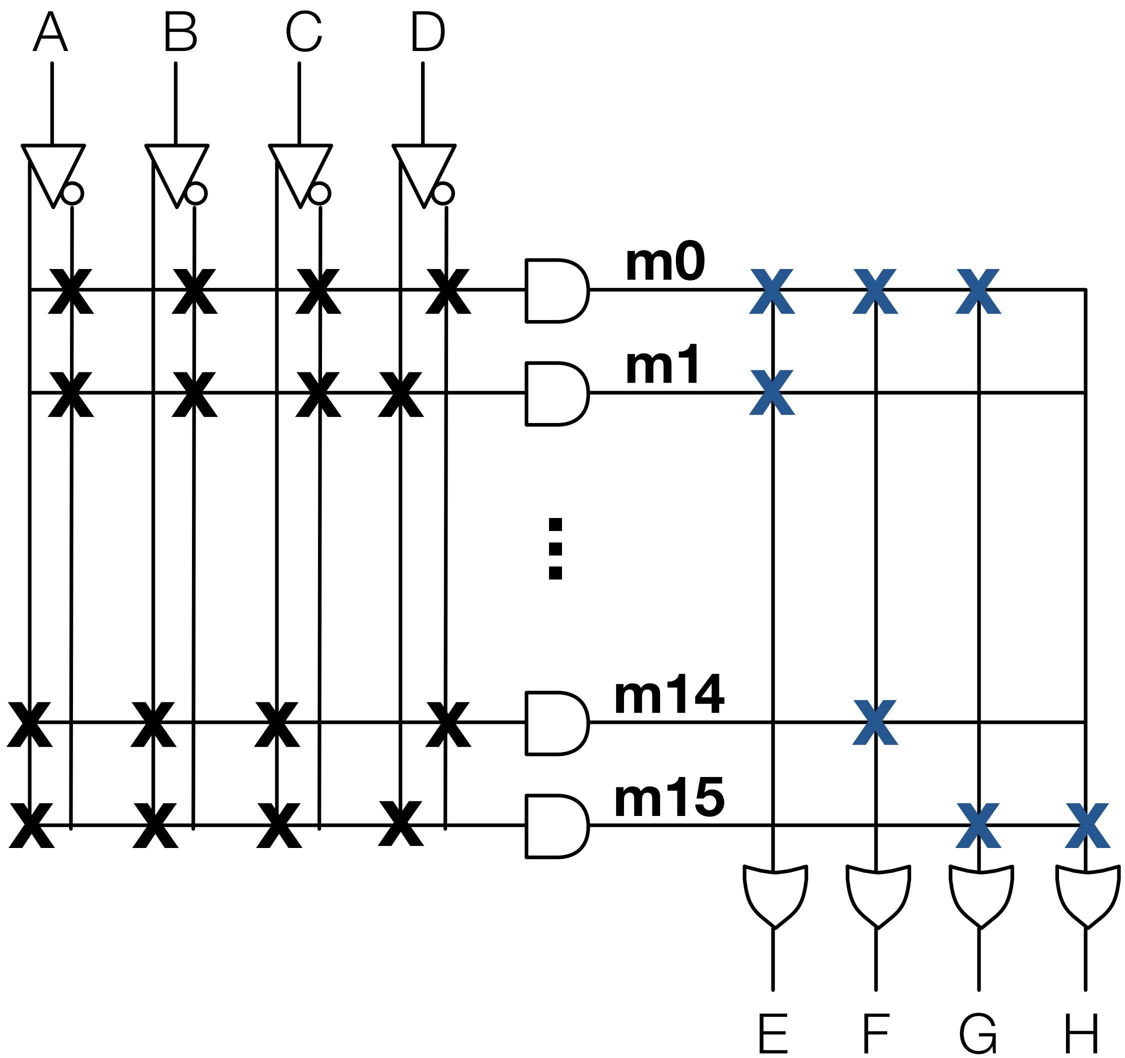


Fixed AND, programmable OR = Programmable ROM (PROM)

Programmable AND, fixed OR = Programmable Array Logic (PAL)

Programmable AND, programmable OR = Programmable Logic Array (PLA)

Programmable ROM (PROM)

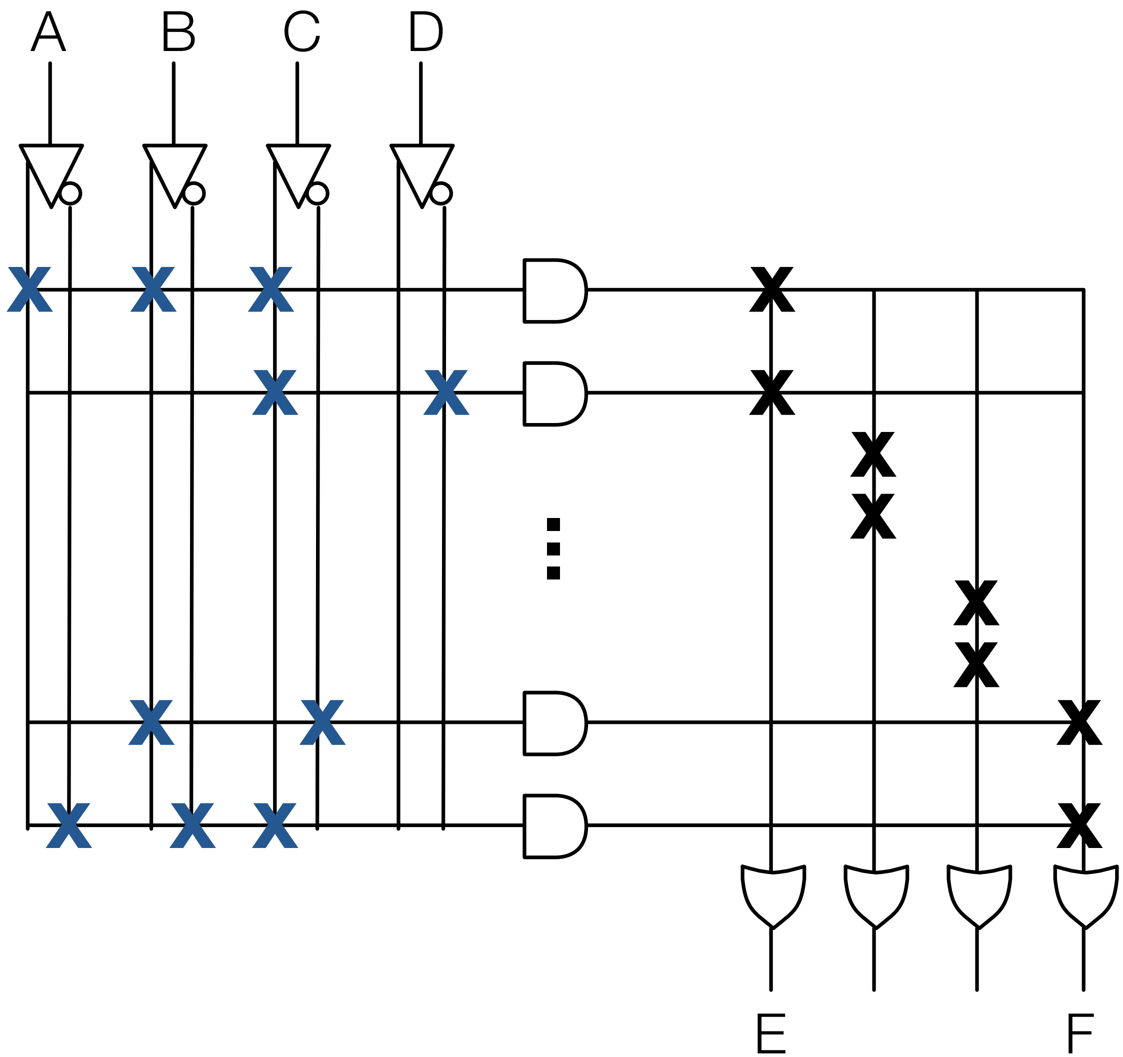


Summing of Minterms!!!

Fixed (**X**) AND, programmable (**X**) OR

e.g., $E = m0 + m1 = \overline{A}\,\overline{B}\,\overline{C}\,\overline{D} + \overline{A}\,\overline{B}\,\overline{C}D$

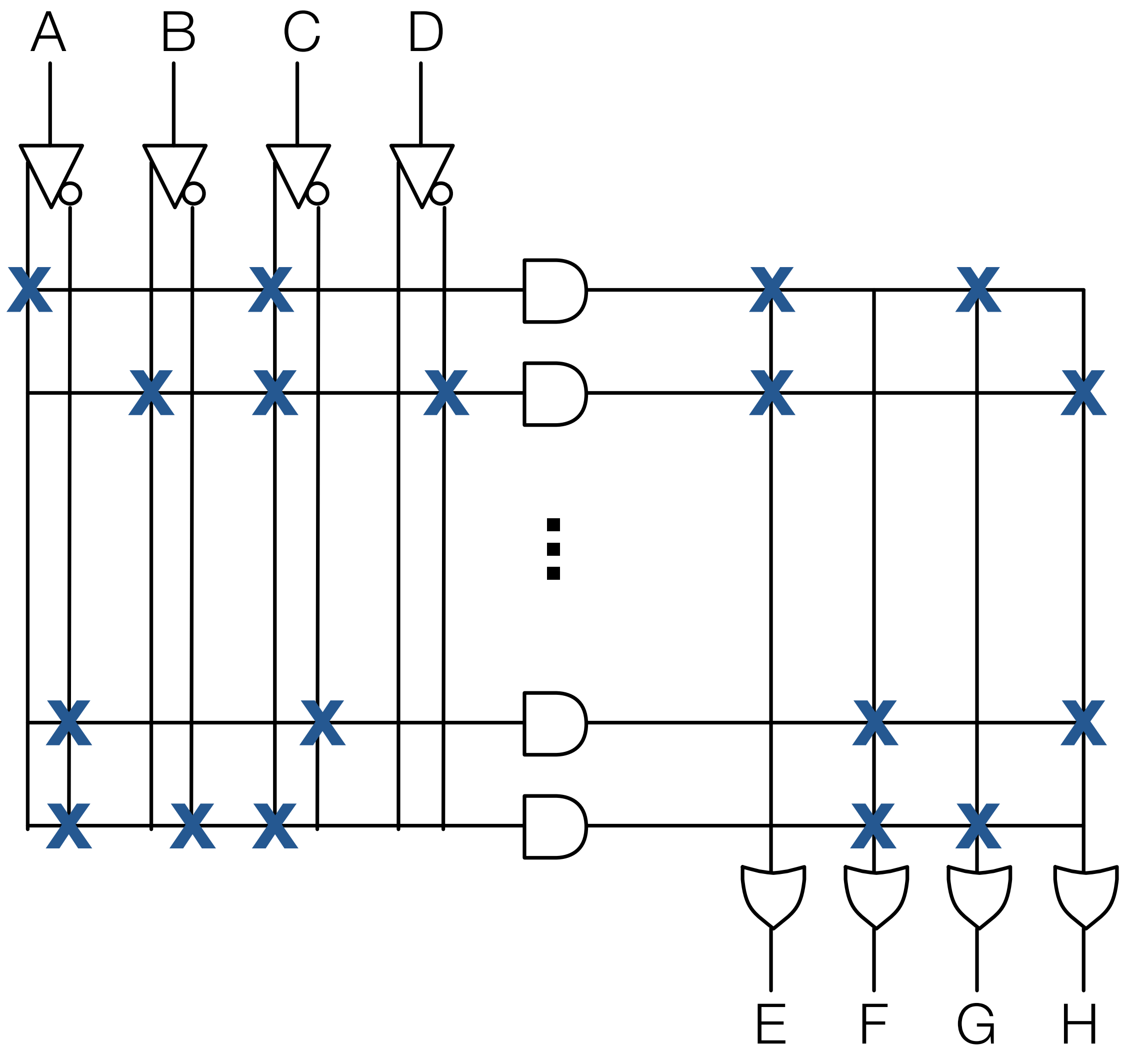
Programmable Array Logic (PAL)



Choose 2 product terms per function

Programmable (**X**) AND, fixed (**X**) OR

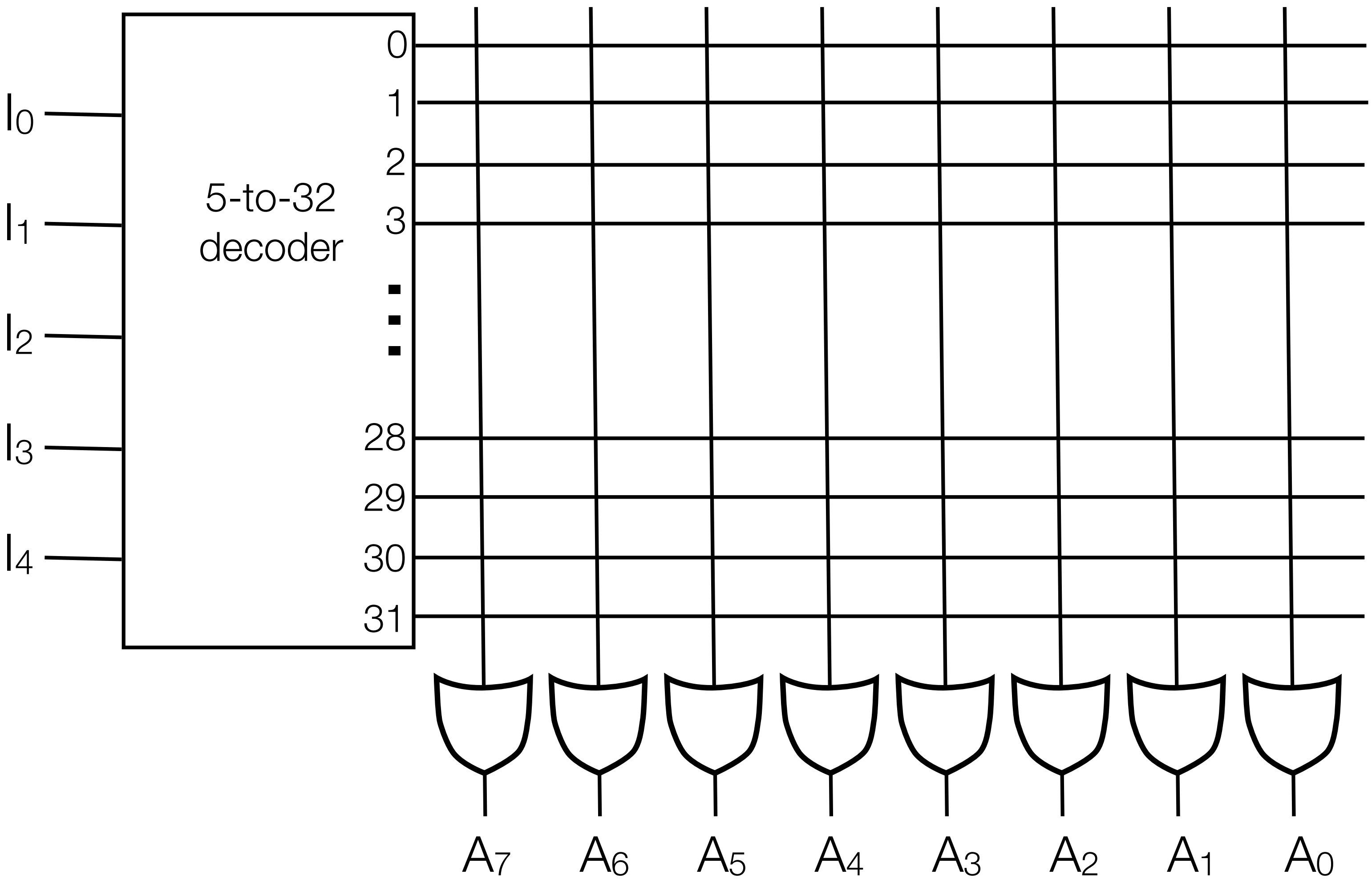
Programmable Logic Array (PLA)



Pick product terms & pick how to combine!

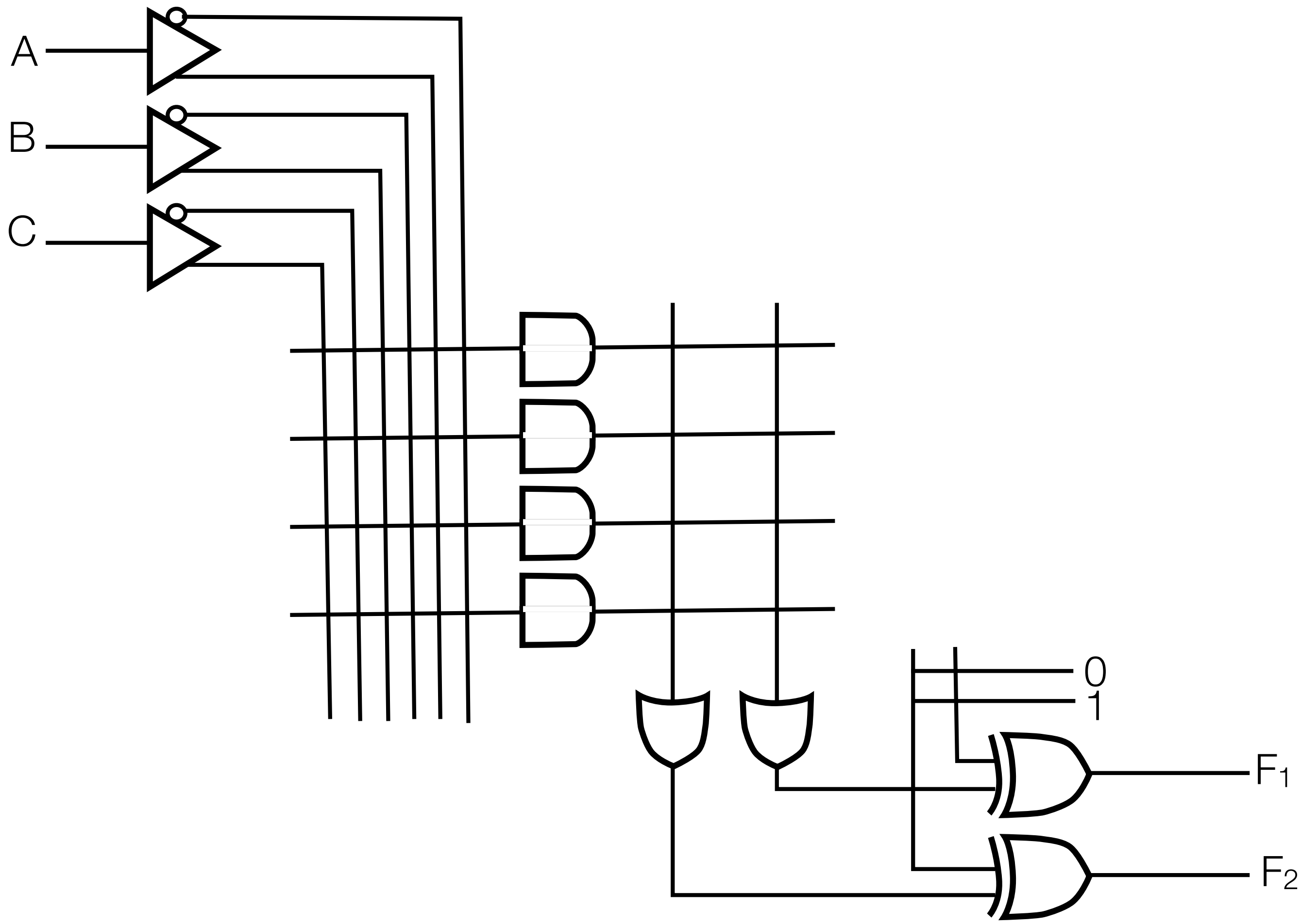
Programmable (X) AND, programmable (X) OR

ROM

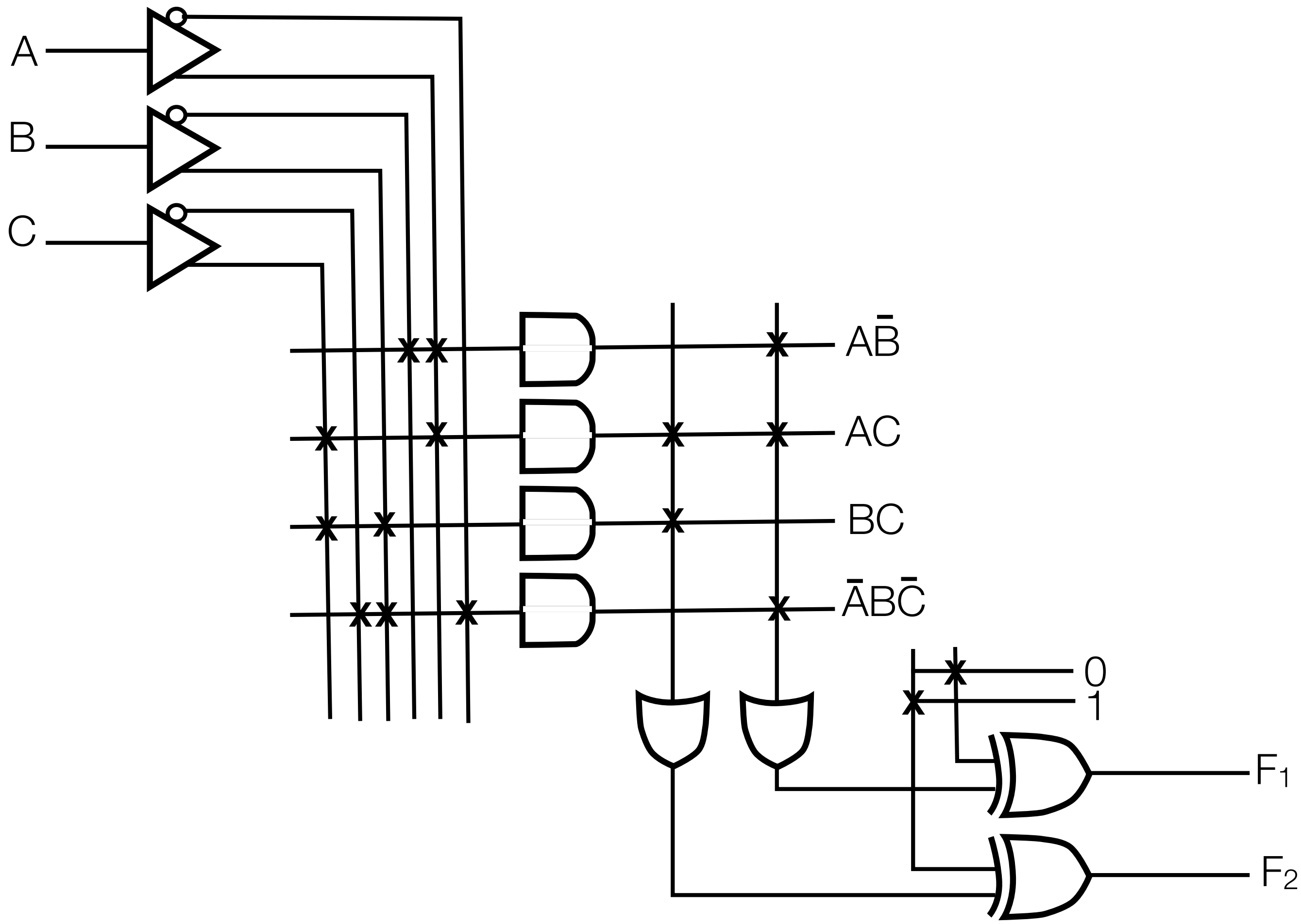


I_0
I_1
I_2
I_3
I_4
5-to-32
decoder
0
1
2
3
28
29
30
31
A_7
A_6
A_5
A_4
A_3
A_2
A_1
A_0

PLA

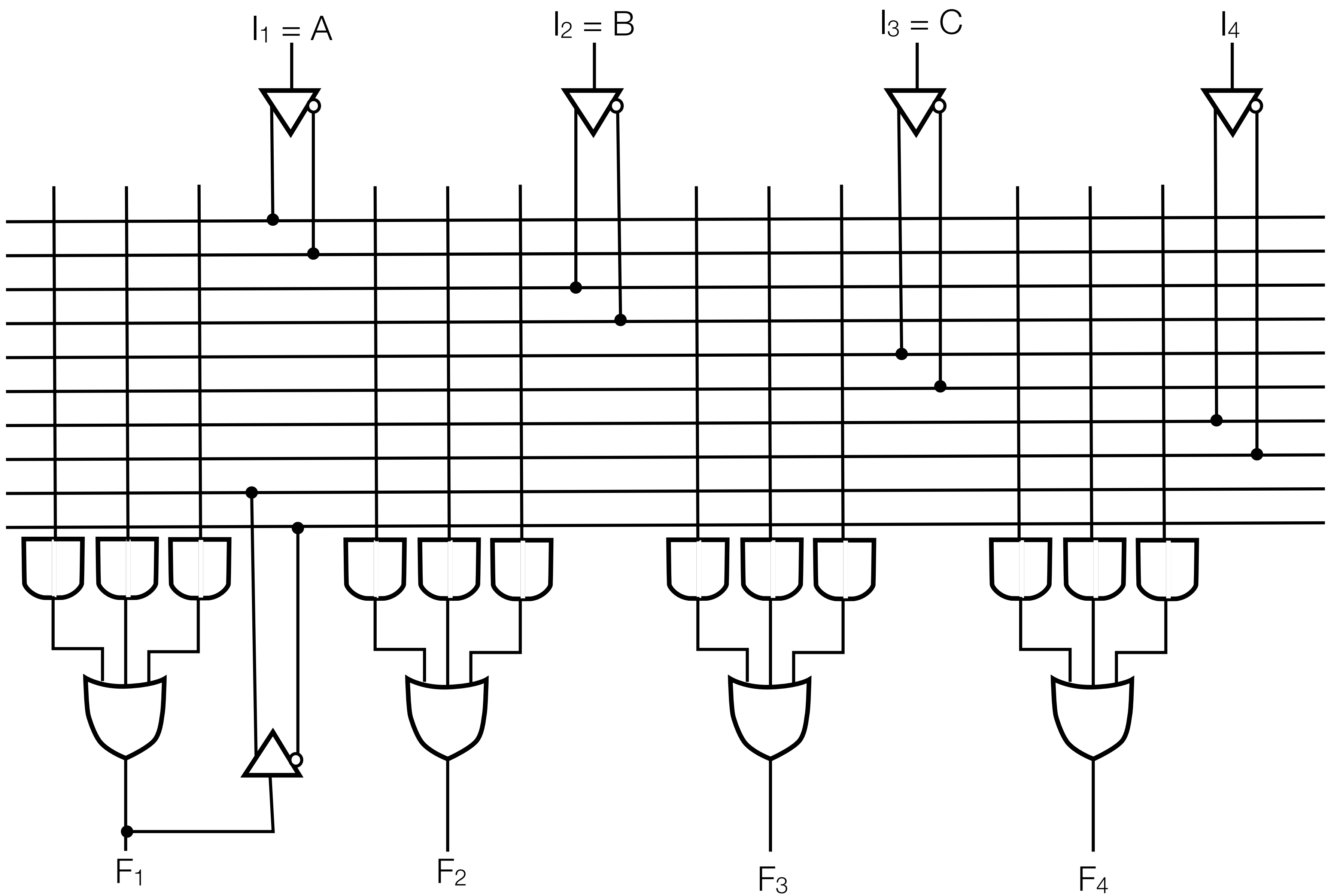


PLA Example

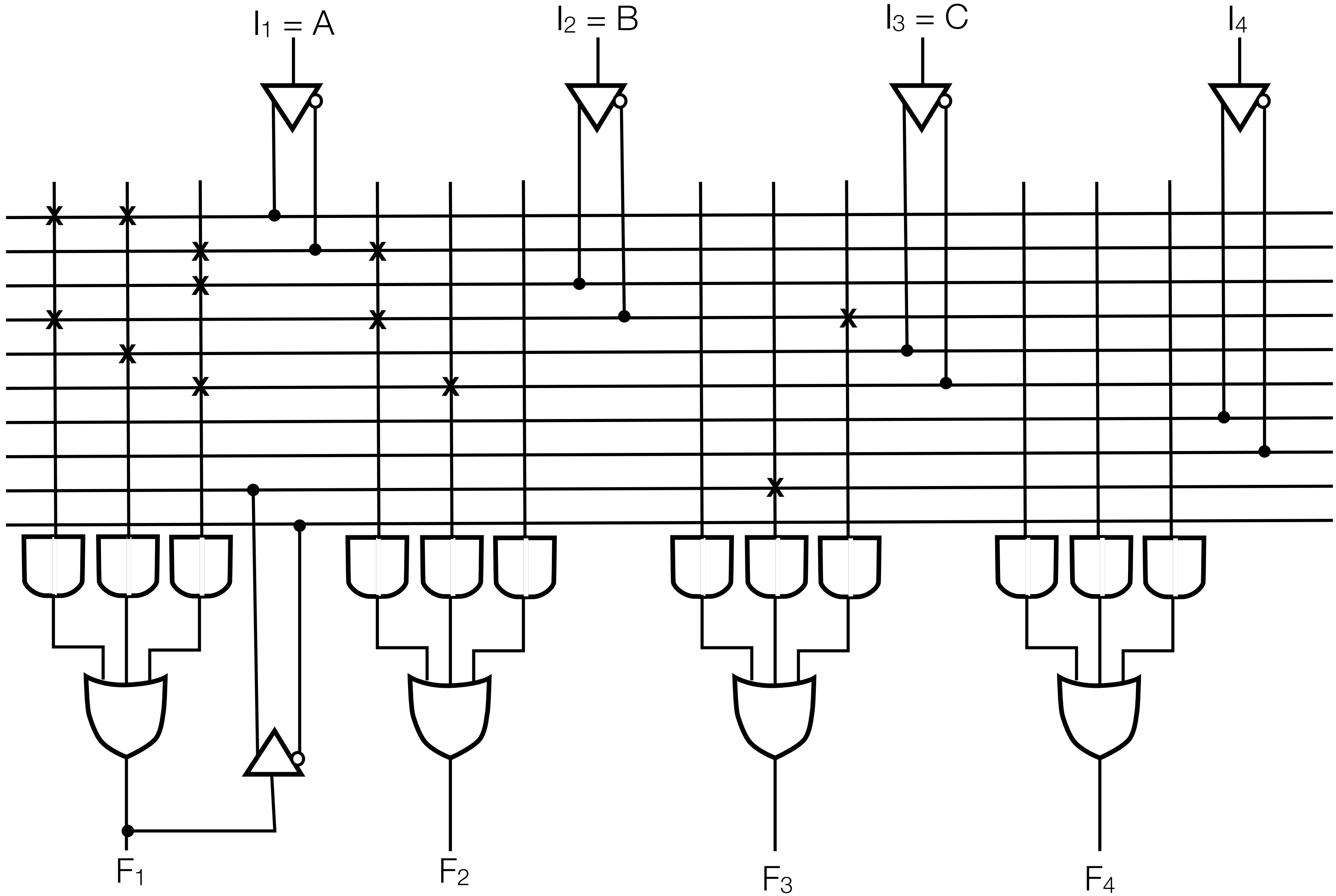


$F_1 = A\bar{B} + AC + \bar{A}B\bar{C}$ $F_2 = (AC + BC) \oplus 1 = \overline{AC + BC}$

PAL



PAL example



$F_1 = A\bar{B} + AC + \bar{A}B\bar{C}$ $F_2 = \bar{A}\bar{B} + \bar{C}$ $F_3 = F_1 + \bar{B}$